Valuing Neighborhood Amenities with Continuous Spatial Difference-in-Differences*

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Abstract: Many studies use difference-in-differences to identify the impact of an event occurring at a location in space on house prices. Empirical papers often estimate the average effect in a ring around the treatment site. However, in the hedonic model the welfare relevant parameter is the price derivative with respect to distance. I develop a formal framework for understanding what can be learned about this parameter through difference-in-differences in settings with heterogeneous treatment effects. I show that the derivative of difference-in-difference estimates with respect to distance identifies the relevant parameter under homogeneity but can be biased under heterogeneity. I introduce an alternative assumption. When effects are concave in distance from the treatment site, a natural restriction in many spatial applications, I show a treatment event's welfare impact is bounded. I find in two applications, polluting industrial plants and low income housing developments, that the estimated bounds are tight.

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1 Introduction

Many studies evaluate the impact of an event occurring at a location in space on house prices. A common research design is a ring difference-in-differences (DiD), which divides space into discrete groups and then estimates level treatment effects. For example, imagine an industrial plant opens at an address and spews pollutants into the air. To estimate the level effect for being within 1 mile of the plant, a researcher might compare the prices among houses within 1 mile of the plant to the prices among houses 1-2 miles away, before and after the plant opens. If prices are lower closer to the source of pollution, that suggests nearby homeowners do not like the plant.

Rosen (1974)'s hedonic model provides a theoretical foundation that relates preferences to prices but this relation hinges on a continuous price function's *derivative*. In equilibrium, the price derivative with respect to a housing characteristic equals household's marginal willingness to pay (MWTP) for that characteristic. In the above example, hedonic theory would say more specifically that when the slope of price is positive with distance from the plant, homeowners are willing to pay to be farther away from the plant.

I identify two issues that lead to a gap between what standard empirical approaches recover and the price derivative. First, most spatial DiD applications discretize space and only estimate level treatment effects, so the estimated parameters do not map cleanly to the price derivative identified as relevant by the theory. An exception is Diamond and McQuade (2019), who use a continuous DiD estimate's derivative with respect to distance as an estimator for homeowner MWTP to live farther away from a treatment site. Second, the recent applied econometrics literature has raised the issue that a DiD estimate's derivative does not identify the average treatment effect derivative when treatment effects are heterogeneous (Callaway et al., 2025, de Chaisemartin and D'Haultfœuille, 2025).

This paper develops a methodological foundation for evaluating the welfare-relevant derivative in the spatial DiD context. I show that under treatment effect homogeneity, the derivative of the Average Treatment Effect on the Treated (ATT) with respect to distance, as estimated in Diamond and McQuade (2019), captures the relevant parameter. However, in practice we might expect house price impacts to be heterogeneous. In this case, building on the recent applied econometrics literature, I show that the derivative of the ATTs is biased and then discuss a possible solution.

To do this, I begin by adapting the setup, notation, and causal parameters from Callaway et al. (2025)'s econometric framework for continuous DiD to the spatial setting, which distinguishes between level effects and the price derivative. In this setup, houses are untreated prior to the treatment event and then receive a continuous dose of exposure indexed by their distance from the treatment site, d. House prices follow a potential outcomes function that is continuous in d. I call the difference moving along the price function from a distance d to a control distance far enough way to be unaffected by the treatment site a *level treatment effect*. I call the difference in price from a marginal increase in distance a *price derivative*. As in binary DiD designs, I show comparisons between houses close to a treatment site and houses at a control distance identify average level price effects (ATTs) under a parallel trends assumption.

I then turn to identification of the average price derivative. When treatment effects are homogeneous, the derivative of the ATTs identifies the average price derivative. However, in many urban settings, we may expect treatment effect heterogeneity. For example, if urban planners choose the sites of polluting plants selectively, houses closer to a plant may differ in their baseline characteristics. Say houses close to a plant tend to be in poorer neighborhoods than houses farther away, and pollution affects the prices of houses in rich neighborhoods more.[1] In a counterfactual where the closer-in houses and the houses farther away were instead exposed to the same amount of pollution, the closer-in houses would lose less value on average.

When treatment effects are heterogeneous, the derivative of the ATTs equals the average price derivative plus an additional selection bias term. Marginal comparisons of ATTs at adjacent distances conflate the change in price from varying distance to the treatment site with the change in effect from varying the composition of the treatment group subpopulation. In the above example, this would conflate the change in price from varying distance to the pollution source with the change in effect from shifting the composition of housing units toward poorer neighborhoods. If the selection bias is large, then the estimated derivative can in principle even have the wrong sign.

In response, I introduce an alternative assumption and estimator to bound the average price derivative. To build intuition, I first show identification under treatment effect heterogeneity if potential outcomes are linear in distance from the treatment site. In this special case, a simple "rise over run" estimator, which divides the level price effect by its distance from the location where spillovers end, identifies the average price derivative. I then relax linearity to a concavity assumption plausible in spatial applications. For example, it may be reasonable to assume that a polluting plant's impact on price is concave in distance from the plant because chemical pollutants exhibit spatial exponential decay with distance from a pollution source. In this case, the "rise over run" estimator is a conservative lower bound on the average price derivative.

The ideas discussed above establish an econometric framework for this setting but do not relate house prices to preferences. In the second part of the theory, I connect the causal parameters and results derived in the econometric framework to the hedonic model.

To start, I extend Rosen (1974)'s insight that homeowner marginal willingness to pay equals the price derivative by connecting this relation to the average price derivative parameter. I show the average MWTP at each value of a housing characteristic equals the average price derivative within the group of houses with that value of the characteristic. In the spatial setting, the average MWTP at a distance from the treatment site equals the average price derivative at that same distance. Applying the results from the econometric framework, the derivative of the ATTs does not identify the average MWTP without imposing additional homogeneity assumptions. However, under the concavity assumption, the "rise over run" estimand lower bounds the average marginal willingness to pay for distance to a treatment site.

I then construct bounds on an aggregate welfare measure by adopting the modeling framework of Diamond and McQuade (2019)—except for their assumption of treatment effect homogeneity

[1] Such effect heterogeneity may be plausible if high income people tend to have a high willingness to pay to avoid pollution. See Banzhaf et al. (2019) for a summary of studies that find higher income people are willing to pay more to avoid pollution.

by distance. I conduct nonmarginal analysis by placing parametric assumptions on homeowners' utility functions (Bajari and Benkard, 2005) and bound the aggregate homeowner willingness to pay for the treatment site. For a polluting plant, this measure is the total amount homeowners in an affected area would be willing to pay for the plant to not exist. Using the "rise over run" estimator provides a conservative lower bound. To complete the bounds, I use a revealed preference argument as in the prior hedonic literature (Ekeland et al., 2004, Banzhaf, 2021) that upper bounds the aggregate WTP. In this paper's empirical applications these bounds are tight.

I illustrate the methodological framework in two empirical applications.[2] In the first, I revisit Currie et al. (2015)'s study of industrial plant openings catalogued in the EPA's Toxic Release Inventory (TRI). I nonparametrically estimate a plant's continuous spillover onto nearby houses' prices. I find a negative effect of 3% directly next to a plant, with effects observable out to 4 kilometers—a larger geographic area than covered by Currie et al. (2015)'s ring specification. Connecting the reduced form estimate to the model, I obtain bounds of [$13.1M, $19.5M] on aggregate homeowner WTP to avoid a polluting plant, compared with a point estimate of $16.4M under the strong assumption of homogeneous price effects.

In the second application, I use the same estimation procedure and modeling framework to estimate spillovers and WTP bounds for Low Income Housing Tax Credit (LIHTC) developments (Diamond and McQuade, 2019). I find a moderate decline in house values around LIHTC developments with houses right next to a development losing 2% of value. In contrast to the previous literature, I find less evidence of heterogeneous price effects by neighborhood income level. I obtain bounds of [$6.7M, $10.5M] on homeowner WTP to avoid a low income housing development. Placing the two applications on a common scale reveals the typical externality from a toxic plant is roughly twice as large as the externality from a low income housing development.

A *Relation to the literature*

Many applied studies use a spatial DiD design, but few articles focus on the econometric framework. Most studies run a standard DiD regression after specifying a binary threshold or a few discrete thresholds to form distance rings. Butts (2023) formalizes this multi-ring approach with a data-driven choice of rings.[3] In contrast, Diamond and McQuade (2019) estimate a continuous price effect and connect the continuous estimate to a structural hedonic model to estimate welfare impacts. This paper also estimates a continuous effect but uses an alternative estimation procedure with data-driven tuning parameter choices following Cattaneo et al. (2024) and Chen et al. (2024). To my knowledge this is the first paper to consider treatment effect heterogeneity by distance to the treated site, which is key to interpreting these estimates in a hedonic model. I provide an econometric framework that permits treatment effect heterogeneity and explicitly connects the estimands to the hedonic model.

This paper's econometric framework builds off the recent applied econometrics literature on DiD with a continuous treatment in the presence of heterogeneous treatment effects. I start by showing how Callaway et al. (2025)'s framework fits this spatial setting. As in recent work by

[2] I also estimated local price effects from sex offender move ins similar to Linden and Rockoff (2008) and Pope (2008) but found evidence of parallel trends violations in my sample.

[3] See Butts (2023) for more description of earlier studies on multi-ring approaches.

D'Haultfœuille et al. (2023) and de Chaisemartin et al. (2025), I then exploit a connection between DiD and an older literature on panel data correlated random coefficient models (Chamberlain, 1982, 1992, Graham and Powell, 2012).[4] This older literature shows the structural derivative can be identified under heterogeneous treatment effects if the researcher imposes functional form restrictions. While Callaway et al. (2025) do not consider functional form restrictions on potential outcomes, I show that if price effects are linear in distance to a treatment site, then average marginal effects are identified. I then further develop a bounding argument if linearity is violated but price effects are concave in distance from the plant, a natural assumption in many spatial settings.

A few other recent papers consider similar bounding arguments based on shape restrictions (D'Haultfœuille et al., 2023, de Chaisemartin et al., 2025). Most closely related, de Chaisemartin et al. (2025) consider an "average of slopes" estimand similar to the rise over run estimator described above and note that under concavity the average of slopes bounds counterfactual treatment effects. I use a similar insight to lower bound average marginal effects, justifying concavity by using the context-specific assumption of spatial decay with distance. In contrast to de Chaisemartin et al. (2025), I also construct an upper bound by exploiting additional restrictions from the economic model's equilibrium conditions.

This paper also contributes to a large literature on hedonic property valuation, originating with Rosen (1974). First, I formalize how DiD estimates relate to homeowner MWTP in Rosen (1974)'s first order condition under treatment effect heterogeneity. Then, I employ a structural hedonic model closely following Diamond and McQuade (2019)'s model who calculate the total willingness to pay of nearby homeowners for a LIHTC development. Relative to Diamond and McQuade (2019), I connect these welfare estimands explicitly to the causal parameters defined in an econometric framework that allows treatment effect heterogeneity. Another important study of hedonics in difference-in-differences settings is Banzhaf (2021) who shows that reduced form price effects can measure an informative lower bound on the general equilibrium welfare change. However, it is worth noting that adding up reduced form price effects to form a lower bound is anti-conservative when price changes are negative (because both quantities are negative, the inequality reverses for absolute values). This paper shows how the hedonic modeling of both Diamond and McQuade (2019) and Banzhaf (2021) form complementary statistics that bound both sides of a welfare measure in the presence of treatment effect heterogeneity. Finally, by quantifying aggregate price and welfare impacts, this paper's method allows for comparison of hedonic values across completely different kinds of amenities.

The rest of this paper proceeds as follows. Section 2 develops the econometric framework and discusses identification of the price derivative. Section 3 presents the structural hedonic model. Section 4 outlines the estimating procedure. As an example, Section 5 applies the procedure to estimate the local welfare impact from a polluting industrial plant (Currie et al., 2015). Section 6 compares price and preference estimates across applications. Section 7 concludes.

[4]See Arkhangelsky and Imbens (2024) for more details on the equivalence between panel data models and the recent applied econometrics literature on DiD.

2 Econometric framework

I now develop an econometric framework for the spatial DiD setting. Section 2.A introduces the setup, notation, and estimands. Section 2.B establishes sufficient assumptions to identify average level treatment effects. Section 2.C develops intuition and shows identification of the derivative with homogeneous treatment effects. Section 2.D discusses challenges with identifying the average treatment effect derivative when there is treatment effect heterogeneity. Section 2.E shows the average derivative can be lower bounded under shape restrictions.

A *Setup and notation*

I follow the notation in Callaway et al. (2025). Though this notation applies to many settings, I use the effect of toxic plant openings on local house prices as a running example to fix ideas.

For ease of exposition, I focus on the problem of estimating the impact of a single plant opening. I denote by period $t = 1$ the observation prior to the plant opening and $t = 2$ the period afterwards. In practice, in my applications, I observe many different plant openings. I therefore apply the approach described here for each of the plant openings separately and then aggregate them, as I discuss in more detail in Section 3.B.

A house's location determines how exposed it is to the polluting plant. In the post period, an industrial plant opens at a specific location and spews toxic chemicals into the air. Housing units receive a dose of exposure to the plant indexed by their distance from the toxic plant, $D = d$ (while the index is increasing, intensity of exposure actually decreases with distance). Let $\mathcal{D}$ denote the continuous support of D.

Define $Y_{j,t}(d)$ as the potential outcome for house j in time period t. This is the house price (or log transformed house price) if house j was d distance away from a treatment site in time period t. The realized outcome is $Y_{j,t} = Y_{j,t}(D_j)$. In contrast to most DiD applications, in the spatial context we often do not know ex ante which units are untreated, so to define a control group we must assume the plant has no effect on house prices beyond some distance.

Assumption 1 (Limit on the spatial extent of treatment effects). *Assume there is a known constant $\overline{d} \in (0, \infty)$ that bounds the distance that a treatment site affects house prices, so units farther than $\overline{d}$ from a plant are untreated in every time period. That is, for any $d \geq \overline{d}$, $Y_{j,t}(d) = Y_{j,t}(\overline{d})$.*

Later, I discuss how researchers can corroborate this assumption by visually inspecting the observed price effects. Henceforth, I denote a unit's untreated potential outcome as $Y_{j,t}(\overline{d})$. Lastly, denote $\Delta Y_j = Y_{j,2} - Y_{j,1}$ as the change over time for a housing unit.

Callaway et al. (2025) define two treatment effect estimands of interest: the *level treatment effect* $Y_{j,t}(d) - Y_{j,t}(\overline{d})$ and the *causal response* $\frac{\partial Y_{j,t}(d)}{\partial d}$, the derivative of the potential outcomes function. In the DiD framework, we consider averages of these types of effects. For treated units,

$$ATT(d|d') = \mathrm{E}[Y_t(d) - Y_t(\overline{d})|D = d']$$

is the average effect induced by being d distance from the plant in time period t for units that are located d' from the plant. When $d = d'$, this is the observed ATT. Similarly, define the average causal response parameter as

$$ACRT(d|d') = \frac{\partial ATT(l|d')}{\partial l}|_{l=d} = \frac{\partial \mathrm{E}[Y_t(l)|D=d']}{\partial l}|_{l=d}$$

The average causal response is the ATT function derivative while holding the treatment group subpopulation fixed. For identifying individuals' preferences according to the hedonic model's equilibrium condition, we will see the object of interest is the average price derivative, so $ACRT(d|d)$ is the key estimand.

B *Identification of the ATT*

In this subsection, I show the ATT is identified under standard DiD assumptions following Callaway and Sant'Anna (2021) and Callaway et al. (2025). First, I set up the data as repeated cross sections of a geographic area across time periods, i.e., a distance ring surrounding a treatment site in a given period, which is common in practice. Let X denote a house's pretreatment covariates. The following assumption ensures identification with repeated cross sections:

Assumption 2 (Random sampling for repeated cross sections). *Conditional on $T = t$, the data are independent and identically distributed from the distribution of (Y_t, D, T, X), for $t = 1$ and $t = 2$ with (D, X) invariant to T.*

This assumption rules out changes to the composition of housing characteristics during the study period, which is often cited as an important threat to identification in the context of housing sales.

Assumption 3 (Support and continuously differentiable treatment). *(a) The support of D is $\mathcal{D} = [0,d_U]$ with $0 < \overline{d} < d_U$, where $d_U < \infty$ is an upper bound on the spatial extent considered. Furthermore, $\mathrm{P}(D > \overline{d}) > 0$ and $dF_D(d) > 0$ for all $d \in [0,\overline{d}]$ where $dF_D(d)$ is the density of housing units by distance. (b) Assume $\mathrm{E}[\Delta Y|D = d]$ is continuously differentiable in d on $[0,\overline{d}]$.*

Assumption 4 (No anticipation and observed outcomes). *For all units, $Y_{j,t=1} = Y_{j,t=1}(\overline{d})$ and $Y_{j,t=2} = Y_{j,t=2}(D_j)$.*

Assumption 5 (Parallel trends). *For all $d \in \mathcal{D}$,*

$$\mathrm{E}[Y_{t=2}(\overline{d}) - Y_{t=1}(\overline{d})|D = d] = \mathrm{E}[Y_{t=2}(\overline{d}) - Y_{t=1}(\overline{d})|D = \overline{d}].$$

Assumption 3 defines the support of treatment. Assumption 4 says that units do not anticipate treatment in ways that affect their outcomes, so before treatment we observe units' untreated potential outcomes, and after treatment we observe the potential outcome for the actual dose that unit j experienced. Assumption 5 is a parallel trends assumption. In contrast to a binary DiD, this parallel trends assumption must hold across the continuum of distances $\mathcal{D}$, requiring that the average paths of untreated potential outcomes are the same for all distances.

Under these assumptions, the Average Treatment Effect on the Treated $ATT(d|d)$ is identified. As in Callaway et al. (2025),

Theorem 2.1 *Under Assumptions 1 to 5, $ATT(d|d)$ is identified.*

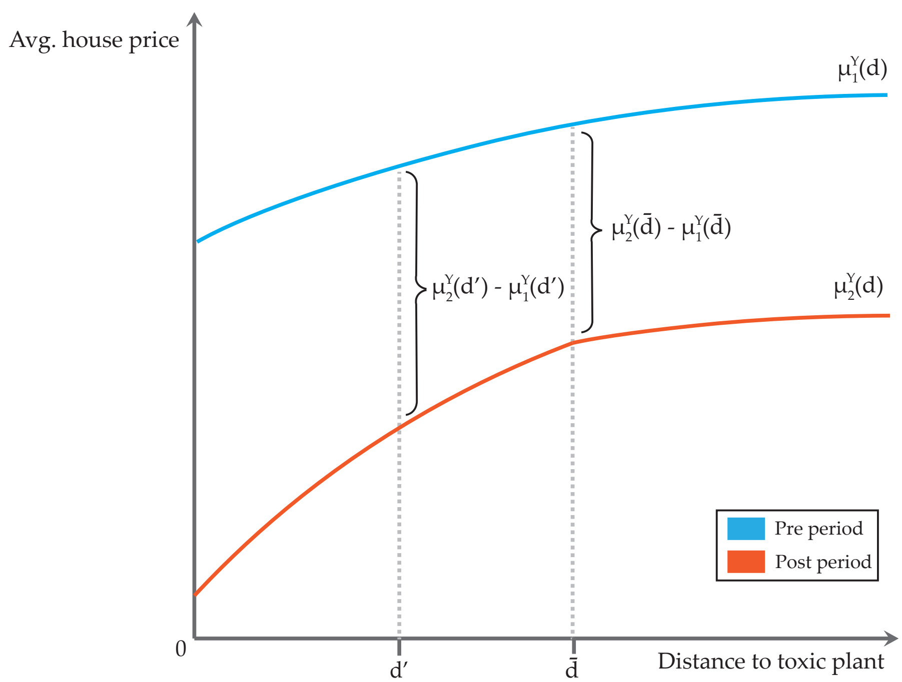
An explicit expression and all proofs for this section are in Appendix B.

C *Intuition and derivative identification with homogeneous treatment effects*

To build intuition, I motivate graphically how a continuous spatial DiD estimator gives the $ATT(d|d)$ function discussed above. I then consider identification of the average price derivative in the simpler case with homogeneous treatment effects and a fully separable price function. In this case, the derivative of the DiD estimand is the hedonic price function derivative and therefore identifies homeowners' MWTPs.

Figure 1 presents a stylized diagram of average housing prices before and after a plant's opening as a function of distance to the plant's site. The blue line plots housing prices in the pre period and the orange line plots prices in the post period. Housing prices are lower close to the site even before the plant opens reflecting that the plant is sited in an area with systematically lower residential land quality. In the post period, houses depreciate in value everywhere (perhaps owing to macroeconomic conditions), but those closer to the plant depreciate more, with the additional contribution of pollution from the plant to housing values.

Figure 1: Continuous spatial difference-in-differences



Note: This figure illustrates the construction of a price effect from a continuous spatial difference-in-differences. The figure plots average house price on distance to the toxic plant's site, before and after the plant's opening. The blue line shows the gradient in the pre period and the orange line shows the gradient in the post period. The vertical distance between the gradients, marked by curly braces, shows the price change over time at two different locations d' and $\bar{d}$, a distance at which pollution from the plant is no longer relevant. The DiD estimator at d' is $DiD(d') = (\mu_2^Y(d') - \mu_1^Y(d')) - (\mu_2^Y(\bar{d}) - \mu_1^Y(\bar{d}))$.

The black braces illustrate the construction of a price effect estimate for houses located d' from the plant using a DiD strategy. Denote average house prices at a distance d by $\mu_t^Y(d) = \mathrm{E}[Y_t|D = d]$. The average price depreciation $(\mu_2^Y(d') - \mu_1^Y(d'))$ for houses at d' is larger in magnitude than the price depreciation for houses at $\bar{d}$ because d' is closer to the new plant. Take the difference of the two price changes and assume parallel trends between the two locations:

$$\begin{aligned} DiD(d') &= (\mu_2^Y(d') - \mu_1^Y(d')) - (\mu_2^Y(\overline{d}) - \mu_1^Y(\overline{d})) \\ &= (\mathrm{E}[Y_2(d')|D = d'] - \mathrm{E}[Y_1(\overline{d})|D = d']) - (\mathrm{E}[Y_2(\overline{d})|D = \overline{d}] - \mathrm{E}[Y_1(\overline{d})|D = \overline{d}]) \\ &= \mathrm{E}[Y_2(d') - Y_2(\overline{d})|D = d'] \quad \text{(by parallel trends).} \end{aligned} \tag{1}$$

Then, the final expression is the Average Treatment Effect on the Treated, $ATT(d'|d')$. This is the average price effect induced by being d' away from the plant among the subpopulation of housing units that are d' away from the plant. Evaluating $DiD(d')$ at each location $d' \in [0, \overline{d}]$, generates a continuous $ATT(d|d)$ function over the interval $[0, \overline{d}]$.

For a house j, consider the level price effect induced by having a polluting plant d away, $\tau_j(d) := Y_{j,2}(d) - Y_{j,2}(\bar{d})$. For now, assume homogeneous price effects, so $\tau_j(d) = \tau(d)$ for all houses j. Given this homogeneity assumption, $ATT(d|d)$ is $\tau(d)$:

$$\begin{aligned} ATT(d|d) &= \mathrm{E}[Y_2(d) - Y_2(\bar{d})|D = d] \\ &= \mathrm{E}[\tau(d)|D = d] \quad \text{(by definition of } \tau(d)) \\ &= \tau(d) \quad \text{(by treatment effect homogeneity).} \end{aligned}$$

In this subsection, we take $\tau(d)$ as the hedonic price function. Later, I define the separability and parallel trends assumptions needed for this to hold formally.

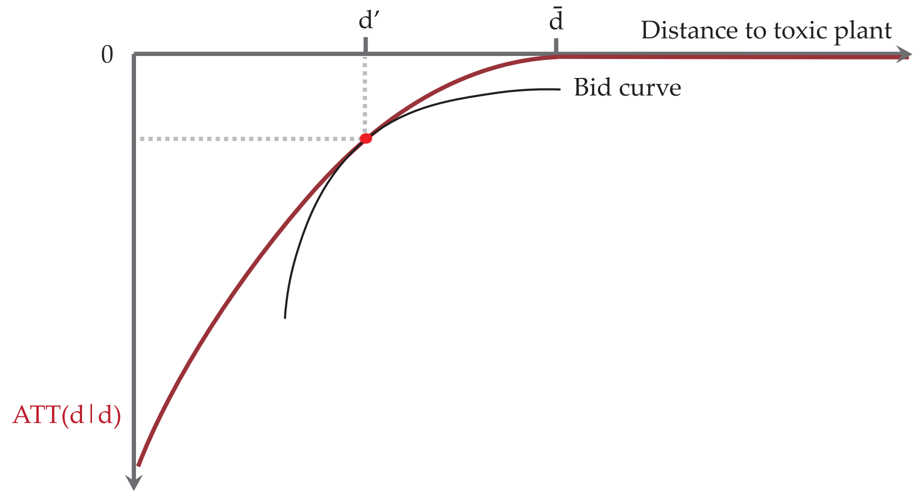
The red line in Figure 2 shows the $ATT(d|d)$ function from (1), based on the differences between the pre and post period pricing gradients in Figure 1. The key insight of Rosen (1974)'s model is that homeowners' bid (indifference) curves lie tangent to the hedonic price function at each value of a housing characteristic. Figure 2 assumes that the difference of the two pricing gradients recovers the hedonic price function and illustrates this tangency condition for one evaluation point d'. The black line shows the bid curve of homeowners who live d' from the plant in the post period lying tangent to the price function at d'. As a result, the derivative of $ATT(d|d)$ evaluated at a given d' equals homeowners' marginal willingnesses to pay for distance to the treatment site,

$$\frac{\partial ATT(d|d)}{\partial d}|_{d=d'} = \frac{\partial \tau(d)}{\partial d}|_{d=d'} = \mathrm{MWTP}|_{d=d'}.$$

Note also that in this case the average causal response estimand defined confers with the derivative of ATTs and we have $ACRT(d|d) = \frac{\partial \tau(d)}{\partial d}$. In this paper, the derivative of the price function is the key estimand. A more detailed formulation of Rosen (1974)'s tangency condition is presented in Section 3.

In this subsection, we assumed that $\tau(d)$ was homogeneous. However, in practice we may expect treatment effect heterogeneity, i.e. $\tau_j(d)$ may vary by house j. In that case, the argument above breaks down: any systematic dependence between $\tau_j(d)$ and d implies differentiating $ATT(d|d) = \mathrm{E}[\tau(d)|D = d]$ does not yield the average price derivative $\mathrm{E}[\frac{\partial \tau(d)}{\partial d}|D = d]$. As a result, the derivative of $ATT(d|d)$ may not equal the welfare relevant parameter. I consider this case in detail next.

Figure 2: DiD identifies the ex post hedonic derivative under treatment effect homogeneity



Note: This diagram shows the hypothetical ATT function constructed from the pre and post period pricing gradients in Figure 1. The red line shows price effects from the DiD plotted on distance to the toxic plant. With homogeneous treatment effects, DiD identifies the hedonic price function in the ex post equilibrium. The black line shows the bid (indifference) curve of homeowners who live d' from the plant in the post period lying tangent to the pricing function at that point.

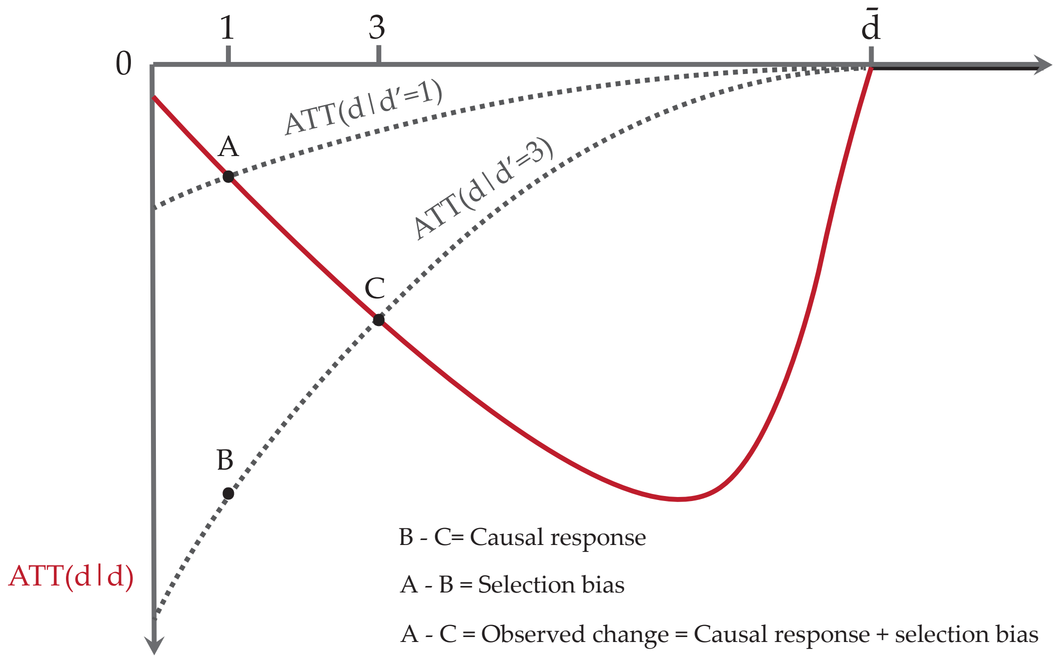
D *Nonidentification of the $ACRT(d|d)$ under heterogeneity*

In a continuous DiD framework, Callaway et al. (2025) discuss the need for additional restrictions on treatment effect heterogeneity along the treatment dosage level to identify the $ACRT(d|d)$. In this section, I first review the key threat to identification discussed by Callaway et al. (2025). I then discuss the *strong parallel trends* assumption they propose and why it may not hold in many spatial contexts.[5] Unfortunately, without this assumption, the ATT function's derivative no longer has a clear interpretation as a structural object. However, in the next subsection I suggest methods to bound the welfare measure under alternative assumptions.

Figure 3 illustrates the identification issue when there is treatment effect heterogeneity by distance. As in Figure 2, the red line plots the ATT function observed from difference-in-differences. The two gray dotted lines show counterfactual average treatment effects at various exposure levels. The top left gray line shows the average effect on housing prices for units who are actually 1 kilometer from the plant had they counterfactually been at different distances from the plant. The middle gray line is similar but for the subpopulation of units 3 kilometers away. The black points A and C show where the counterfactual curves intersect the observed ATT function, i.e., where for $ATT(d|d')$ we have $d = d'$.

[5] Callaway et al. (2025) state explicitly, "Before doing that, it is worth mentioning that we are not proposing [strong parallel trends] as an assumption that empirical researchers should readily adopt; in fact, in many applications, [Strong parallel trends] may be a strong or implausible assumption. Rather, our aim is to clarify that many natural target parameters in DiD applications with a continuous treatment require stronger assumptions than parallel trends..."

Figure 3: Non-identification of the *ACRT* with treatment effect heterogeneity



Note: This figure illustrates that the ATT derivative does not identify the $ACRT$ under treatment effect heterogeneity. The red line plots the ATT function observed from difference-in-differences. The two gray dotted lines show counterfactual average treatment effects at various exposure levels. The top left gray line shows the average effect on housing prices for units who are actually 1 kilometer from the plant had they counterfactually been at different distances from the plant. The middle gray line is similar but for the subpopulation of units 3 kilometers away. The three black points, A,B, and C, illustrate how the difference in observed $ATT(d|d)$ values does not estimate the causal response for moving a unit at 3 km to a 1 km level of pollution exposure because the vertical change from C to B does not equal the change from C to A.

On average any counterfactual exposure level affects the housing units 1 kilometer away less than units 3 kilometers away. In the figure, the top left gray curve is smaller in magnitude at any d than the middle gray curve. To give a concrete example, say urban planners only site an industrial plant in poor neighborhoods, which tend to have smaller houses. The prices for the houses farther away may be more affected by pollution because the individuals who buy large houses are high income and income may be correlated with a higher WTP to avoid pollution.[6] Then, at any given exposure level, pollution affects houses closer to the plant less. In this scenario, the average effect on houses 1 kilometer away is less than the counterfactual effect if the houses at 3 kilometers were exposed to the level of pollution 1 kilometer away.

While the ATT is identified, this effect heterogeneity poses a problem for comparisons across treated levels of d. Say we want to know the average effect from moving a unit at 3 kilometers to 1 kilometer. That is, we want to know the vertical movement along the middle counterfactual gray curve, from C to B. However, we can only observe the movement along the red observed ATT

[6] Lower income households may also prefer less pollution but may prefer to prioritize necessities like food and clothing (Banzhaf et al., 2019).

function, from C to A. In this example, the true causal response is negative while the difference between the observed ATT values is positive.

This issue also applies to marginal comparisons. The red line's derivative is negative at most distances; the gray curves' derivatives are always positive. A researcher interpreting the observed ATT function's derivative as homeowners' MWTPs would find that most homeowners prefer to live closer to the plant, despite all homeowners in this example preferring to live farther away. So, it is possible to imagine scenarios in which treatment effect heterogeneity causes the observed ATT derivative to have the wrong sign.[7]

More formally, I present one of the primary theorems in Callaway et al. (2025) on non-identification of the $ACRT$ from the ATT derivative.

Theorem 2.2 *Under Assumptions 1 to 5, the ATT derivative recovers a mix of causal effect parameters and selection bias terms. In particular, for $d \in [0,\overline{d}]$,*

$$\frac{\partial \mathrm{E}[\Delta Y \mid D = d]}{\partial d} = \frac{\partial ATT(d \mid d)}{\partial d} = ACRT(d \mid d) + \underbrace{\left.\frac{\partial ATT(d \mid l)}{\partial l}\right|_{l=d}}_{\textit{local selection bias}}.$$

So estimating the $ACRT$ by the derivative of the ATT curve results in the $ACRT$ plus an additional selection bias term.

What should a researcher do if they want to use DiD price effect estimates to make statements about welfare? As suggested in Callaway et al. (2025), one option to recover a structural interpretation is to invoke a stronger assumption that limits treatment effect heterogeneity along the distance margin, which they call the "strong parallel trends" assumption.

Assumption 6 (Strong parallel trends). *For all $d \in \mathcal{D}$,*

$$\mathrm{E}[Y_t(d) - Y_t(\overline{d})] = \mathrm{E}[Y_t(d) - Y_t(\overline{d})|D = d].$$

Assumption 6 says that units at all distance levels will have the same ATT response function or $ATT(d|d) = ATT(d)$. Effectively, this rules out treatment effect heterogeneity that is systematically correlated with distance, e.g. selection on gains, which might not be plausible in many settings. However, with this assumption, the ATT derivative equals the $ACRT$.

Theorem 2.3 *Under Assumptions 1 to 6, the $ACRT(d|d)$ is identified. That is, for $d \in [0,\overline{d}]$,*

$$\begin{aligned}\frac{\partial \mathrm{E}[\Delta Y|D = d]}{\partial d} = \frac{\partial ATT(d|d)}{\partial d} &= ACRT(d|d) + \underbrace{\frac{\partial ATT(d|l)}{\partial l}|_{l=d}}_{=0} \\ &= ACRT(d|d).\end{aligned}$$

The discussion above suggests it is not hard to imagine situations where treatment site selection will imply heterogeneity by distance. In this case, a researcher may not be comfortable imposing the strong parallel trends assumption. In the next subsection, I discuss how researchers can still bound the $ACRT$ under an alternative assumption.

[7]Figure 3 also highlights an issue if the researcher incorrectly measures the spatial extent where treatment effects end. Say a researcher stops measuring treatment effects before $\overline{d}$. By only comparing ATT estimates at treated distance levels, the researcher would incorrectly estimate a positive effect from the treatment because of selection bias, e.g., by looking at the change from C to A.

E *Lower bound on the* $ACRT$

In this subsection, I first motivate the bounding argument graphically. I then show that the $ACRT$ is point identified under restricted forms of treatment effect heterogeneity. In particular, if potential outcomes are linear in distance, then a simple estimator identifies the $ACRT$. I then weaken the linearity assumption to concavity. In this case, the simple estimator is a bound on the $ACRT$.

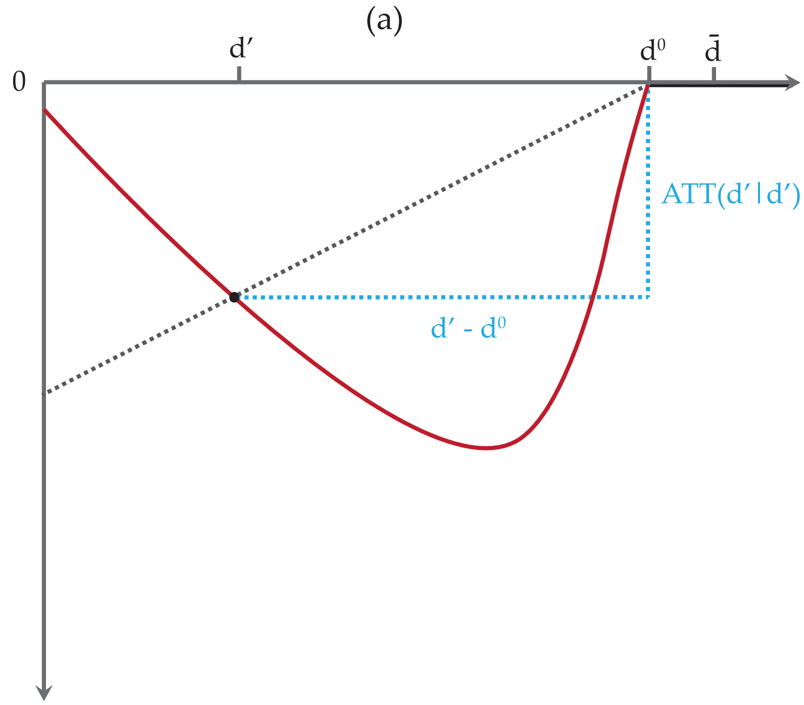
I will consider a "rise over run" estimator that takes the observed ATT estimate and divides by its distance from the exact location where the spillovers from the plant end, which I denote d^0. Figure 4 illustrates the intuition. As in Figure 3 the red line is the observed ATT function. The gray dotted line is the counterfactual ATT response function for units in the subpopulation d' away from the treatment site. The blue dotted lines illustrate the rise over the run at d' relative to the point d^0.

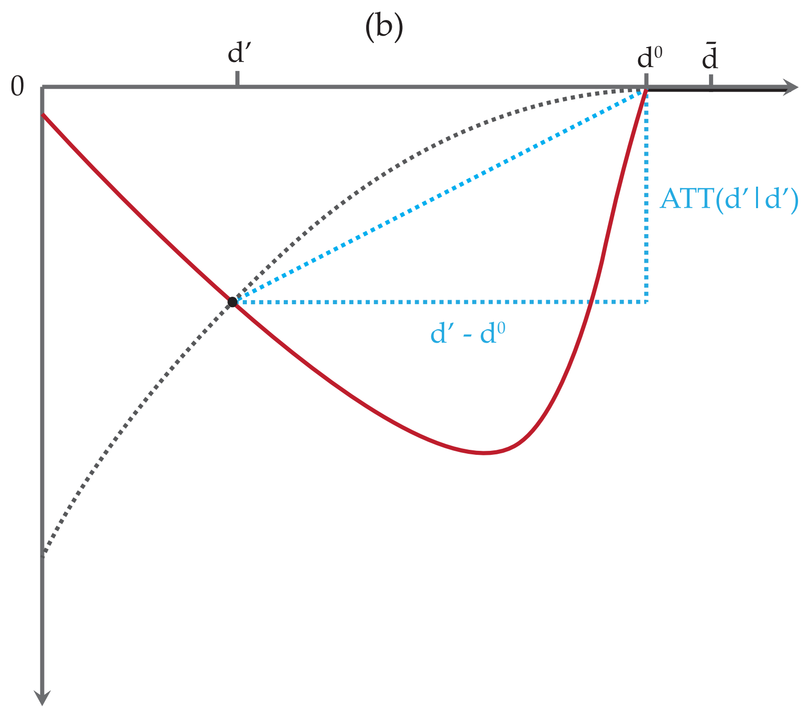
To start, Panel (a) imagines the true counterfactual response function $ATT(d|d')$ as linear in d. In this special case, the observed $ATT(d'|d')$ identifies the causal response because we can "connect the dots" between 0 treatment effect at d^0 and the $ATT(d'|d')$ at d'. The $ACRT$ or the price derivative is then simply the average rise over the common run: $\frac{ATT(d'|d')}{d'-d^0}$. Unfortunately, we may expect a linearity assumption to be restrictive. As an equilibrium object, theory suggests the hedonic price function is generically nonlinear (Ekeland et al., 2004).

However, this rise over run estimator for the $ACRT$ can still be useful. Panel (b) shows the counterfactual ATT response function again as a nonlinear function. The price derivative is no longer identified by the rise over the run, but in this example with extreme selection bias the "connect the dots" approach yields a price derivative estimate that is much closer to the true price derivative than the observed ATT function's derivative. Furthermore, a weaker assumption of concavity implies the rise over run estimator will always underestimate the $ACRT$. In Figure 4, this can be seen by the diagonal blue dotted line's slope being flatter than the gray dotted line's slope.

I argue a concavity assumption is plausible in many spatial DiD applications, where the ring DiD design was ex ante motivated by the expectation that treatment effects dissipate with distance from the treatment site. For example, imagine a house is several kilometers away from an industrial plant. The counterfactual price loss if the plant was sited a few blocks closer to the house may not be very large. On the other hand, imagine a house starts a few blocks away from an industrial plant. Moving the plant directly next to the house could have a significant negative effect on its value. In other words, we may expect pollution to impose convex costs getting closer and closer to the treatment site, which implies that the price effect is concave in distance from the plant. It should be caveated that concavity is also an assumption and it might not hold depending on the application. For example, a subway station opening increasing prices for houses within walking distance of the station but decreasing prices for houses a little farther away would violate concavity. Still, for many spatial applications, the researchers' initial motivation for using the ring DiD research design was because they expected treatment effects to dissipate moving away from the treatment site.

Figure 4: Linear approximation to the $ACRT$





Note: This figure illustrates that the rise over the run identifies the $ACRT$ when potential outcomes are linear in Panel (a) and is a conservative estimate when the counterfactual ATT response functions are concave in Panel (b). As in Figure 3, the red line is the observed ATT function. The gray dotted line is the counterfactual ATT response function for units of the type at d'. The blue dotted lines illustrate the rise over the run at d' relative to d^0.

The next few results formalize the intuition from Figure 4.

Assumption 7 (Linear potential outcomes). *Suppose there exists a known constant $d^0 \in (0, \overline{d}]$ such that d^0 is the minimum distance where spillovers from the plant no longer impact house prices and assume a housing unit j's potential outcomes is piecewise linear in distance to the treatment site. That is, for $D_j \in \mathcal{D}$ we have*

$$Y_{j,t} = Y_{j,t}(d^0) + \gamma_j \min\{D_j - d^0, 0\}$$

where γ_j is a unit-specific slope coefficient.

This assumption says that housing units' potential outcomes follow a linear correlated random coefficients model. Under this model, the average of the unit-specific slope coefficients γ_j is the average causal response $ACRT$.

The following theorem shows that under Assumption 7 an alternative "rise over run" estimator identifies the $ACRT$.

Theorem 2.4 *Under Assumptions 1 to 5 and Assumption 7, the $ACRT$ is identified. In particular, for $d \in [0, d^0)$,*

$$\frac{ATT(d|d)}{d - d^0} = ACRT(d|d).$$

While similar estimators for the causal response are proposed in D'Haultfœuille et al. (2023), de Chaisemartin et al. (2025) and in an earlier literature on panel data correlated random coefficients models (Chamberlain, 1982, 1992, Graham and Powell, 2012), in this paper I show that this argument also holds in the framework of Callaway et al. (2025). Hence, identifying or bounding the causal response in this way may be of independent interest for DiD applications building on Callaway et al. (2025).

Note that Assumption 7 technically requires the researcher to know d^0, the distance at which treatment effects become nonzero. Previously, we only required the researcher to know an upper bound $\overline{d}$ at which treatment effects are nonzero. However, if we look at Figure 4(a), intuitively d^0 is identified by the location where, starting from $\overline{d}$ and moving in the direction of the treatment site, the ATT first becomes nonzero. In Lemma B.1 in the appendix, I show that if one knows an upper bound $\overline{d}$, then the minimum distance d^0 is identified by the ATT function under mild regularity conditions, so that a researcher need not know d^0 ex ante. Define $d^0_* := \sup_d \{d \in \mathcal{D} | ATT(d|d) \neq 0\}$ as this distance where the ATT function becomes nonzero. In the next result, I will show d^0_* is useful for bounding the $ACRT$ under a weaker concavity assumption.

Assumption 8 (Concave ATT response functions). *Assume the ATT functions for each treatment group subpopulation $d' \in [0, \overline{d}]$ are concave on the interval $[d', \overline{d}]$. That is, for any $\lambda \in [0,1]$ we have*

$$ATT((1-\lambda)d' + \lambda\overline{d}|d') \geq (1-\lambda)ATT(d'|d') + \lambda ATT(\overline{d}|d').$$

Recall in this spatial setting the treatment intensity is decreasing in d, so the x-axis is effectively inverted. Another way to view Assumption 8 is that it says the costs of pollution are convex in the amount of pollution exposure. For disamenities with $ATT(d|d) \leq 0$, concavity of $ATT(d|d')$ may be a natural assumption because it implies $|ATT(d|d')|$ is convex, i.e. price effects get stronger and stronger approaching the treatment site.

The next theorem shows the rise over the run is less than the $ACRT$ if the ATT response functions are concave. This follows from a standard theorem for 1-dimensional concave functions.

Theorem 2.5 *Suppose Assumptions 1 to 5 and Assumption 8. Then, for $d \in [0, d_*^0)$, we have*

$$\frac{ATT(d|d)}{d - \bar{d}} \leq \frac{ATT(d|d)}{d - d_*^0} \leq ACRT(d|d).$$

Hence, under concavity a rise over run estimator lower bounds the true $ACRT$. A researcher could use $\bar{d}$ as the reference distance in the denominator, but using d_*^0 provides a more informative bound. In empirical applications, d_*^0 can be estimated from the ATT function.

For a disamenity, a lower bound is a conservative bound. Prices are higher farther from the plant, so the derivative with respect to distance and the rise over the run are positive. Hence, the above inequality implies the rise over the run is smaller in magnitude than the true $ACRT$. While I have motivated this section using a disamenity as an example, under the analogous shape restriction for an amenity, the rise over the run is still a conservative bound. In the case with $ATT(d|d) \geq 0$, it is natural to impose convexity instead of concavity because convexity is now the restriction that implies price effects get stronger and stronger approaching the treatment site. Under convexity, the second inequality in the above result flips, but, because the derivative with respect to distance and the rise over the run are now negative, the rise over the run is still smaller in magnitude, i.e. $|\frac{ATT(d|d)}{d-d_*^0}| \leq |ACRT(d|d)|$.

In Section 3, I will use Assumption 8 and Theorem 2.5 to bound homeowners' willingnesses to pay. Interpreted in the hedonic model, a conservative bound on the price derivative underestimates homeowners' MWTP to avoid proximity to an industrial plant. I then pair this result with a revealed preference argument as in the prior hedonic literature (Ekeland et al., 2004, Banzhaf, 2021) to bound both sides of the welfare measure.

3 Structural hedonic model of housing choice

In this section, I present a structural hedonic model of housing choice for proximity to a spatially targeted treatment following the setup in Diamond and McQuade (2019) closely and maintaining most of their modeling assumptions. However, relative to Diamond and McQuade (2019), I impose weaker assumptions on the hedonic price function that allow for heterogeneous treatment effects and parallel shifts in the price function over time.

First, I introduce the canonical setup from Rosen (1974) and show formally how Rosen (1974)'s tangency condition relates average homeowner MWTP to the $ACRT$ estimand in Section 2. I then characterize nonmarginal WTPs by relying on a parametric assumption on homeowner's utility functions following Bajari and Benkard (2005). Finally, I construct bounds on local homeowners' aggregate willingness to pay for the plant not to exist using results from Section 2. As in Section 2, I discuss how weaker assumptions affect the interpretation of the welfare estimand.

In addition, a common empirical practice is to estimate the price change in logs and then multiply the ATT estimates by a baseline price level to convert from logs back to levels, as in this paper's empirical applications. However, when allowing for heterogeneous price effects, this does not identify the level price change without an additional assumption. In the final part of this section, I formalize the required assumption to identify preferences in dollars using the ATT for the log-transformed house price.

Setup: In Rosen (1974)'s classic model, a differentiated good is described by a vector of product characteristics. For a house, these characteristics may consist of structural attributes such as the lot size or the number of bedrooms, or neighborhood qualities such as proximity to a toxic industrial plant. I take the vector of characteristics determining price as $(d, \mathbf{z})$ where the first characteristic is radial distance to a toxic plant and $\mathbf{z} = (z_2, z_3, \ldots, z_k)$ is a vector collecting all other housing characteristics. In the previous section, house prices $Y_{j,t}$ were considered in a potential outcomes framework. Now, for a house j with characteristics $(d, \mathbf{z})$ in time period t, the potential outcomes $Y_{j,t}(d)$ are defined by a mapping from all housing characteristics

$$Y_{j,t}(d) = Y_t(d, \mathbf{z})$$

where as before $t \in \{1,2\}$ represents the pre versus the post period. In each period, the equilibrium matchings of consumers and producers leads to the observed function between house prices and characteristics, termed the hedonic price function.

MWTP: For each homeowner $i \in \mathcal{I}$, her utility maximization depends on her house's characteristics and consumption of a numeraire good, c.

$$\max_{d,\mathbf{z},c} u_{i,t}(d,\mathbf{z}) + c \;\; s.t. \;\; w_{i,t} - Y_t(d,\mathbf{z}) - c = 0$$

where $w_{i,t}$ is homeowner wealth. Maximization of the utility function implies that in equilibrium a consumer i chooses a house with distance to an industrial plant satisfying

$$\frac{\partial u_{i,t}(d_i^*, \mathbf{z}_\mathbf{i}^*)}{\partial d} = \frac{\partial Y_t(d_i^*, \mathbf{z}_\mathbf{i}^*)}{\partial d}. \tag{2}$$

Thus, a consumer's marginal willingness to pay (MWTP) for distance to an industrial plant is the hedonic price function's partial derivative evaluated at her equilibrium product choice. Equation (2) is the key insight of Rosen (1974) and is the formal statement of the tangency condition suggested in Figure 2. It is worth noting that an implicit assumption of this first order condition is that homeowners choose from a continuum of housing choices and therefore can always solve for an interior solution.[8]

However, before we can relate DiD estimates to homeowners' MWTPs, we still require an additional assumption on the price function.

Assumption 9 (Price function separability). *Assume the price function takes the following form:*

$$Y_t(d, \mathbf{z}) = m(d, \mathbf{z}) + h_t(\mathbf{z})$$

where $m(.)$ is the contribution to price from proximity to an industrial plant and $h_t(.)$ is the contribution from the other housing characteristics.

[8]Bajari and Benkard (2005) discuss weakening the assumption of complete, continuous product choice, and show that it also leads to bounds instead of point identification on preferences.

This assumption says the change in the price function over time follows an additive fixed effects model. For example, such an assumption would hold if a house's price is determined by its individual fixed housing characteristics and aggregate macro shocks that shift all prices by the same amount. Similar assumptions on the data generating process are common in the panel data literature[9] and Diamond and McQuade (2019) assume a stronger separability assumption. $Y_t(\mathbf{z})$ admits limited forms of nonseparable treatment effect heterogeneity through $m(d, \mathbf{z})$, whereas a completely additively separable price function, i.e. $m(d,\mathbf{z}) = m(d)$, would imply strong parallel trends.[10]

With this assumption, homeowners' average marginal willingnesses to pay in the post period relates to the $ACRT$. The following proposition formalizes the connection between Rosen (1974)'s tangency condition and DiD estimates. Proofs for this section are in Appendix D.

Proposition 1 (Rosen (1974)'s first order condition and the $ACRT$). *Suppose Assumptions 1 to 5, Assumption 9, and the standard hedonic model hold. Then*

$$\mathrm{E}\Big[\frac{\partial u_{t=2}}{\partial d}|D = d\Big] = \mathrm{E}\Big[\frac{\partial Y_{t=2}}{\partial d}|D = d\Big] = ACRT(d|d).$$

Furthermore, Section 2 suggested three alternative assumptions for identifying or bounding the $ACRT$:
(a) If strong parallel trends holds (Assumption 6), then

$$\mathrm{E}\Big[\frac{\partial u_2}{\partial d}|D = d\Big] = \frac{\partial ATT}{\partial d}(d|d).$$

(b) If instead linear potential outcomes holds (Assumption 7), then

$$\mathrm{E}\Big[\frac{\partial u_2}{\partial d}|D = d\Big] = \frac{ATT(d|d)}{d - d^0}.$$

(c) If instead concave ATT functions holds (Assumption 8), then

$$\mathrm{E}\Big[\frac{\partial u_2}{\partial d}|D = d\Big] \geq \frac{ATT(d|d)}{d - d^0_*}.$$

Proposition 1 says that, in the post period equilibrium, the average marginal willingness to pay at each distance d is equal to the $ACRT$ at d. As discussed in Section 2, the ATT function's derivative only identifies the $ACRT$ under strong parallel trends, but the alternative rise over run estimator lower bounds the $ACRT$ under concavity. Note that once the price function allows for heterogeneous treatment effects we can no longer identify individual-level marginal willingnesses to pay, instead only identifying averages within each distance level.

Nonmarginal WTP: Although Rosen's first order condition in (2) relates each individual's MWTP with the hedonic price function's derivative at the chosen equilibrium bundle, in order to conduct nonmarginal welfare analysis, we need to characterize individuals' entire bid curves. Bajari and Benkard (2005) suggest making a parametric assumption on individuals' utility functions. Then, for each individual, her entire bid curve is recoverable from the one product choice made

[9]See for example Graham and Powell (2012).

[10]To see this, consider the definition of the ATT under Assumption 9: $ATT(d|d') = \mathrm{E}[Y_{t-2}(d) - Y_{t-2}(\overline{d})|D = d'] = \mathrm{E}[m(d, \mathbf{z})|D = d'] = \mathrm{E}[m(d)]$ for any d'.

in equilibrium. In particular, they suggest a utility function that is log linear in the product characteristics.

For individual i, assume their utility maximization problem in time period t is:

$$\max_{d,\mathbf{z},c} u_{i,t} = \beta_{i,d} \log(1+d) + \beta_{i,2} \log(z_2) + ... + \beta_{i,k} \log(z_k) + c \\ \text{s.t. } Y_t(d,\mathbf{z}) + c \leq w_{it} \tag{3}$$

where $1+d$ is used so that utility is defined at zero.[11] Then individual i's taste parameter for distance to the toxic plant $\beta_{i,d}$ can be solved by

$$\beta_{i,d} = (1+d_i^*) \frac{\partial Y_t(d_i^*, \mathbf{z_i^*})}{\partial d}. \tag{4}$$

We can now consider homeowners' utility change under counterfactual pollution exposures. Consider the counterfactual where a homeowner is no longer exposed to the industrial plant but their house is otherwise the same. For a homeowner in the affected geographic area, her willingness to pay for the plant not to exist is

$$WTP_i = \beta_{i,d} \log\left(\frac{1+d_*^0}{1+d_i^*}\right).$$

For homeowners farther than d_*^0 away, their willingness to pay for the plant not to exist is $WTP_i = 0$. By assumption, an immediate consequence of Proposition 1 and Equation (4) is that the $ACRT$ identifies the average nonmarginal willingnesses to pay at each distance d.

Lemma 3.1 *Suppose Assumptions 1 to 5, Assumption 9, the standard hedonic model hold, and homeowners' utility functions are log linear according to Equation (3). Then*

$$\mathrm{E}[WTP|D=d] = (1+d)ACRT(d|d) \log\left(\frac{1+d_*^0}{1+d}\right).$$

As in Proposition 1(a),(b), and (c), depending on the assumption imposed, one of the two $ACRT$ estimators from Section 2 used in place of the $ACRT$ in the right hand side expression will identify or lower bound $\mathrm{E}[WTP|D=d]$.

A *Totaling to aggregate welfare impacts*

To create an aggregate welfare measure, we can add up individual homeowners' willingnesses to pay for the plant not to exist:

$$\text{Agg WTP} = \sum_i WTP_i = \sum_i \beta_{i,d} \log\left(\frac{1+d_*^0}{1+d_i^*}\right).$$

This sum can be rewritten as the number of affected homeowners times the average willingness to pay by integrating over $[0, d_*^0]$. Define N_H as the number of affected housing units and $f(d)$ as the density of housing units. Then,

$$\text{Agg WTP} = N_H \int_0^{d_*^0} \mathrm{E}[WTP|D=d] f(d)\, \mathrm{d}d.$$

[11] For all empirical results, I assess sensitivity to this choice of constant in Appendix Table A3. Differences in WTP estimates across different choices of constant are marginal.

Plugging in Lemma 3.1 yields an expression for the aggregate willingness to pay in terms of the $ACRT$.

Proposition 2 (Aggregate willingness to pay). *Suppose Assumptions 1 to 5, Assumption 9, the standard hedonic model hold, and homeowners' utility functions are log linear according to Equation (3). Then the aggregate willingness to pay for the plant not to exist is*

$$\text{Agg WTP} = N_H \int_0^{d_*^0} (1+d) ACRT(d|d) \log\left(\frac{1+d_*^0}{1+d}\right) f(d)\, \mathrm{d}d. \tag{5}$$

As in Proposition 1 and Lemma 3.1, depending on the assumption imposed, one of the two $ACRT$ estimators from Section 2 used in place of the $ACRT$ in the right hand side expression will identify or lower bound Agg WTP.

In Section 2, I argued that a concavity assumption was more plausible in many spatial applications than either a homogeneous treatment effects or a linear potential outcomes assumption. Under concavity, the right hand side expression evaluated using the rise over run as the $ACRT$ estimator is a lower bound on the aggregate willingness to pay.

We can also bound the aggregate willingness to pay on the other side using a similar argument as in the prior hedonic literature. Banzhaf (2021) shows the total of the reduced form price effects from a policy change is a lower bound on the aggregate welfare change. Note the sign reversal as I defined WTP_i as the willingness to pay for the plant *not* to exist. The aggregate price change is

$$\Delta\text{Price} = N_H \int_0^{d_*^0} ATT(d|d) f(d)\, \mathrm{d}d. \tag{6}$$

Reversing signs, $-\Delta$Price is an upper bound on the aggregate willingness to pay for the plant not to exist. It is worth noting that the aggregate price change is not necessarily conservative. While for amenities the aggregate price change will underestimate the magnitude of the welfare change, for disamenities like pollution exposure it will overestimate the magnitude.

The following result formalizes the bounding argument for homeowners' aggregate WTP.

Theorem 3.2 (Bounds on Agg WTP). *Suppose Assumptions 1 to 5, Assumption 9, the standard hedonic model hold, and homeowners' utility functions are log linear according to Equation (3). Under concave ATT functions (Assumption 8), homeowners' aggregate willingness to pay is bounded:*

$$\begin{aligned} &N_H \int_0^{d_*^0} (1+d) \frac{ATT(d|d)}{d-d_*^0} \log\left(\frac{1+d_*^0}{1+d}\right) f(d)\, \mathrm{d}d \quad \leq \text{Agg WTP} \\ &\leq -N_H \int_0^{d_*^0} ATT(d|d) f(d)\, \mathrm{d}d. \end{aligned}$$

Theorem 3.2 says homeowners' aggregate willingness to pay for the plant not to exist is bounded between evaluating the willingness to pay using the rise over run estimator and the negative aggregate price change.

Here, I provide an intuition for the proof. First, the lower bound follows from the previous results and the properties of integrals because $\frac{ATT(d|d)}{d-d_*^0} \leq ACRT(d|d)$ and all terms are positive. For the upper bound, the proof relies on a revealed preference argument exploiting the hedonic model's equilibrium conditions. In the canonical hedonic model, homeowners maximize over a complete, continuous set of housing choices. In this setting, homeowners in the affected geographic area could have chosen a house with characteristics $(\bar{d},\mathbf{z_i^*})$ but instead chose $(d_i^*,\mathbf{z_i^*})$ close to the plant. If the price difference was less than their willingness to pay for the plant not to exist, they would have a profitable deviation by choosing a house with $(\bar{d},\mathbf{z_i^*})$. Hence, $Y_t(\bar{d},\mathbf{z_i^*}) - Y_t(d_i^*,\mathbf{z_i^*}) \geq WTP_i$ for every homeowner i with d_i^* in the affected area to be in equilibrium. Observe that the ATT is the negative of the average of the left hand side expression conditioning on treatment group subpopulation $D = d$. As a result, $-ATT(d|d) \geq E[WTP|D = d]$. The upper bound then follows by the definition of the Agg WTP and the properties of integrals. An explicit proof is given in Appendix Section D.

The aggregate willingness to pay estimand is a useful welfare measure with a clear interpretation. It is the amount homeowners' living close to the plant in the post period would pay for the plant not to exist. This result holds under weaker assumptions than those in the prior literature: Diamond and McQuade (2019) assume treatment effect homogeneity and a constant hedonic price function over time.

However, without stronger assumptions that the price function is fully separable and time constant,[12] the aggregate willingness to pay is not a complete measure of all possible effects on homeowner welfare. Homeowners may reoptimize their optimal bundle of other housing characteristics after the plant opens. For example, a homeowner might optimally decide to move close to the plant to save money, and also buy a house with a smaller yard because they will not use their yard as much. The utility loss from a smaller yard would not be accounted for in their willingness to pay for just proximity to the plant. While for interpreting the estimand researchers face a tradeoff between stronger conclusions and more credible assumptions, in practice the expression for the estimand does not change. After reversing signs, Equation (5) above coincides with the solution of Diamond and McQuade (2019)'s model, which under the stronger assumptions on the price function is an aggregate homeowner welfare loss, point-identified by using the ATT derivative as the $ACRT$.

Appendix E considers the bounds' sharpness. For Theorem 2.5 in the previous section, I show the lower bound on the $ACRT$ is sharp when taking prices as primitives. For any observed data, one can construct distributions of potential outcomes (price functions) such that the lower bound holds with equality. However, establishing sharpness is analytically difficult when taking preferences in the model as the primitives, since prices are now determined as solutions to a system of partial differential equations (Ekeland et al., 2004). Nevertheless, I conjecture that the bounds remain sharp in Theorem 3.2 because the model does not obviously restrict the set of attainable price functions.

Log vs. level house prices: Before proceeding to the empirical applications, I discuss using DiD estimates for log-transformed house prices to estimate monetary measures of homeowners' preferences. There are several good reasons researchers often use log-transformed house prices instead of level prices as the outcome. For example, the parallel trends and separability assumptions may be more plausible using proportional changes, and in estimation it may limit

[12] That is, strengthen Assumption 9 so the price function is $Y(d, \mathbf{z}) = m(d) + h(\mathbf{z})$ in both time periods.

the influence of outliers. A common way to get a price effect in dollars from a log-transformed outcome is to multiply the estimates by average house prices. However, this transformation requires an additional assumption. Below, I formalize the conditions under which this works.

Throughout the remaining analysis, let $\widetilde{ATT}(d|d)$ and $\widetilde{ACRT}(d|d)$ refer to the analogous DiD estimands for the log-transformed outcome. That is,

$$\widetilde{ATT}(d|d) = \mathrm{E}[\log Y_2(d) - \log Y_2(\bar{d})|D = d], \quad \widetilde{ACRT}(d|d) = \frac{\partial \widetilde{ATT}(l|d)}{\partial l}|_{l=d}.$$

Suppose separability (Assumption 9) instead holds for the log house price:

$$\log Y_t(d, \mathbf{z}) = m(d,\mathbf{z}) + h_t(\mathbf{z}).$$

The house price derivative with respect to d is

$$\frac{\partial Y_t(d, \mathbf{z})}{\partial d} = \frac{\partial m(d,\mathbf{z})}{\partial d} Y_t(d, \mathbf{z})$$

and the $\widetilde{ACRT}(d|d)$ is

$$\begin{aligned}\widetilde{ACRT}(d|d) &= \frac{\partial}{\partial l}\mathrm{E}[\log Y_2(l) - \log Y_2(\bar{d})|D = d]|_{l=d} \\ &= \mathrm{E}[\frac{\partial \log Y_2(d)}{\partial d}|D = d] = \mathrm{E}[\frac{\partial m(d, \mathbf{z})}{\partial d}|D = d].\end{aligned}$$

Rosen's first order condition relates to the log house price. From utility maximization the first order condition for a homeowner i is

$$MWTP_i = \frac{\partial Y(d_i^*, \mathbf{z_i^*})}{\partial d} = \frac{\partial m(d_i^*,\mathbf{z_i^*})}{\partial d} Y(d_i^*, \mathbf{z_i^*}).$$

First, notice that the average causal response using the log-transformed outcome identifies homeowner average MWTP as a *percentage* of house price. Dividing both sides by the house price,

$$\frac{MWTP_i}{Y(d_i^*, \mathbf{z_i^*})} = \frac{\partial m(d_i^*,\mathbf{z_i^*})}{\partial d}.$$

Then, as before take conditional expectations on both sides:

$$\mathrm{E}[\frac{MWTP}{Y(d, \mathbf{z})}|D = d] = \mathrm{E}[\frac{\partial m(d,\mathbf{z})}{\partial d}|D = d] = \widetilde{ACRT}(d|d).$$

Even without additional assumptions, $\widetilde{ACRT}$ is an interpretable quantity which tells us the average marginal willingness to pay to be farther away from the plant as a percentage of house value.

However, without an additional assumption the average MWTP in dollars is not identified from multiplying $\widetilde{ACRT}(d|d)$ by the mean house price.[13] The average MWTP in dollars is

$$\mathrm{E}[MWTP|D = d] = ACRT(d|d) = \mathrm{E}[\frac{\partial m(d,\mathbf{z})}{\partial d} Y(d, \mathbf{z})|D = d]$$

where the right hand side is not identified without the following assumption.

[13] Issues with estimating an average treatment effect by first estimating the effect in logs and multiplying by a baseline mean are discussed in more detail in Chen and Roth (2024), Remark 8.

Assumption 10 *(Uncorrelated proportional response) For all $d \in \mathcal{D}$,*

$$\mathrm{E}[\frac{\partial m(d,\mathbf{z})}{\partial d} Y(d,\mathbf{z})|D=d] = \mathrm{E}[\frac{\partial m(d,\mathbf{z})}{\partial d}|D=d]\mathrm{E}[Y(d,\mathbf{z})|D=d].$$

This assumption says that, conditional on the distance d, percent changes in house price from the plant are uncorrelated with level house prices. With this additional assumption,

$$\mathrm{E}[MWTP|D=d] = ACRT(d|d) = \widetilde{ACRT}(d|d)\mathrm{E}[Y(d,\mathbf{z})|D=d]$$

where $\mathrm{E}[Y(d,\mathbf{z})|D=d]$ is the average post period house price d distance away from a plant.

If Assumption 10 holds, then the average causal response in logs multiplied by the average house price is the average causal response in levels. As a result, $\widetilde{ACRT}(d|d)\mathrm{E}[Y(d,\mathbf{z})|D=d]$ can be substituted for $ACRT(d|d)$ throughout the propositions and theorems developed in this section, with the other assumptions (1-9) instead applied to the log-transformed house price.

I now write equations for the aggregate willingness to pay and price change measures in terms of the DiD estimands for the log-transformed house prices. Using the log-transformed $\widetilde{ACRT}(d|d)$, the estimating equation for the aggregate willingness to pay is

$$\text{Agg WTP} = N_H \int_0^{d_*^0} (1+d)\widetilde{ACRT}(d|d)\mathrm{E}[Y(d,\mathbf{z})|D=d] \log\left(\frac{1+d_*^0}{1+d}\right) f(d)\,\mathrm{d}d. \tag{7}$$

Using the log-transformed $\widetilde{ATT}(d|d)$, the aggregate price change is

$$\Delta\text{Price} = N_H \int_0^{d_*^0} \widetilde{ATT}(d|d)\mathrm{E}[Y(d,\mathbf{z})|D=d] f(d)\,\mathrm{d}d \tag{8}$$

where $ATT(d|d) = \widetilde{ATT}(d|d)\mathrm{E}[Y(d,\mathbf{z})|D=d]$ under Assumption 10 and log changes approximating percent changes.

Equations (7) and (8) are the estimating equations I take to the data. In the primary specifications, I use log-transformed house prices as the outcome, which imposes Assumption 10. For each application in this paper, I report three estimates: (i) a lower bound using the rise over run estimator as the $ACRT$ in Equation (7), (ii) an upper bound using the aggregate price change, and (iii) a benchmark estimate assuming homogeneous treatment effects by using the ATT derivative as the $ACRT$. In addition, I report homeowner's average willingness to pay as a percentage of house price, which does not impose Assumption 10. All terms in these equations can be estimated given data on treatment site locations, house sales, and the density of housing units where the key inputs to these equations are simple transformations of $\widetilde{ATT}(d|d)$.

B *Aggregating across multiple treatment sites and time periods*

Thus far, the theory has considered the case with one treatment site and two time periods. In practice, researchers often have multiple treatment sites and time periods. However, we can apply the results discussed above by considering ATTs separately by treatment site and then aggregating across treatment sites and time periods. Let $g \in \mathcal{G}$ index distinct industrial plant openings and let

$t \in \{1,2,\ldots,T\}$ to allow for more than two time periods. Consider the analogous $\widetilde{ATT}$ estimand that is treatment site and time period specific:

$$\widetilde{ATT}(g,t,d|g,d) = \mathrm{E}[\log Y_t(d) - \log Y_t(\overline{d})|G = g, D = d]$$

Following Callaway et al. (2025), among units that ever participate in treatment, we can consider aggregations of the form:

$$\widetilde{ATT}_{agg}(d|d) = \sum_{g\in\mathcal{G}}\sum_{t=2}^{T} w(g,t,d)\widetilde{ATT}(g,t,d|g,d)$$

where the weights are nonnegative and sum to one.

A key takeaway from Callaway and Sant'Anna (2021) and Callaway et al. (2025) is that by constructing the subgroup ATT estimates explicitly, the researcher can now choose the weighting of effects across subgroups rather than having the weights be determined by OLS, which at best does not target a sensible summary parameter and at worst might lead to well known problems with negative weighting.[14] There are several possible reasonable weighting choices. For example, the simplest choice is to weight the ATT estimates equally. In the next section, I discuss alternative weighting choices in more detail.

4 Estimation strategy

I now outline an estimating procedure a researcher could apply to housing sale microdata that yields a continuous ATT function estimate. This procedure has two stages. In the first stage, house sales are grouped into different rings or bins by distance to a polluting plant, giving an average sale price at each bin. Binning the data forms repeated cross sections at different distances from the treatment site. I then construct 2×2 DiD comparisons from the bin level averages yielding price effect estimates at various d from a treatment site. In the second stage, I trace out a smooth relationship using a nonparametric sieve regression of the DiD comparisons on d. In both stages, this procedure employs data-driven tuning parameter choices.

First stage: Figure 5 returns to the intuition for the continuous DiD estimator in the simple example in Figure 1 but now imagines estimating pre and post period pricing gradients using housing sale microdata. The left panel shows individual house sales distributed around the pre and post period pricing gradients with some unit-level noise. Similar to Butts (2023), in the right panel, the sales have been grouped into different bins by distance to the plant's site and then aggregated to house price averages indicated by the constant fit lines.

An advantage of aggregating to distance bins is that we can compare averages of the same bin over time forming repeated cross sections that fit into a standard DiD framework. Let $\overline{Y}_{g,t,d}$ be the log sample average house price in the bin at distance d away from the toxic plant g's site in time period t. In this example, $t \in \{1,2\}$. Consider the first bin closest to the toxic plant as the bin at distance d' and the bin farthest away from the plant at distance $\overline{d}$. The price depreciation for the d' bin is $\Delta Y_{g,d'} = (\overline{Y}_{g,2,d'} - \overline{Y}_{g,1,d'})$ and the price depreciation at $\overline{d}$ is $\Delta Y_{g,\overline{d}} = (\overline{Y}_{g,2,\overline{d}} - \overline{Y}_{g,1,\overline{d}})$.

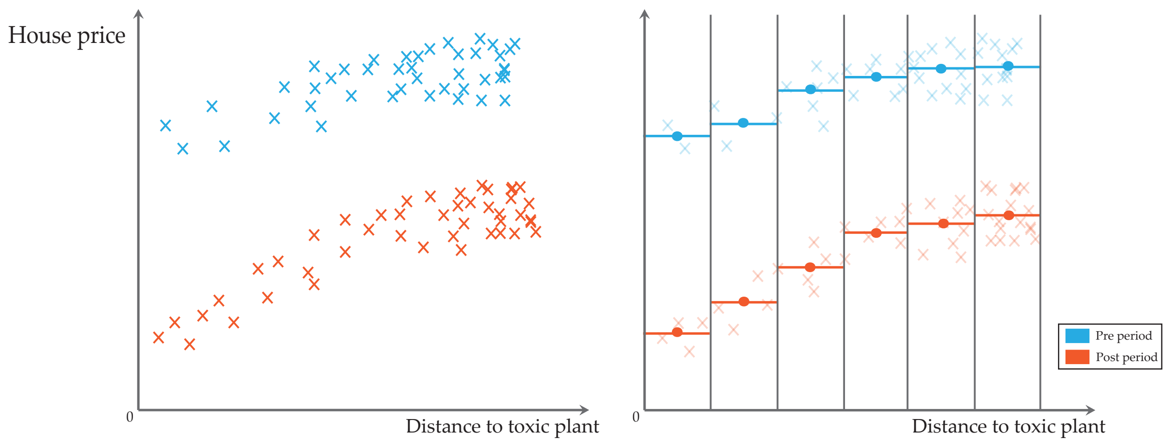
[14]See for example Borusyak and Jaravel (2017), de Chaisemartin and D'Haultfoeuille (2020), and Goodman-Bacon (2021), among others.

Then, a measure of the treatment effect for being d' away from the plant is given by

$$\widehat{\widetilde{ATT}}(g,2,d'|d') = (\overline{Y}_{g,2,d'} - \overline{Y}_{g,1,d'}) - (\overline{Y}_{g,2,\bar{d}} - \overline{Y}_{g,1,\bar{d}}).$$

For this hypothetical plant, we have six DiD comparisons, one for each of the six distance bins where the control bin $\bar{d}$ is 0 by construction. In the actual empirical applications, I choose the optimal number of bins following Cattaneo et al. (2024).

Figure 5: Binning diagram



Note: This diagram conceptualizes how binning data by its distance to the plant site forms a nonparametric approximation of the pre and post period pricing gradients. The left plot shows hypothetical realizations of housing sale microdata in the pre and post periods. On the right, the microdata has been aggregated to distance bin averages indicated by the constant fit lines.

In the application in Section 5, I apply this binning procedure across a large number of polluting plants. I estimate pre and post period pricing gradients around over 11,000 industrial plants. Binning the data around these plants forms hundreds of thousands of ATT estimates at various distances from the plants' sites, which I use to trace out a smooth relationship between price and distance to a polluting plant.

Second stage: After constructing ATT estimates, estimating a smooth price effect is now a 1-dimensional regression of the price changes on distance to the treatment site. For this regression, I follow Callaway et al. (2025)'s recommendation to use the nonparametric sieve estimator from Chen et al. (2024). This estimator avoids functional form assumptions and has desirable statistical properties.[15] Chen et al. (2024) compute the optimal sieve dimension $\hat{K}$ for regressions on flexible $\hat{K}$-dimensional transformations of d, which I denote by $\psi^{\hat{K}}(d)$. Then, the second stage regression of the $\widetilde{ATT}$ estimates on d is

$$\widehat{\widetilde{ATT}}(g,2,d|d) = \psi^{\hat{K}}(d_{g,d})'\gamma_{\hat{K}} + \epsilon_{g,d}, \tag{9}$$

and the estimator for $\widetilde{ATT}_{agg}(d|d)$ is

$$\widehat{\widetilde{ATT}}_{agg}(d|d) = \psi^{\hat{K}}(d_{g,d})'\hat{\gamma}_{\hat{K}}. \tag{10}$$

[15] Chen et al. (2024) chooses the sieve dimension such that $ATT(d|d)$ converges at the fastest possible rate in sup-norm and yields uniform confidence bands that are asymptotically more precise than undersmoothing. More details are in Callaway et al. (2025) and Chen et al. (2024).

One could also use a simpler parametric regression or other nonparametric estimators to smooth the price effect estimates.

A *Aggregation weights*

Equation (9) weights the ATT estimates equally, but there are many reasonable weighting choices for aggregating the treatment effect parameters. In this particular context, I argue that it is feasible to gain significant improvements in efficiency by weighting by the ATT estimates' inverse variances. In the empirical applications, the number of observations per bin can vary dramatically both across and within treatment sites, especially for bins that are equally spaced and for bins close to the treatment site.[16] For this regression, we have knowledge of the data generating process by calculating the bin means from the house sale microdata. Consider the case above with two time periods. An estimator of $\widetilde{ATT}$ in site g using the bin means is

$$\widehat{\widetilde{ATT}}(g,2,d) = (\hat{\overline{Y}}_{g,2,d} - \hat{\overline{Y}}_{g,1,d}) - (\hat{\overline{Y}}_{g,2,\overline{d}} - \hat{\overline{Y}}_{g,1,\overline{d}})$$

where $\hat{\overline{Y}}_{g,t,d}$ is a sample bin mean with an estimated variance $\hat{\sigma}^2_{g,t,d}$. An estimator of the ATT variance is $var(\widehat{\widetilde{ATT}}(g,t,d)) = \hat{\sigma}^2_{g,2,d} + \hat{\sigma}^2_{g,1,d} + \hat{\sigma}^2_{g,2,\overline{d}} + \hat{\sigma}^2_{g,1,\overline{d}}$ where all terms can be estimated using the standard errors of the means in the data. Then, the normalized weights are

$$\hat{w}(g,2,d) = \frac{1/var(\widehat{\widetilde{ATT}}(g,2,d))}{\sum_{g \in \mathcal{G}} 1/var(\widehat{\widetilde{ATT}}(g,2,d))},$$

and we can estimate $\widetilde{ATT}_{agg}(d|d)$ as in Equation (9) using $\hat{w}(.)$ as regression weights.

Because there are multiple reasonable weighting choices, in the appendix I show all price effect estimates under several different weighting choices. In general, estimates based on equal weighting are similar but less precise, and this discrepancy looks to be driven by estimates derived from small numbers of house sales.

5 Application: Impact of a polluting plant (Currie et al., 2015)

In this section, I illustrate the implications of the framework in an empirical example by returning to Currie et al. (2015) who study the effect of toxic industrial plant openings on house values. Overall, I find similar price effect estimates within the same spatial range as the original paper but the continuous DiD reveals a much wider spatial extent of these price effects. Finally, I extend their analysis by incorporating these price effects into the structural hedonic model to estimate bounds on local homeowners' aggregate willingness to pay for a polluting industrial plant not to exist.

A *Data*

Estimating the aggregate welfare measures requires only three data sources. The main data sources are a list of treatment events with geographic coordinates/dates and housing transaction

[16] Appendix C discusses the rationale for this weighting choice in more detail. In particular, it shows empirically the variation in how densely populated bins are with housing sales and formally tests for heteroskedasticity around polluting industrial plants.

microdata. Additionally, getting an estimate of the number of affected houses requires a complete count of housing units in these areas.

Toxic plants (EPA): As in Currie et al. (2015), I identify plants that emit toxic pollutants using the EPA's Toxic Release Inventory (TRI) database. The EPA requires industrial plants to self report their emissions if they handle more than a threshold amount of toxic chemicals. While the self reported emission quantities themselves are unreliable, the TRI is useful as a registry to identify plants that emit pollutants with their precise geographic locations.

Currie et al. (2015) are able to measure plants' exact operating dates by linking the TRI data to the U.S. Census Longitudinal Business Database. Unfortunately, this database is confidential and not available to me. Instead, I rely on imputation using when a plant first appears in the TRI database as the event date and restricting to plants that appear in the TRI database throughout the duration of a plant's post period. On the other hand relative to Currie et al. (2015), I have housing transaction data from more states over a longer time frame, implying my estimates are based on many more plant openings and housing transactions. Nevertheless, in comparable specifications to those in Currie et al. (2015), I find similar price effect estimates.

Housing sales (CoreLogic): I use data on single family home sales from the CoreLogic Deed database as the primary outcome. CoreLogic is a private data aggregator that provides detailed records on the near universe of single family home transactions in most metropolitan areas. This housing transaction microdata includes a rich collection of parcel-level housing characteristics and the exact coordinate location and date of each sale. I use transactions from 1990 to 2020 restricting to arms length transactions and dropping extreme outliers.[17] All house sales are adjusted to 2023 dollars.

Housing unit counts (U.S. Census): Not all houses in an area will transact during the study time period, so to get an estimate of the total affected housing units I add up the housing unit counts in 2010 U.S. census blocks surrounding the toxic plants. I retrieve counts separately by whether housing units are owner or renter occupied. In this paper, because I do not use rental listing data, I focus on the density of owner occupied housing units.

Summary statistics: Panel A of Table 1 presents summary statistics on the plants used in the analysis sample. In total, the price effect estimates are based on 11,849 distinct plants that first appeared in the TRI data from 1995-2015 with the typical plant first appearing in 2005. The final row shows the average annual self reported emissions for these plants was around 18,457 pounds per year. Although the self reported quantities are unreliable, this average suggests that these plants do emit significant amounts of pollutants over the study time period.

Panel B shows the average house prices by distance from the toxic plants within 6 kilometers of

[17]Specifically, I drop sales recorded as less than $10,000 or greater than $10 million dollars as in Currie et al. (2015). I also log transform prices to limit the influence of outliers.

the plants' sites.[18] On average, house values are substantially higher farther away from the plant site, suggesting the locations selected for industrial plants are of systematically lower residential land quality. The final two rows of panel B show how aggregating to distance bin averages reduces the large number of individual housing transactions to a more manageable analysis dataset.

Panel C tabulates totals of owner occupied housing units in nearby census blocks by distance to the toxic plant site. A key consideration for evaluating a spatial treatment's impact is that there are typically many more housing units farther away than close in, as the area of a circle grows geometrically with its radius. For example, around all treatment site locations in my sample there are only 3.9 million owner occupied housing units within 1 kilometer, while there are 55.4 million owner occupied housing units 5-6 kilometers away from the treatment sites. As a result, even small price effects far away from a treatment site can still have a big economic impact, as they affect large numbers of housing units.

Table 1: Plant characteristics and housing sales / units by distance to plant site

Panel A. Plant characteristics						
Number of plants	11,849					
Average year of opening	2005					
Avg. annual emissions (lbs)	18,457					
	Distance to toxic plant site (km)					
Panel B. Housing sales	0-1	1-2	2-3	3-4	4-5	5-6
Avg. house price (2023 $)	$321,537	$350,266	$360,961	$375,839	$382,229	$393,012
Num. plant × house sales	3,029,638	12,556,605	21,862,672	30,144,086	37,186,807	43,141,011
Num. plant × distance bins	27,894	51,819	54,881	56,036	56,315	56,315
Panel C. Housing units						
Num. owner occ. units	3,936,235	15,902,543	27,660,928	38,031,320	47,101,033	55,373,611

Note: This table presents summary statistics on treatment sites used in the sample and the average sale prices and owner occupied housing unit counts of nearby houses by distance to the treatment site. Panel A describes 11,849 distinct plants that first appeared in the TRI data from 1995-2015. Panel B shows the average house price in real 2023 dollars by distance to the toxic plant site. Panel C tabulates totals of owner occupied housing unit counts in nearby census blocks by distance to the toxic plant site.

B *Results*

In this section, I first show the construction of DiD bin comparisons following the procedure outlined in Figure 5 for a single plant. I then present estimates using all of the 11,849 plants, which results in 303,260 DiD bin comparisons at various distances from a plant site. Finally, I

[18] It is not obvious how far away from a treated site a researcher should consider potential price effects ex ante. For TRI plant sites, I began by considering houses within 2 miles or roughly 3.2 kilometers of the treated sites following the choice of Currie et al. (2015). However, in the price effect estimates I observed a strong gradient at least out to 3.5 kilometers leading me to extend the range considered to 6 kilometers where the gradient flattens. An advantage of approaches like those outlined in Diamond and McQuade (2019), Butts (2023) or in this paper, is that they allow visual inspection of the treatment effect curve to assess where the treatment effects decay to 0.

smooth these binned comparisons with a nonparametric sieve regression following Chen et al. (2024).

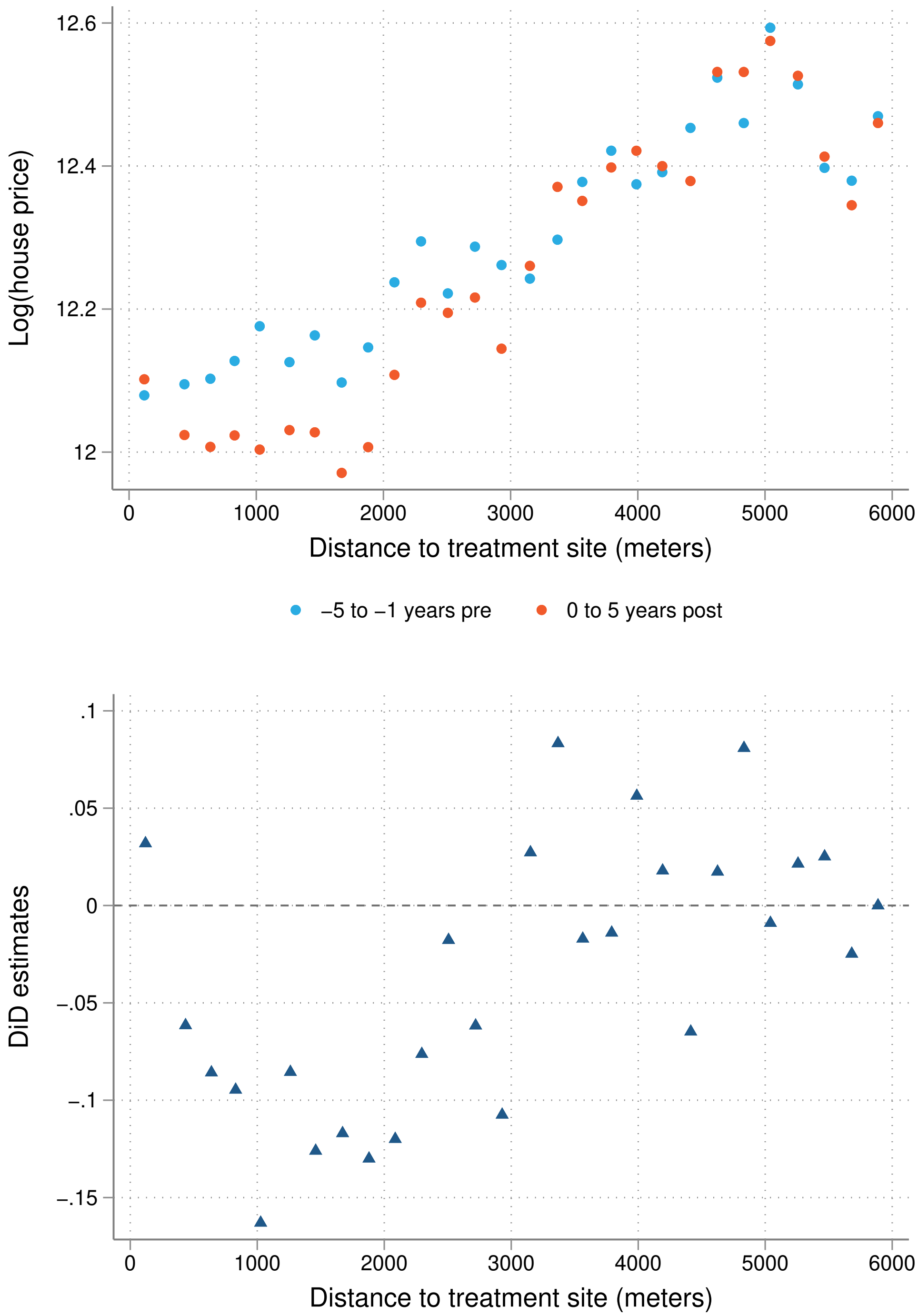
Figure 6 illustrates the binning procedure for just one of the treatment sites. The top graph plots log house price averages by distance to an industrial plant located in the Providence, RI metro area which first appeared in the TRI database in 1996. The blue dots show the averages by bin in the pre period from 1991 to 1995 and the orange dots show the averages in the post period from 1996 to 2001. The bins were chosen following Cattaneo et al. (2024) as the IMSE optimal number of equally spaced bins in the pre period.[19,20] In the bottom graph, the blue triangles show comparisons at each bin from taking the difference between the orange and the blue dots and then subtracting the analogous difference in the bin farthest away from the plant (rightmost bin). This procedure yields a distinct difference-in-differences estimate for each bin. Estimates using only comparisons around a single plant are noisy, but suggest a gradient with larger price declines closer to this plant.

I repeat this procedure separately for each of the 11,849 plants in the sample allowing for a different choice of bins at every treatment site. Performing this procedure separately by treatment site ensures that the comparisons are between housing units with the same treatment timing in the same neighborhood. Comparing within treatment timing group avoids issues with negative weighting in staggered adoption designs well documented in the literature. Comparing within the same neighborhood lends more credibility to the parallel trends assumption. The output of this binning process across all sites is a large number of price effect estimates at various distances d from the plants. In a final step, I regress these price effect estimates on d to trace out a continuous ATT function.

[19] Using quantile bins, which is standard for bin scatter plots and suggested in Butts (2023), leads to similar results. However, because the density of housing sales increases moving away from the treatment site the number of bins close to $d = 0$ is often sparse using quantile bins. For each type of spatial treatment, I also present estimates using quantile bins in Appendix A1.

[20] Software for this method is available at `https://nppackages.github.io/binsreg/`.

Figure 6: Binning example



Note: The top graph plots log house price averages by distance to a single industrial plant located in the Providence, RI metro area that first appeared in the TRI data in 1996. The blue dots show the averages by bin from 1991 to 1995 while the orange dots show the averages from 1996 to 2001. In the bottom graph, the blue triangles show the comparisons at each bin generated by taking the difference between the orange and the blue dots and then subtracting the analogous difference in the bin farthest away from the plant (rightmost bin). This procedure yields a distinct difference-in-differences estimate for each distance bin.

Price effects for all plants

Figure 7 shows price effect estimates as a smooth function of distance to the toxic plant site using comparisons from all 11,849 plants. The top graph compares distance bin average house prices in the 0 to 5 years after a plant opens relative to -5 to -1 years before. I call these estimates the "short run" price effects. The bottom graph shows the comparisons 6 to 10 years after a plant opens again relative to -5 to -1 years before, or the "long run" price effects.[21] In both plots, the red line is the fit line from a nonparametric sieve regression of the 303,260 DiD bin comparisons on distance to the plant site. This regression uses the data-driven choice of sieve dimension of Chen et al. (2024) and weights by the DiD estimates' inverse variances.[22,23] The shaded red areas are 95% uniform confidence bands. As a second check, blue dots are bin scatter plots of the 303,260 comparisons following Cattaneo et al. (2024), in which the data is now binned for a second time for visualization.

In both the short run and long run estimates, an industrial plant opening causes house prices to decline. Furthermore, plants' negative externalities affect houses closer to the plant more. In the "short run", which considers house prices 0 to 5 years after the opening of a plant, houses within a kilometer of the plant site lose 1% of house value with negative estimated effects smaller in magnitude extending out to 3–3.5 kilometers. In the "long run", which considers house prices 6 to 10 years after the opening of a plant, the relationship becomes more pronounced. Houses within a kilometer of the plant site lose 2-3% of house value with estimated effects extending out to around 4 kilometers.

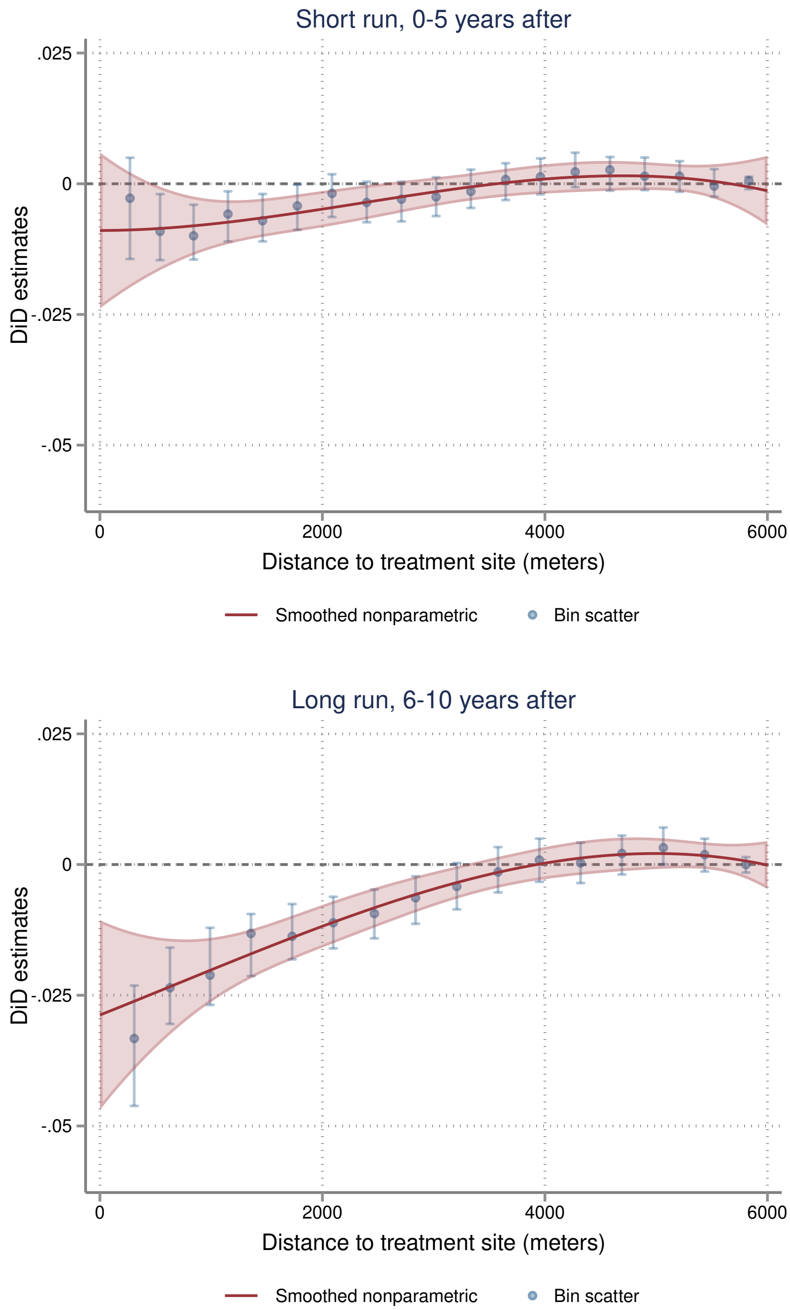
In this paper, I interpret the long run price effect estimate or the bottom plot in Figure 7 as the main price effect estimate. Hence, I use the red line as the key input to characterize aggregate local welfare and price impacts according to Equations (7) and (8).

[21] Pooling the observations into three time periods creates some disconnect from a standard DiD regression which includes year fixed effects. However, the procedure can be adapted to include treatment site×sale year fixed effects by absorbing year fixed effects in the estimation of the bin level averages. This is similar to the aggregation correction suggested in Section IV.C of Bertrand et al. (2004). Appendix A8 shows price effects residualizing treatment site×sale year fixed effects. These estimates are very similar to the estimates without year fixed effects.

[22] Appendix A1 shows binscatter plots of the long run price effects using alternative binning and weighting choices. In particular, A1 compares bin scatter plots with equal weighting of the ATT estimates, equal weighting but dropping ATT estimates based on small numbers of observations, weighting by the minimum number of observations in a bin, and weighting using inverse variance weights under both equal and quantile spaced binning strategies. In all cases, I find similar price effects for TRI industrial plants though equal spaced binning without weighting is less precise. While quantile binning is standard for binscatter plots and yields precise estimates far away from the site, estimation close to the site is poor because the number of bins becomes sparse as there are far fewer housing sales close in than far away.

[23] Software for this method is available at `https://rdrr.io/github/JeffreyRacine/npiv/man/npiv-package.html`.

Figure 7: Price effects as a function of distance to the toxic plant

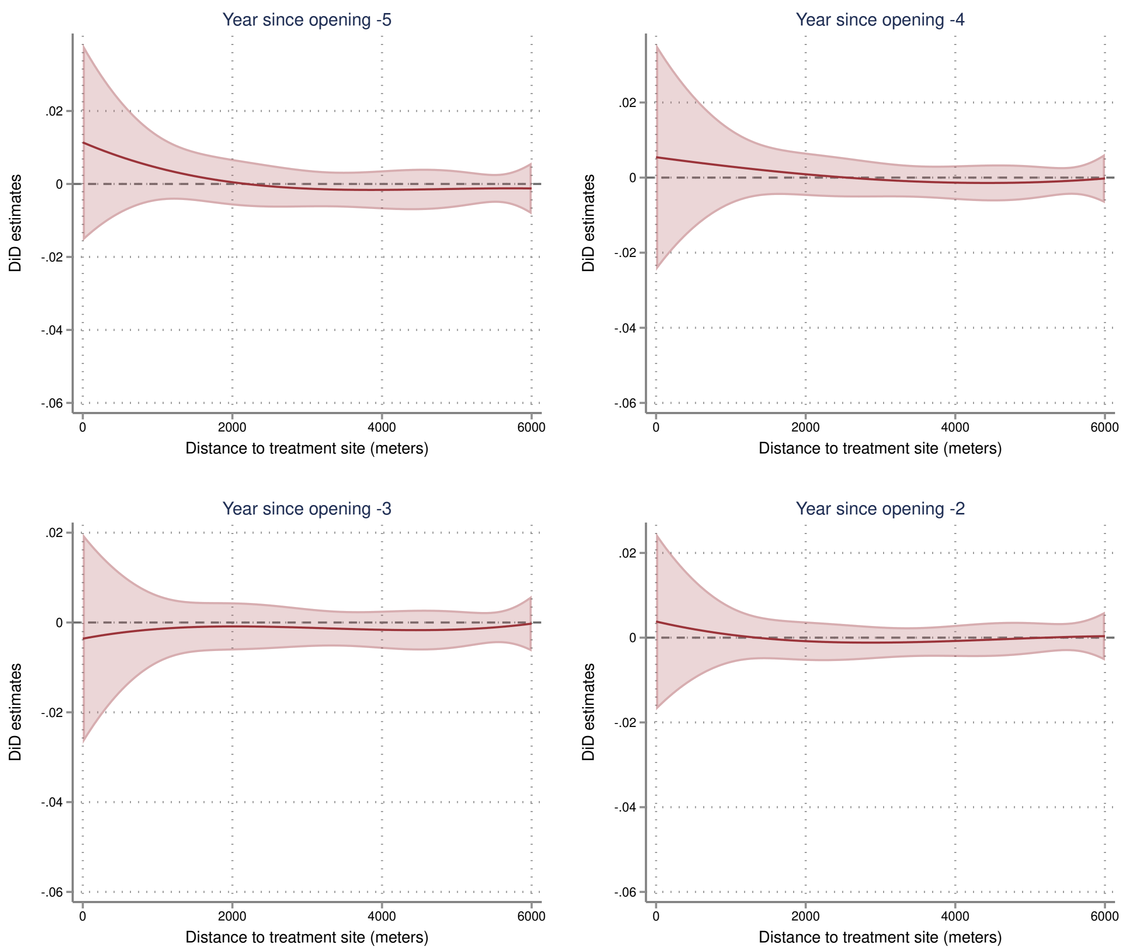


Note: This figure shows price effect estimates as a smooth function of distance to the toxic plant site. The top graph compares averages prices in the distance bins 0 to 5 years after a plant opens relative to the prices in the same distance bin -5 to -1 years before. The bottom graph shows the analogous comparisons 6 to 10 years after a plant opening relative to -5 to -1 years before. In both plots, the red line is the fit line from a nonparametric sieve regression of 303,260 DiD bin comparisons surrounding 11,849 plants on distance to the plant site following Chen et al. (2024) and weighting by the DiD estimates' inverse variances. The shaded red areas are 95% uniform confidence bands. Blue dots are bin scatter plots of the 303,260 comparisons (binning the data a second time) with confidence intervals clustered by plant site following Cattaneo et al. (2024). For the bin estimates, the intervals are not centered exactly on each bin because they are robust bias-corrected intervals.

Event studies

Before proceeding to the welfare calculation, I first present event study plots. The key assumption of the difference-in-differences strategy is the parallel trends assumption. That is, in absence of treatment the price trend for houses close to the site would have been the same as the price trend for the houses slightly farther away but unaffected by the toxic plant. However, with a continuous treatment, the price effect is now estimated along a continuum of different levels, thereby implying a continuum of possible pre trend checks. Below, I present a generalization of the event study to the continuous setting.

Figure 8: Continuous DiD event study



Note: This figure is a pre trends check by performing a similar analysis as in Figure 7 but using data prior to the treatment date. Specifically, for each year of the pre period, I form DiD comparisons at every distance bin relative to average house prices in the year before treatment. I then regress these comparisons on distance to the treatment site using the same nonparametric sieve regression procedure as in Figure 7. In Appendix Figure A9 I show the analogous plots for all years in the post period.

To check for pre trends, Figure 8 performs a similar analysis as in Figure 7 but using data prior to the treatment date. Specifically, for each year of the pre period, I form DiD comparisons at every distance bin relative to average house prices in the year before treatment. I then regress these comparisons on distance to the treatment site using the same nonparametric sieve regression procedure as in Figure 7. This pretrend check does not show evidence of significant parallel trends violations. If the treated and control units were on parallel trends in the pre period, we would see no relationship between price and distance in the pre period years. In years -4, -3, and -2, the estimated relationship is essentially a flat line. Year -5 shows a moderate decreasing gradient, but given the confidence band a null effect can not be ruled out. Additionally, in each pre period year, the estimate is small in magnitude relative to the price effect observed in the post period. Appendix Figure A9 shows the analogous plots for all years -5 to +10, which shows negative price effects in the post periods similar in magnitude to the price effects in Figure 7.

I report two other event study plots in the appendix as additional checks. In Appendix A13, I show an event study plot that pools together the ATT estimates into two distance groups to reduce the number of partitions following Callaway et al. (2024). In Appendix A14, I show the results from a standard binary stacked DiD event study regression. These figures also do not show evidence of parallel trends violations with similar negative price effects to Figure 7.

Aggregate price and welfare impact estimates

Recalling Equation (7) from Section 3,

$$\text{Agg WTP} = N_H \int_0^{d_*^0} (1+d)\widetilde{ACRT}(d|d)\mathrm{E}[Y(d,\mathbf{z})|D=d] \log\left(\frac{1+d_*^0}{1+d}\right) f(d)\,\mathrm{d}d.$$

Each term is as defined before:

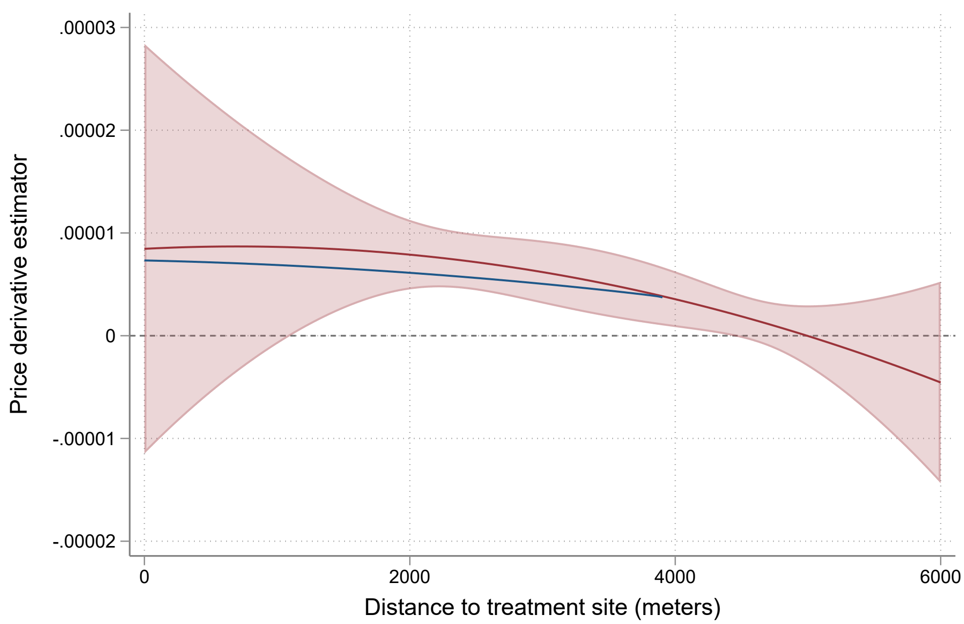
N_H	–	Total number of owner occupied housing units affected
d_*^0	–	Spatial extent of price effects
$\widetilde{ACRT}(d\|d)$	–	Log price function derivative
$\mathrm{E}[Y(d,\mathbf{z})\|D=d]$	–	Average house price at distance d
$f(d)$	–	Density of owner occupied housing units at distance d.

All terms can be estimated from the data. $\widetilde{ACRT}(d|d)$ is the primary estimand, which I estimate based on the red line in the bottom graph of Figure 7. Depending on the assumption imposed, I use either the red line's derivative $\frac{\partial \widetilde{ATT}(d|d)}{\partial d}$ or its rise over run $\frac{\widetilde{ATT}(d|d)}{d-d_*^0}$. Because I measure price changes in logs rather than levels, I multiply by the average house price at each distance d, denoted $\mathrm{E}[Y(d,\mathbf{z})|D=d]$. To estimate $\mathrm{E}[Y(d,\mathbf{z})|D=d]$, I regress post period housing prices on distance to the treated site using the same nonparametric sieve regression procedure (Chen et al., 2024). For the other terms, I estimate $\hat{d}_*^0 = 3{,}928$ meters by the point where the long run price effect first crosses 0 in Figure 7. Within the $\hat{d}_*^0$ meters of a plant, on average there are $\hat{N}_H = 6{,}939$ owner occupied housing units in nearby U.S. Census blocks affected per site. Finally, I estimate $\hat{f}(d)$ by standard kernel density estimation using owner occupied housing unit counts in nearby census blocks.

Figure 9 shows estimates for each of the terms used in the aggregate welfare calculation. Panel (a) shows estimates for the price derivative. The red line is the $\widetilde{ATT}(d|d)$'s derivative with uniform confidence bands constructed following Chen et al. (2024). A nice feature of this estimation method is that the data driven choice of sieve dimension for the ATT is the same for its derivative inheriting the desirable statistical properties outlined in Chen et al. (2024). The derivative estimate is always positive over the affected area suggesting that homeowners are willing to pay to be farther away from a toxic industrial plant. Assuming homogeneous treatment effects, this estimate would imply a homeowner near the treatment site's MWTP is $\sim$1% of their house's value to be 1 kilometer farther away from a plant (0.00001 $\times$ 1,000). The blue line shows the price derivative estimate using the rise over the run, which is a lower bound on the MWTP even in the presence of treatment effect heterogeneity if concavity as defined in Assumption 8 holds.[24] In this application, the rise over the run is similar to the ATT derivative but moderately smaller in magnitude.

Figure 9: TRI, components of aggregate welfare calculation

(a) Price derivative estimates $\frac{\partial \widetilde{ATT}}{\partial d}(d|d)$ and $\frac{\widetilde{ATT}(d|d)}{d-\hat{d}_*^0}$



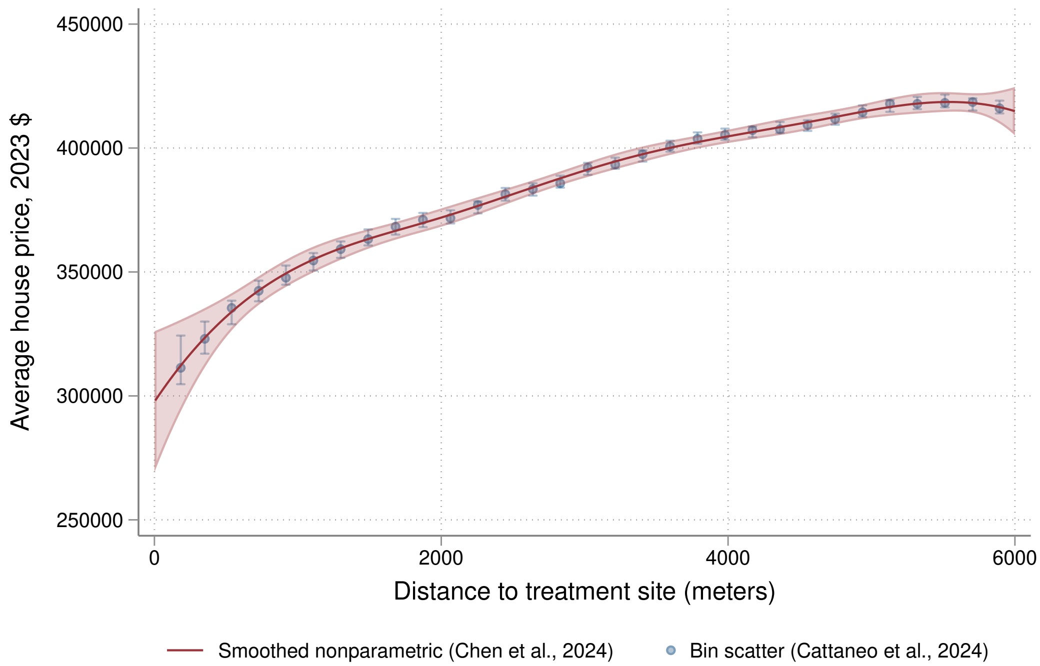
Panel (b) estimates the relationship between the average house price and distance to a plant site in the post period. I regress average house prices in the post period on distance to the treatment site and plant-specific fixed effects.[25] As expected, house prices are significantly lower near the site of a plant, likely reflecting in large part the impact of other factors on property values other

[24]This estimate is trimmed close to d_*^0. As described in Graham and Powell (2012), an average partial effect using the rise over the run is only irregularly identified. This is because for d close to d_*^0 the denominator, or the run, is close to 0. To deal with small denominator effects, I follow Graham and Powell (2012) and trim the distance range close to d_*^0 by following the rule of thumb bandwidth in Graham and Powell (2012), which in this application is only 24 meters.

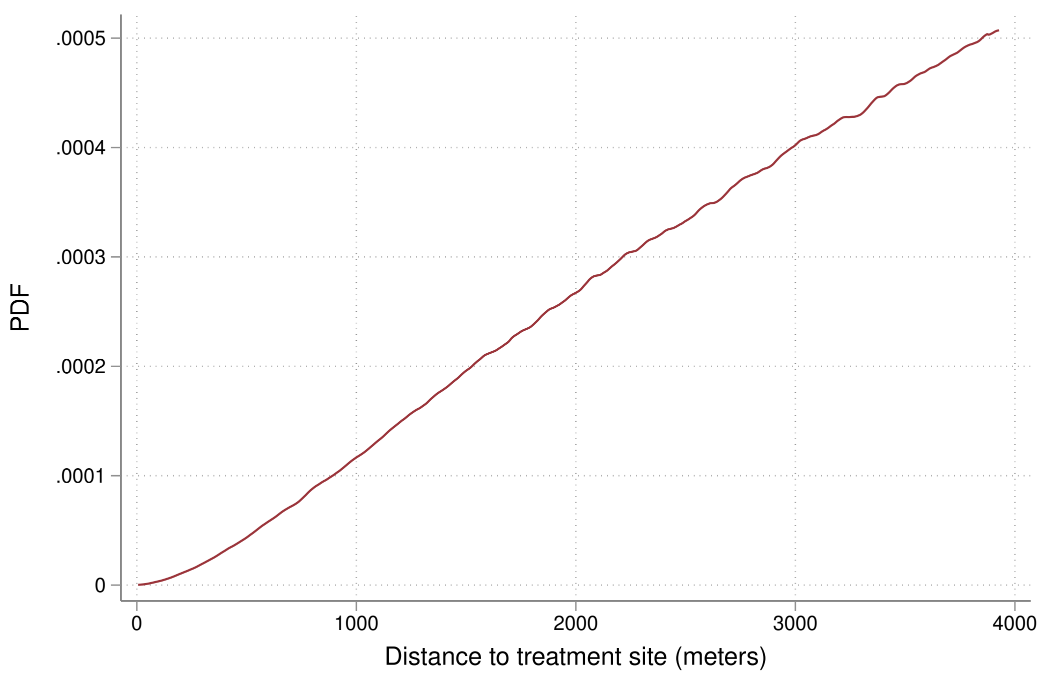
[25]To control for plant fixed effects, I follow the procedure in Chen et al. (2024)'s application. Specifically, I regress housing prices on distance to the treatment site and plant-specific fixed effects under an initial basis choice. I then apply Chen et al. (2024) as before but using house prices minus the estimated plant fixed effects from the initial regression as the outcome variable. As an additional check, the blue scatter plot also shows the estimates from `binsreg` with covariate adjustment following Cattaneo et al. (2024).

Figure 9: continued

(b) Average house price level $\widehat{E[Y_2|D=d]}$



(c) Owner occupied housing unit density $\hat{f}(d)$



Note: This figure plots estimates of each term used in the Aggregate Willingness to Pay Equation (7) for TRI industrial plants as a function of distance to the treatment site. Panel (a) shows the price derivative estimates. Panel (b) shows the average house price in the post period. Panel (c) plots the PDF of owner occupied housing units.

than the plant itself. House prices are $\sim$ \$300,000 close to the plant and increase to $\sim$ \$420,000 at 6 kilometers away. Finally, Panel (c) shows the empirical density of owner occupied housing units on distance to the plant site. This plot confirms the intuition from the summary statistics that there are many more housing units far away from the plant's site than located close in.

After plugging in each term, Equation (7) is a 1-dimensional numerical integration problem that I solve using standard methods. To start, as a baseline I evaluate Equation (7) using $\frac{\partial \widetilde{\widehat{ATT}}(d|d)}{\partial d}$ as the estimator for homeowners' MWTPs. As discussed in Sections 2 and 3, this assumes homogeneous treatment effects. I calculate homeowners would be willing to pay \$16.4 million dollars for the plant not to exist. Strengthening assumptions further to a time constant hedonic price function as in Diamond and McQuade (2019), this estimate says a plant opening causes an aggregate welfare loss of \$16.4 million dollars to nearby homeowners.

However, $\frac{\partial \widetilde{\widehat{ATT}}(d|d)}{\partial d}$ does not identify the price derivative under unobserved treatment effect heterogeneity. Sections 2 and 3 discussed an alternative restriction. In particular, if treatment effects are concave in distance from the plant, the average rise over the common run is a conservative lower bound on the price derivative. Evaluating Equation (7) instead using the rise over the run estimator, I calculate homeowners would be willing to pay \$13.1 million dollars for the plant not to exist. As anticipated, the rise over the run estimate appears to be conservative.

In addition, I calculate the aggregate price loss to homeowners to bound the other side of the welfare measure. By Equation (8),

$$\Delta\text{Price} = N_H \int_0^{d_*^0} \widetilde{\widehat{ATT}}(d|d) \mathrm{E}[Y(d,\mathbf{z})|D=d] f(d)\, \mathrm{d}d.$$

As shown in Section 3, the negative price change is an upper bound on homeowners' willingness to pay for the plant not to exist. In other words, we expect the aggregate price loss to overestimate the magnitude of a toxic plant's welfare impact. Using the long run price effect estimate from Figure 7 as $\widetilde{\widehat{ATT}}(d|d)$ and estimating $\hat{N}_h$, $\widehat{\mathrm{E}[Y|D=d]}$, and $\hat{f}(d)$ as above, I find a toxic plant causes an aggregate price loss of \$19.5 million dollars to local homeowners. Here, consistent with theory, the negative price loss provides an informative but non-conservative bound on the welfare measure.

Taken together, the welfare statistics bound homeowners' willingness to pay between \$13.1 and \$19.5 million dollars for a polluting industrial plant not to exist. The point estimate assuming homogeneous treatment effects falls between these two bounds at \$16.4 million dollars.

6 Additional applications: LIHTC developments and sex offenders

In this section, I use the econometric procedure to evaluate two more key applications in the hedonic literature: low income housing developments and sex offender move ins. First, I briefly review the data sources I use to characterize treatment events for each application. Both applications have significantly more treatment sites than the original studies. Then, I present price effect estimates for low income housing developments. For sex offender move ins, I find evidence of parallel trends violations. Finally, I compare price and welfare impacts between TRI industrial plants and LIHTC developments on a consistent basis.

A *Background*

LIHTC developments

The Low Income Housing Tax Credit is a large government program that incentivizes construction of affordable housing for low income renters. To date, it has funded approximately 1/5 of the multifamily developments in the U.S. and houses 2% of the nation's population. See, e.g. Diamond and McQuade (2019) and Soltas (2024), for a more detailed background on the LIHTC program and its funding allocation process. Briefly, developers receive tax credits for new construction of multifamily units if those units satisfy certain criteria for low income housing. In particular, a certain percentage of the building's tenants must have below the area median gross income (AMGI). Developers usually sell these tax credits to investors to fund the development and those investors receive the credits over a ten year period.

The U.S. Department of Housing maintains a database on LIHTC developments.[26] The LIHTC data provide information on 53,032 developments that have been funded since 1986. I restrict to developments with nonmissing geographic information and with funding allocated from 1995 to 2015. In addition, I can only study sites where I observe nearby housing sales before and after funding allocation. After these restrictions, my final estimation sample has 17,895 LIHTC sites.

Sex offender move ins

Linden and Rockoff (2008) and Pope (2008)'s studies on the effect of sex offender move ins on house prices were some of the ring DiD design's earliest applications. The former study uses data from Mecklenburg County in North Carolina and the latter uses data from Hillsborough County in Florida. Both of these U.S. states are well suited to this analysis in that state law requires sex offenders to register their address to the local sheriff within 10 day of their move in.

I collected the current sex offender registries for the wholes of North Carolina and Florida,[27] which contain all sex offenders' addresses and their move in dates in these states. It is important to note that the current registries reflect only the most recent address change of each sex offender and not a complete history of moves. As a result, my sample is skewed toward sex offenders that have lived in the same location for longer periods of time, whereas the typical sex offender moves frequently. After dropping sex offenders that are in prison, deceased, or no longer in state and restricting to move ins prior to 2015 with nearby housing sales, I am left with 9,193 sex offender move ins in my estimation sample with 1,701 coming from North Carolina and 7,492 from Florida.

Appendix Tables A1 and A2 presents summary statistics on house sales and housing units close to LIHTC developments and sex offender move ins analogous to Table 1.

B *Price effects by application*

LIHTC developments

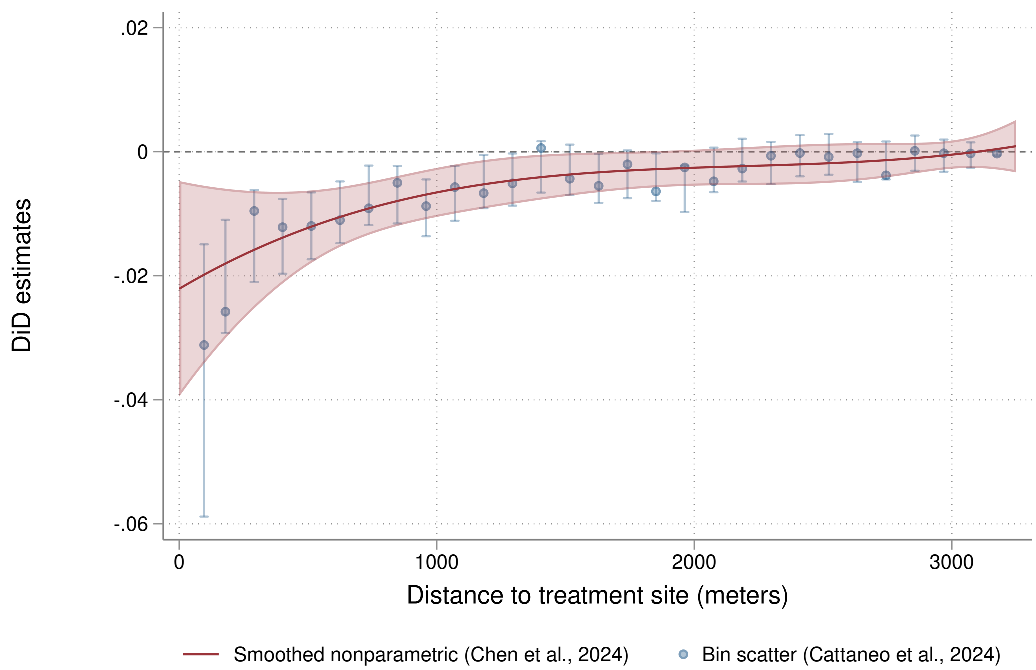
I now present price effect estimates for a LIHTC development following the same procedure as in Section 5. The red line in Figure 10 is the main price effect estimate from comparing average house prices in the 6 to 10 years after a LIHTC development's funding allocation relative to -5

[26] LIHTC data are publicly available at `https://www.huduser.gov/portal/datasets/lihtc/property.html`.

[27] North Carolina's registry is available at `https://sexoffender.ncsbi.gov/stats.aspx`. Florida's registry is available at `https://offender.fdle.state.fl.us/offender/sops/registryDownload.jsf`.

to -1 years before allocation. The red line is the fit line from a nonparametric sieve regression of 324,007 DiD bin comparisons on distance to a LIHTC development around 17,895 LIHTC sites. As before, this regression uses a data-driven choice of sieve dimension following Chen et al. (2024) and weights by the DiD estimates' inverse variances.[28] Blue dots are bin scatter plots following Cattaneo et al. (2024), which provides a second check of the smoothed estimate. I find a small but precise decline in house values of 2% directly next to a development with effects decaying to 0 around 2.5-3 kilometers.

Figure 10: LIHTC developments, long run price effect, all LIHTC sites



Note: This figure shows the long run price effect estimate as a smooth function of distance to a LIHTC development. The red line is the fit line from a nonparametric sieve regression of 324,007 DiD bin comparisons from 17,895 LIHTC developments on distance to the LIHTC site following Chen et al. (2024) and weighting by the DiD estimates' inverse variances. The shaded red areas are the 95% uniform confidence bands. Blue dots are bin scatter plots of the 324,007 comparisons with confidence intervals clustered by treatment site following Cattaneo et al. (2024).

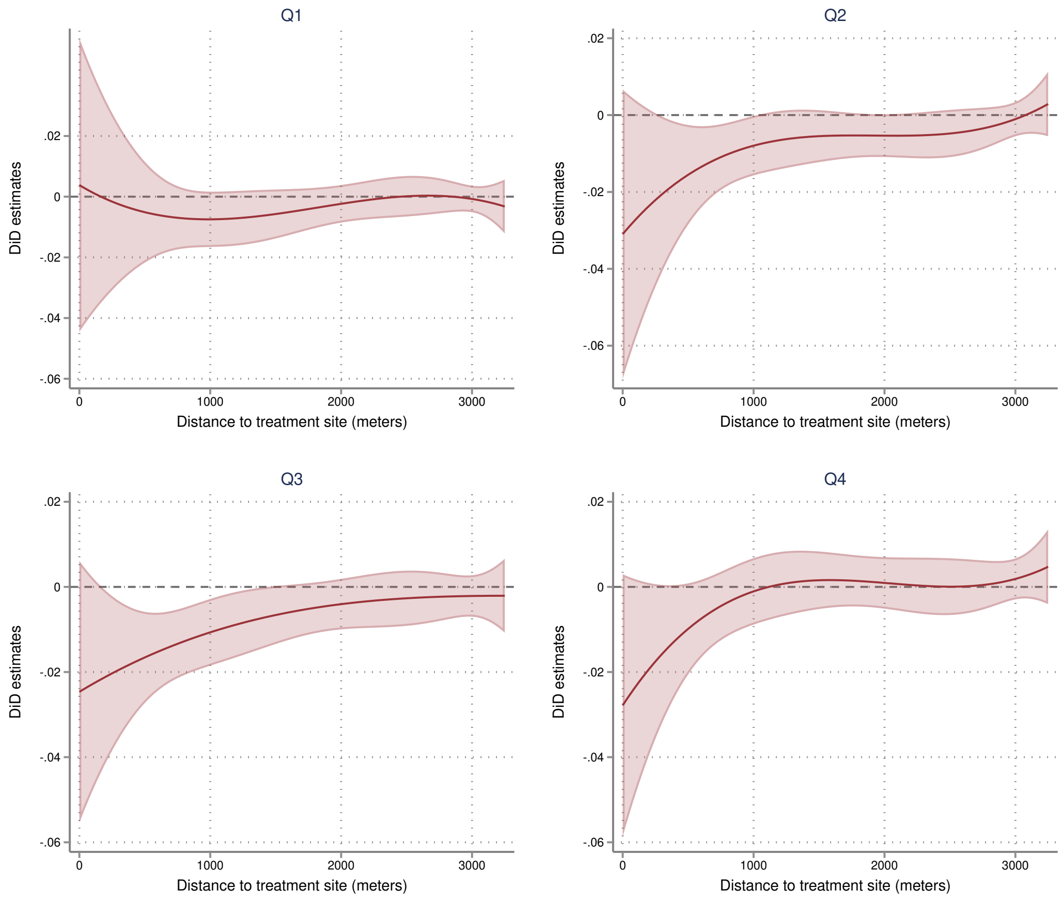
Appendix Figure A10 presents a continuous DiD event study for LIHTC developments analogous to the event study for TRI industrial plants (Figure 8). As in the first application, I do not find strong visual evidence that would suggest parallel trends violations. More discussion is provided in the appendix.

Figure 11 explores heterogeneity by neighborhood household income by splitting the sample of treatment sites by the household income quartile of a LIHTC site's census block group. In the first income quartile, the estimate looks like a null effect. In the second, third, and fourth income

[28] As with TRI industrial plants, I also present long run price effect estimates under an array of alternative weighting and binning schemes, and residualizing treatment site $\times$ sale year fixed effects. These plots are presented in Appendix A2 and Appendix A8.

quartiles, I find a similar price effect estimate showing a moderate price decline of 2-3% directly next to a LIHTC development.

Figure 11: LIHTC developments, price effects by neighborhood household income quartile



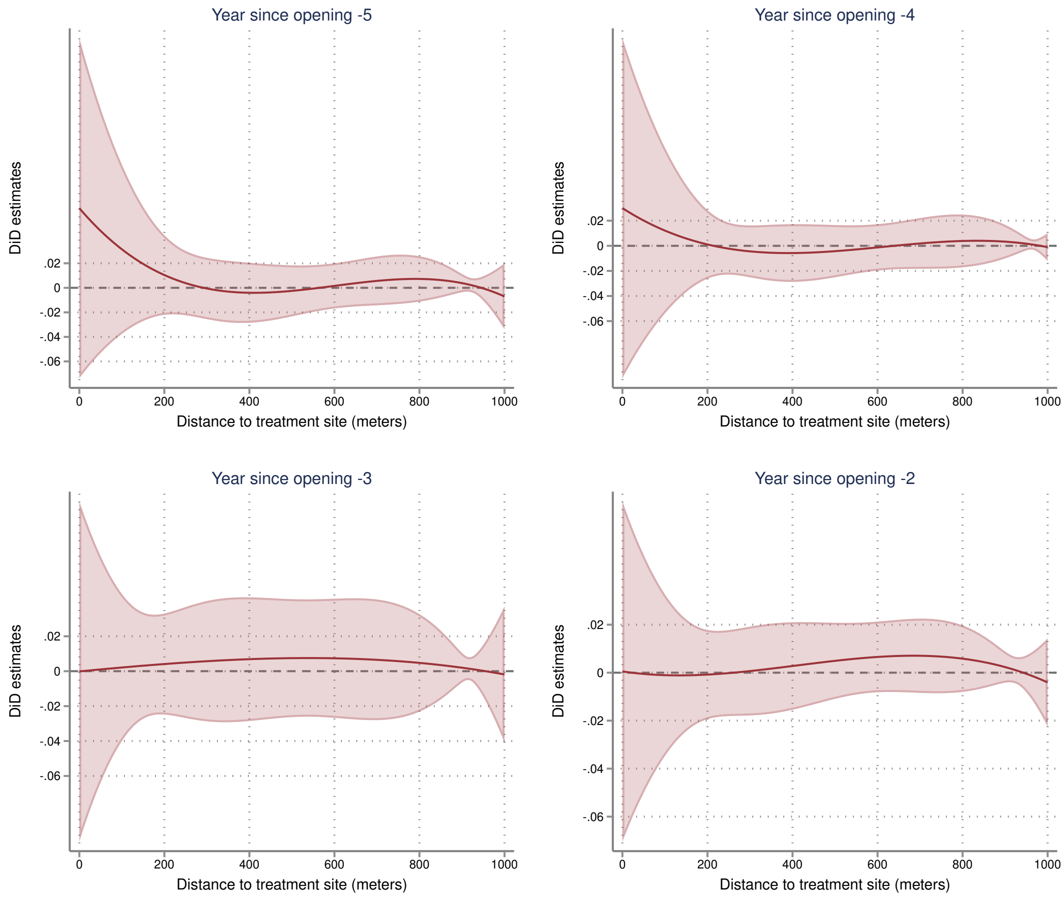
Note: This figure performs a similar analysis as in Figure 10 but splitting the sample of treatment sites into quartiles by the median household income in a site's census block group in the 2008-2012 5-year American Community Survey (ACS). The income quartile cutoffs are $31,657, $45,627, and $64,509 in 2023 dollars.

Sex offender move ins have parallel trends violations

In contrast to TRI plants and LIHTC developments, for sex offender move ins I find evidence of parallel trends violations. Figure 12 is the continuous event study plot for sex offender move ins. While the confidence bands imply there is a lot of uncertainty,[29] prices for housing units close to the sex offender appear to be trending downward even prior to the event date, suggesting sex offenders may be moving into parts of the neighborhood where houses were already depreciating in value.

[29]Precision can be improved by aggregating more comparisons together, e.g. by going back to a binary estimation of the treatment effect and calculating the weighted mean ATT for all bins within 200 meters of the sex offender's address (roughly comparable to the 0.1 mile treated group in Linden and Rockoff (2008)). Evidence of parallel trends violations are also visible in these binary estimates.

Figure 12: Sex offender move ins, event study



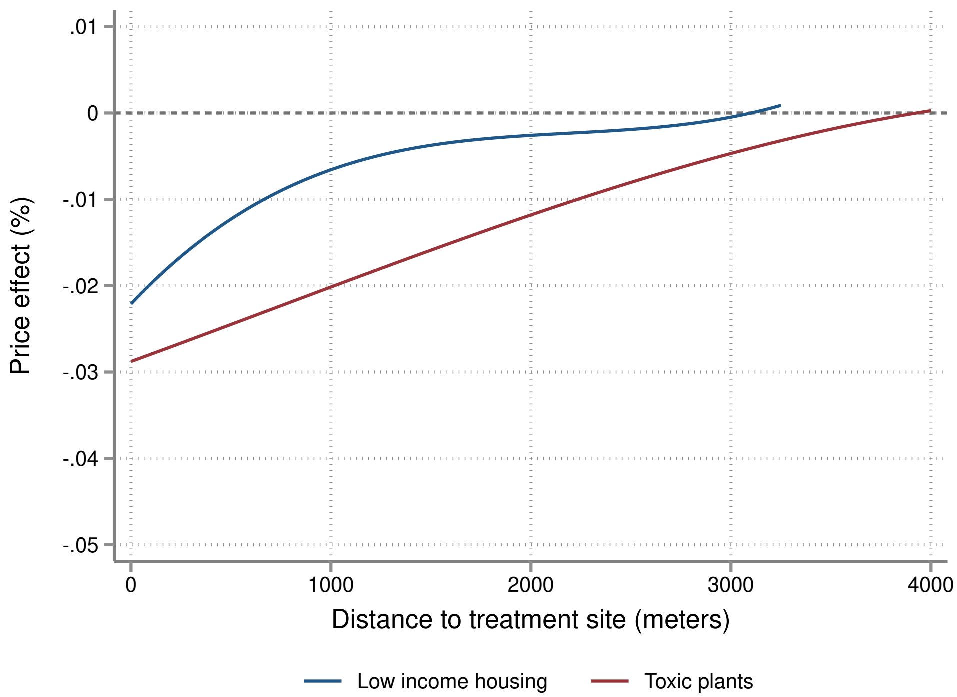
Note: This figure performs an event study similar to Figure 8 for sex offender move ins. Specifically, for each year of the pre period, I form DiD comparisons at every distance bin relative to average house prices in the year before treatment. I then regress these comparisons on distance to the treatment site using nonparametric sieve regression.

C *Comparing price and welfare impacts across treatment types*

I now compare local price and welfare impacts for TRI industrial plants and LIHTC developments on the same scale. I do not report results for sex offenders because the price effect estimates may be due to parallel trends violations.

Figure 13 plots the long run price effect estimates on the same y and x axes. The blue line shows the price effect estimate for LIHTC developments and the red line shows the effect for TRI industrial plants. The price effect from a TRI industrial plant exceeds that of a LIHTC development, in magnitude and the affected geographic area's spatial scale.

Figure 13: Long run price effects on the same scale



Note: This figure shows the price effect estimates from Figures 7 and 10 on the same y and x axes. The blue line shows the price effect estimate for LIHTC developments and the red line shows the effect for TRI industrial plants.

Table 2 presents estimates of the price and welfare impacts on local homeowners from TRI industrial plants and LIHTC developments. The first column reports the estimates for TRI plants and the second column for LIHTC developments. In each panel, the first two rows show heterogeneity-robust bounds on homeowners' willingnesses to pay to avoid the disamenity from the treatment site following the bounding argument in Section 3. Assuming treatment effect concavity, the first row is a lower bound on the willingness to pay from using the rise over the run estimator as the average MWTP. The second row shows the upper bound, which is the negative of the average price change. The third row in each panel compares the bounds to a benchmark that assumes homogeneous treatment effects (strong parallel trends) and uses the ATT derivative as the average MWTP.

The first panel shows the average willingness to pay to avoid the disamenity from the treatment site as a percentage of house value. Following the argument of Theorem 3.2, the average amount a homeowner within the area affected by a toxic industrial plant would be willing to pay to not be exposed to the plant is bounded between 0.5% and 0.76% of their house's value. The analogous figures for LIHTC developments is 0.23% to 0.36%. Assuming homogeneous treatment effects, the estimated average willingness to pay to avoid the treatment site is 0.63% of house value for TRI plants and 0.32% for LIHTC developments.

The second panel is similar but now estimated in dollars by multiplying by average house prices. As noted in Section 3, converting from logs to levels in this way requires Assumption 10. As in the first panel, the first two rows in this panel are the bounds where the first row uses the rise over the run estimator for the price derivative and the second row is the average price change. These correspond to the integrals evaluated in the Aggregate WTP Equation (7) and the Aggregate Price Change Equation (8). The third row in this panel compares to using the ATT derivative as the estimator for homeowner MWTP. The average amount a homeowner within an area affected by a toxic industrial plant would be willing to pay to not be exposed to the plant is bounded between $1,886 and $2,804. The analogous figures for LIHTC developments is $815 to $1,277. Again, the estimates assuming homogeneous treatment effects lie within these bounds at $2,365 and $1,113 for TRI and LIHTC sites, respectively. The final row in this panel shows the estimated number of affected homeowners or $\hat{N}_H$ in equations (7) and (8).

The third panel of the table shows the aggregate impacts per site attained by multiplying the average WTP estimates and price changes in dollars by the number of homeowners, $\hat{N}_H$. Scaling up the figures by the number of homeowners implies that the aggregate homeowner WTP per site is 13.1-19.5 million dollars for TRI industrial plants and 6.7-10.5 million dollars for LIHTC developments. As suggested by the price effect estimates in Figure 13, the welfare loss per site from a TRI industrial plant's externality is significantly larger than that of a LIHTC development. The last row of the third panel has an estimate of the number of treatment sites currently operating in the United States. There are 15,899 TRI sites and 43,466 LIHTC sites in the United States.[30]

The final panel shows back of the envelope estimates of the total externalities nationally based on multiplying the aggregate impacts per site by the estimates for the number of treatment sites currently operating in the United States. Because both types of treatment affect many homeowners and there are numerous sites across the U.S., at the national level these effects appear to be important. The total national welfare externalities are at least $208 billion dollars for TRI industrial plants and $291 billion dollars for LIHTC developments. While a LIHTC development has about half the aggregate price and welfare impact per site compared to a TRI industrial plant, there are more than double the number of LIHTC sites across the nation leading to a larger total externality among local homeowners.

I conclude discussion of this table with two remarks about the estimates assuming homogeneous treatment effects. First, for these two applications, the homogeneous treatment effect estimates are within the heterogeneity-robust bounds, suggesting there may not be a huge bias from assuming homogeneous effects. However, if the degree of selection bias is large, there is no guarantee that in other applications an estimate using the ATT derivative will lie within the bounds. Second, imposing stronger assumptions so that the price function is time constant with homogeneous treatment effects as in Diamond and McQuade (2019) leads to stronger welfare statements. In this case, the third row in each panel is a point identified welfare loss among homeowners. For example, the third row of the third panel implies a polluting industrial plant causes a welfare loss of 16.4 million dollars to homeowners under these stronger assumptions.

[30]The estimate for the number of TRI sites is the number of plants in the TRI database reporting positive pollution emissions in 2022. The estimate for the number of LIHTC sites is the number of developments in the HUD database still being monitored as of April 2024.

Table 2: Total local price and welfare impacts on homeowners

	Toxic industrial plants	Low income housing
LB: Avg. WTP per homeowner (%), $\frac{A\tilde{T}T}{d-d_*^0}$	0.50%	0.23%
	(0.091)	(0.074)
UB: Avg. $-\Delta$Price per homeowner (%)	0.76%	0.36%
	(0.124)	(0.103)
SPT: Avg. WTP per homeowner (%), $\frac{\partial A\tilde{T}T}{\partial d}$	0.63%	0.32%
	(0.095)	(0.066)
LB: Avg. WTP per homeowner (\$), $\frac{ATT}{d-d_*^0}$	\$1,886	\$815
	(296)	(273)
UB: Avg. $-\Delta$Price per homeowner (\$)	\$2,804	\$1,277
	(398)	(373)
SPT: Avg. WTP per homeowner (\$), $\frac{\partial ATT}{\partial d}$	\$2,365	\$1,113
	(313)	(229)
# of affected homeowners per site	6,939	8,219
LB: Agg. WTP per site, $\frac{ATT}{d-d_*^0}$	\$13.1 mil.	\$6.7 mil.
	(3.84)	(2.92)
UB: Agg. $-\Delta$Price per site	\$19.5 mil.	\$10.5 mil.
	(6.18)	(4.11)
SPT: Agg. WTP per site, $\frac{\partial ATT}{\partial d}$	\$16.4 mil.	\$9.1 mil.
	(7.53)	(2.46)
# of sites in the U.S.	15,899	43,466
LB: Total WTP, $\frac{ATT}{d-d_*^0}$	\$208 bil.	\$291 bil.
UB: Total -ΔPrice	\$309 bil.	\$456 bil.
SPT: Total WTP, $\frac{\partial ATT}{\partial d}$	\$261 bil.	\$398 bil.

Note: This table presents estimates of the welfare and price impacts on local homeowners from TRI industrial plants and LIHTC developments. The first column has estimates for TRI industrial plants and the second column has estimates for LIHTC developments. The first two rows in each panel show heterogeneity-robust bounds. The first row shows the lower bound on the avg. willingness to pay among homeowners as a percentage of house value for a treatment site to not exist from using the rise over the run as the avg. MWTP. The second row shows the upper bound, which is the negative of the avg. price change as a percentage of house value. The third row compares to an estimate that assumes homogeneous treatment effects (strong parallel trends) and uses the ATT derivative as the avg. MWTP. Rows 4-6 convert to monetary estimates by multiplying by the avg. house price at each distance. In the next panel, aggregate WTP and price changes are calculated by multiplying the avg. WTP and price change by the # of affected homeowners. The final three rows are back of the envelope estimates of the total externalities nationally based on multiplying the aggregate impacts by an estimate for the number of treatment sites currently in the U.S. (in the row directly above). Standard errors were calculated by bootstrap resampling treatment sites for 1,000 replicates.

D *Discussion*

It is important to discuss the limitations on what is not captured by these local welfare impact estimates.

In particular, while these estimates are well suited as an estimate of the spillovers onto nearby homeowners, for both types of treatment event there is reason to believe the estimates mainly capture costs without consideration of the potential benefits. In the case of industrial plants, these plants may provide jobs to the larger geographic area, which might not show up in a localized pricing gradient, but could increase welfare overall.[31] For LIHTC developments, the obvious beneficiaries are low income tenants who now have greater access to affordable housing, but whose preferences will not be captured by the valuation of nearby owner occupied houses.

In addition, the structural model in Section 3 involves several assumptions. In this paper, I focused on relaxing one assumption—the strong parallel trends assumption to identify the hedonic price derivative. I then provided statistics to bound an aggregate welfare measure without this assumption. While these statistics appear to be informative, their derivation required other assumptions on separability of the hedonic pricing function, homeowners' utility functions, price effect concavity, and other assumptions in the standard hedonic model.

With these limitations in mind, these estimates may be valuable for local municipal policymakers who frequently have to weigh tradeoffs in decisions such as whether to site an industrial plant in their town. In this case, an industrial plant has both positive and negative externalities—while benefits like the projected number of jobs provided by a plant are often available to policymakers, the value of non-marketed neighborhood characteristics like pollution are harder to quantify. This paper fills this gap by connecting the pricing gradients around TRI industrial plants to a structural hedonic model.

For LIHTC developments, in contrast to the previous literature I did not find as much evidence that low income housing developments raise property values in poor neighborhoods (Baum-Snow and Marion, 2009, Diamond and McQuade, 2019). While this result warrants further study, it potentially has important implications to the debate over LIHTC's efficacy as a policy program. Soltas (2024) models developer behavior and estimates that LIHTC subsidies add few net units to the housing stock. On the other hand, low income households do benefit from these subsidies. However, if LIHTC development can neither be justified as place-based policy nor as increasing the housing stock, that provides greater evidence that comparable programs such as housing vouchers might provide similar benefits to low income tenants at less fiscal cost.

Finally, quantifying welfare impacts using this procedure facilitates comparison of values across different amenities: Local homeowners would be willing to pay at least 13.1 million dollars for a polluting industrial plant to not exist, while the analogous figure for LIHTC developments is 6.7 million dollars. Given the large number of TRI industrial plants and LIHTC developments in the U.S., these externalities are important. A back of the envelope calculation estimates national externalities of at least 208 billion dollars for TRI industrial plants and 291 billion dollars for

[31] As noted in Currie et al. (2015), a fact suggestive of the potentially large size of these benefits is that the typical investment in a TRI industrial plant can be in excess of 500 million dollars, dwarfing the 13.1-19.5 million dollar negative externality reported in Table 2.

LIHTC developments.[32] For context, in the most recent Interstate Highway Cost Estimate in 1991, the total cost to build the entire U.S. interstate highway system was estimated at 129 billion dollars or 285 billion in 2023 dollars.[33] Hence, systematic evaluation of non-marketed housing characteristics, which encompasses many environmental and neighborhood amenities, may be useful for local, state, or national accounting.

7 Conclusion

A large literature in economics uses ring DiD designs to characterize the effect of neighborhood amenities on house prices. In hedonics, the price derivative characterizes welfare. By adapting Callaway et al. (2025)'s econometric framework for continuous DiD to this setting, I highlight an implicit assumption of homogeneous treatment effects needed for a DiD derivative to identify the price derivative. I then develop an approach that does not assume treatment effect homogeneity. Instead, I combine a shape restriction consistent with spatial decay and the hedonic model's equilibrium conditions to bound a treatment event's welfare impact. I illustrate this approach revisiting two key applications with more treatment sites and observations.

I conclude by noting that this framework and estimating procedure can apply to many more types of spatial treatments, allowing for comparison of values across different amenities. For example, it could be used to evaluate policies such as EPA cleanup sites including Superfund and brownfield remediation sites or other EPA locations of interest[34] where concrete cost estimates might allow a complete cost-benefit analysis. Other applicable examples include power plants, transit stations, windmills, and so on.[35] In general, there is now a large literature of spatial DiD estimates with reduced form price effects, but these estimates often do not make a clear connection to an underlying economic model. The framework in this paper can be used to better align empirical estimates with the welfare-relevant parameters.

[32] While due to evidence of parallel trends violations I do not report estimates for sex offender move ins in the main paper, the external effects from sex offenders could still be economically important. Even assuming a price impact of approximately -1.3% over just 200 meters, sex offenders still cause 190 billion dollars in externalities nationally. Despite small price impacts per treatment site, there are many more sex offenders in the U.S. than there are either TRI industrial plants or LIHTC developments.

[33] `https://www.fhwa.dot.gov/infrastructure/50estimate.cfm`

[34] For studies of EPA sites, see Greenstone and Gallagher (2008), Gamper-Rabindran and Timmins (2013), Haninger et al. (2017) and Marcus (2021), among others.

[35] See, e.g., Davis (2011), Gibbons and Machin (2005), and Dröes and Koster (2016).

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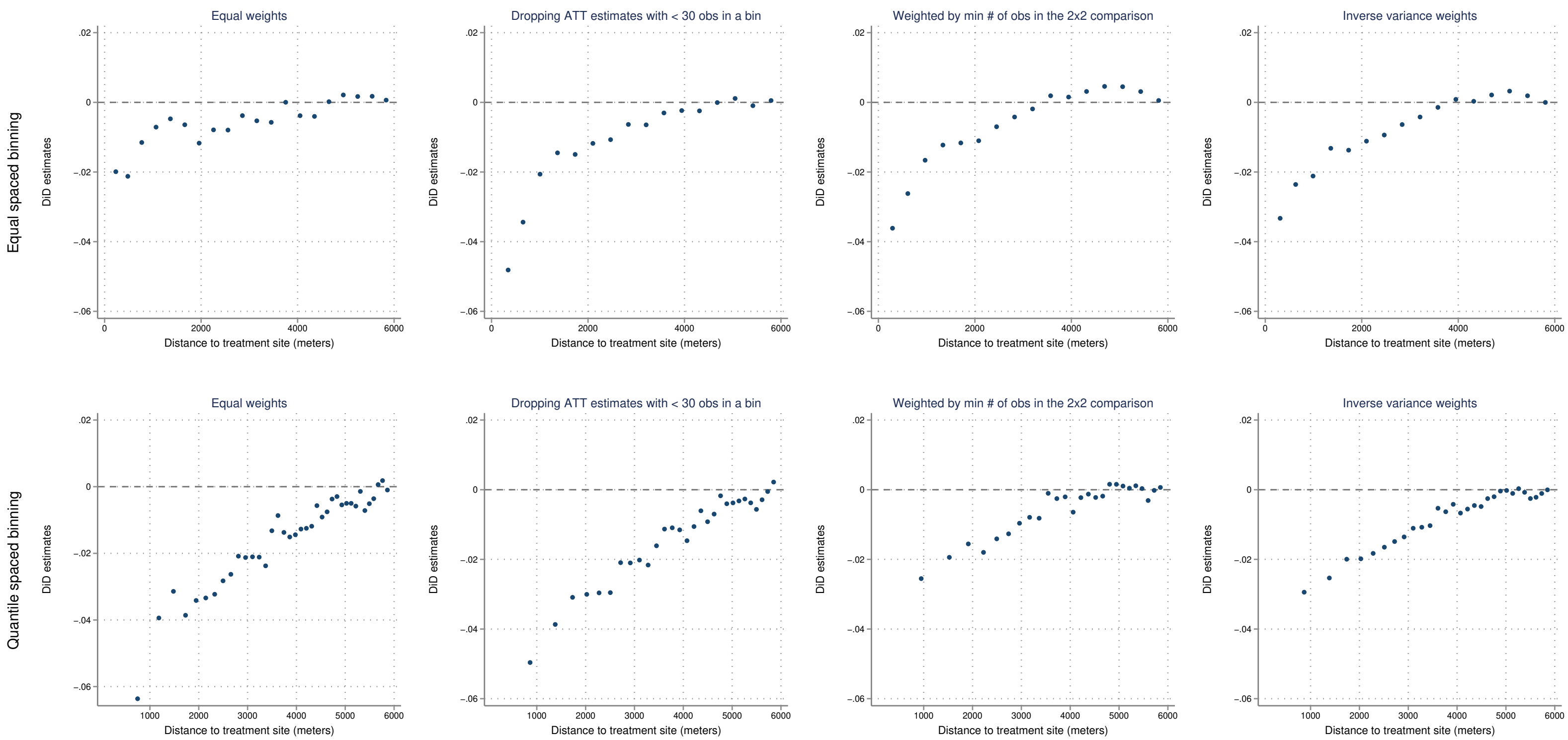
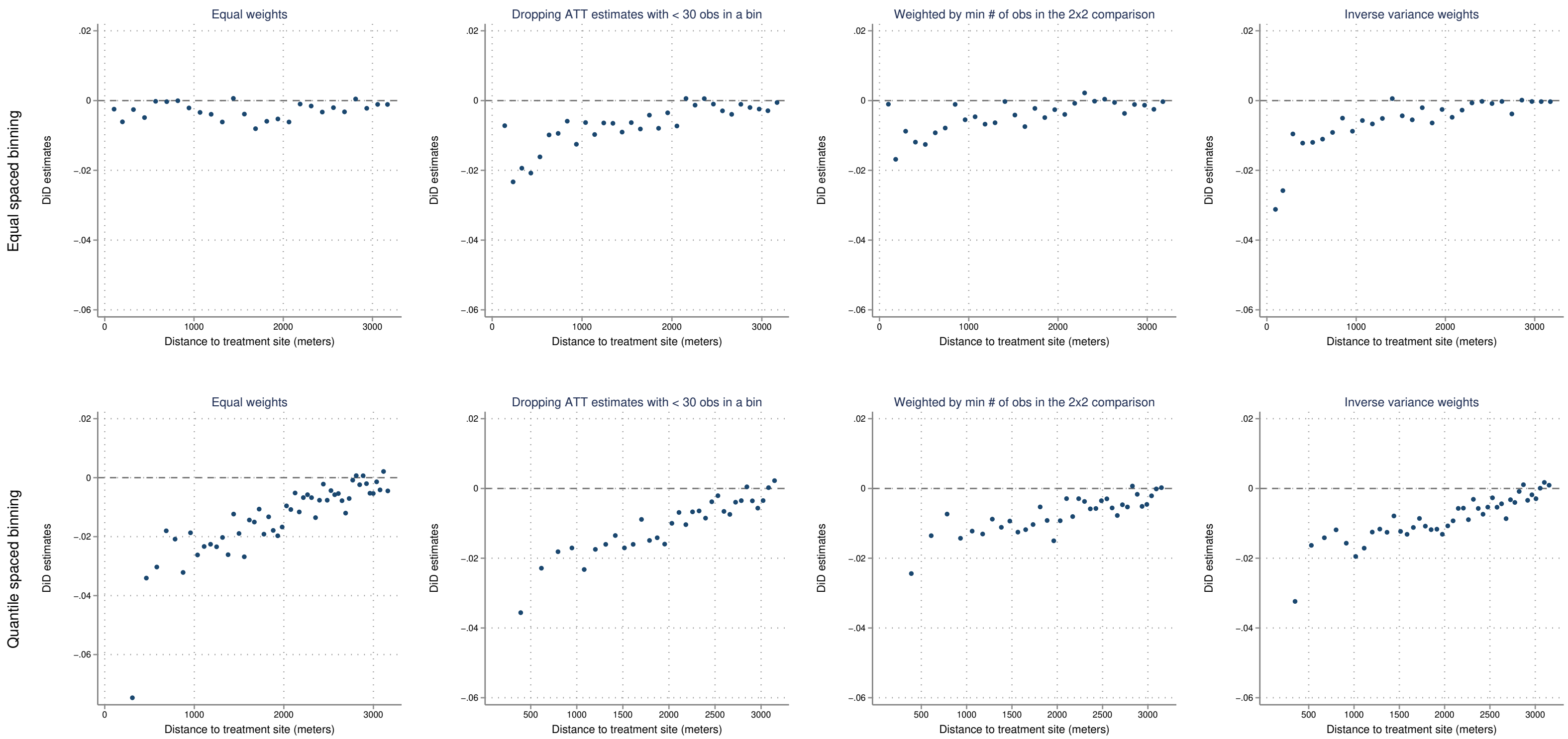
Appendix A Appendix figures

Figure A1: TRI long run price effects under alternative binning and weighting choices



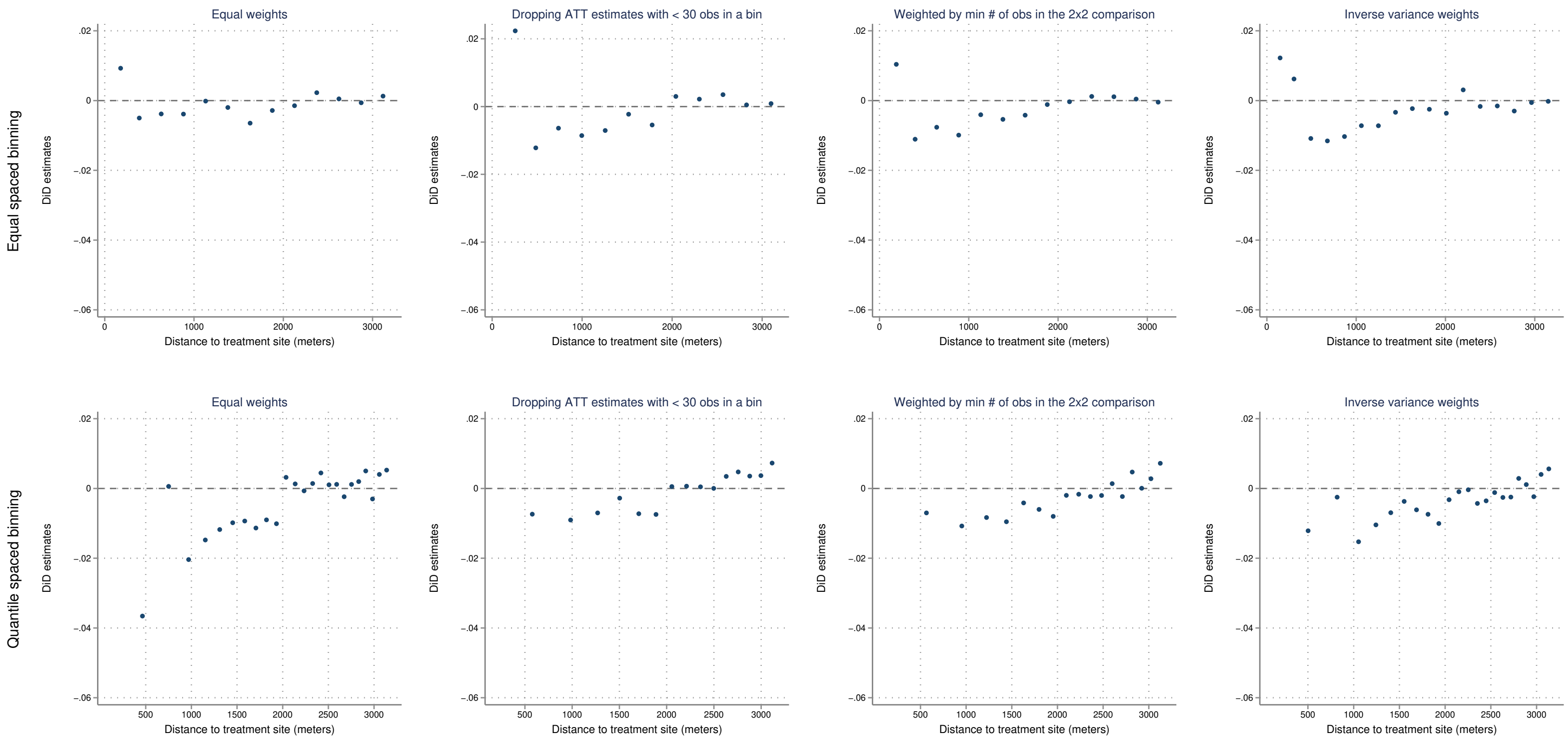
Note: This figure shows the ATT estimates as in Figure 7 under alternative binning and weighting choices. The top row show bin scatter plots using equally spaced bins and the bottom row shows plots using quantile spaced bins. The first column weights each ATT estimate equally. The second column also weights ATT estimates equally but drops ATT estimates with less than 30 observations in any of the 2×2 sample averages. The third column weights by the minimum number of housing sales across the 2×2 sample averages. The fourth column weights ATT estimates by their inverse variances.

Figure A2: LIHTC long run price effects under alternative binning and weighting choices



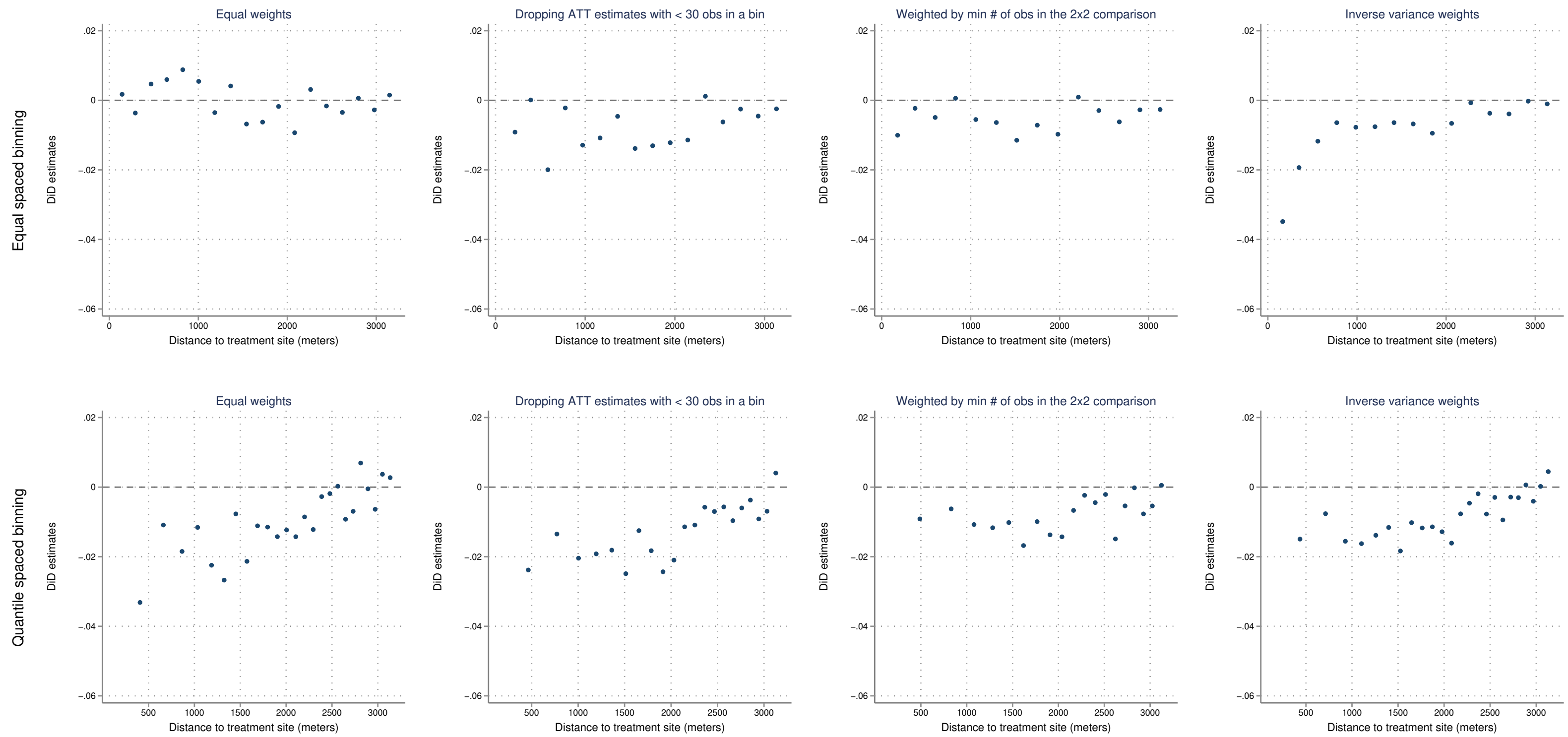
Note: This figure shows the ATT estimates as in Figure 7 under alternative binning and weighting choices. The top row show bin scatter plots using equally spaced bins and the bottom row shows plots using quantile spaced bins. The first column weights each ATT estimate equally. The second column also weights ATT estimates equally but drops ATT estimates with less than 30 observations in any of the 2×2 sample averages. The third column weights by the minimum number of housing sales across the 2×2 sample averages. The fourth column weights ATT estimates by their inverse variances.

Figure A3: LIHTC Q1 long run price effects under alternative binning and weighting choices



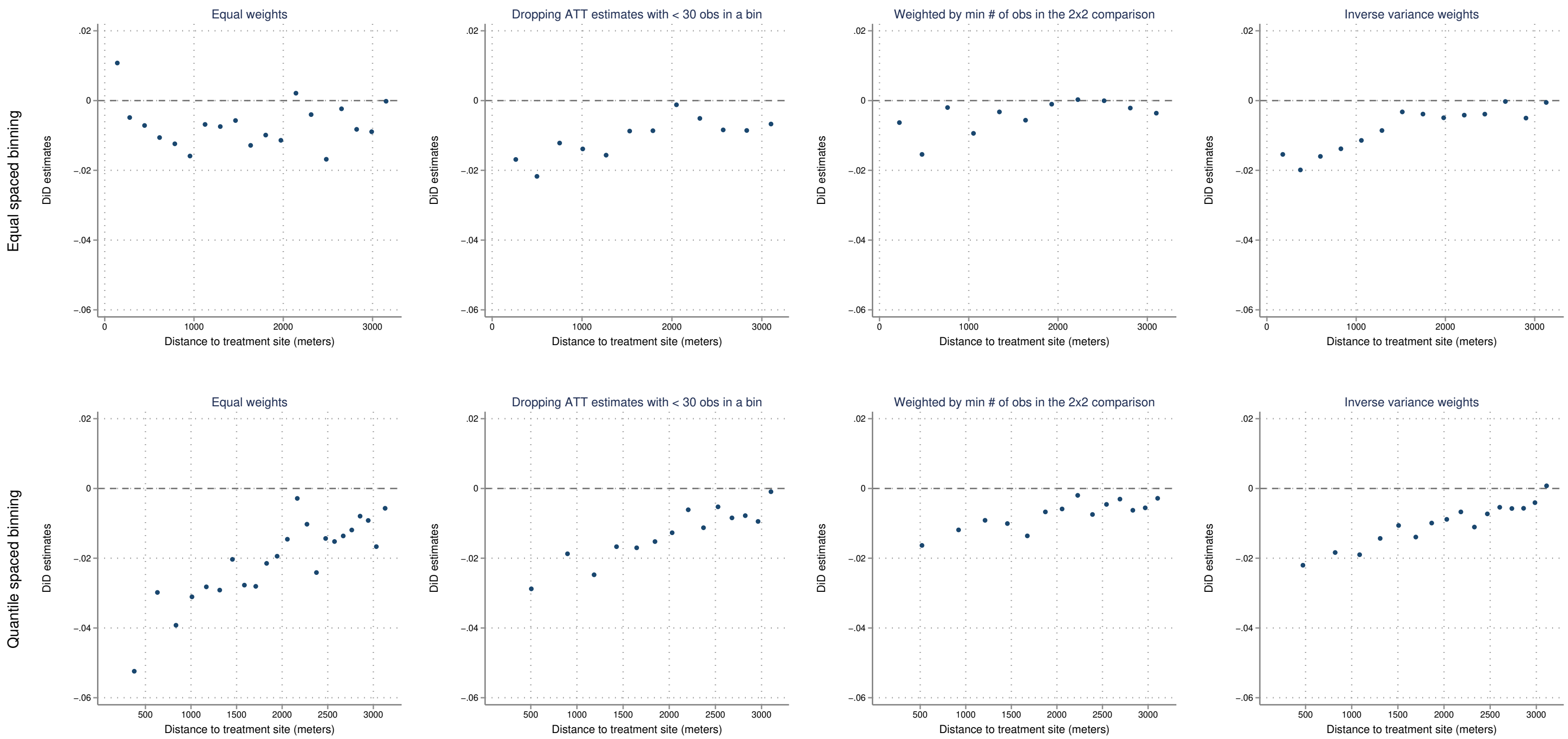
Note: This figure shows the ATT estimates as in Figure 7 under alternative binning and weighting choices. The top row show bin scatter plots using equally spaced bins and the bottom row shows plots using quantile spaced bins. The first column weights each ATT estimate equally. The second column also weights ATT estimates equally but drops ATT estimates with less than 30 observations in any of the 2×2 sample averages. The third column weights by the minimum number of housing sales across the 2×2 sample averages. The fourth column weights ATT estimates by their inverse variances.

Figure A4: LIHTC Q2 long run price effects under alternative binning and weighting choices



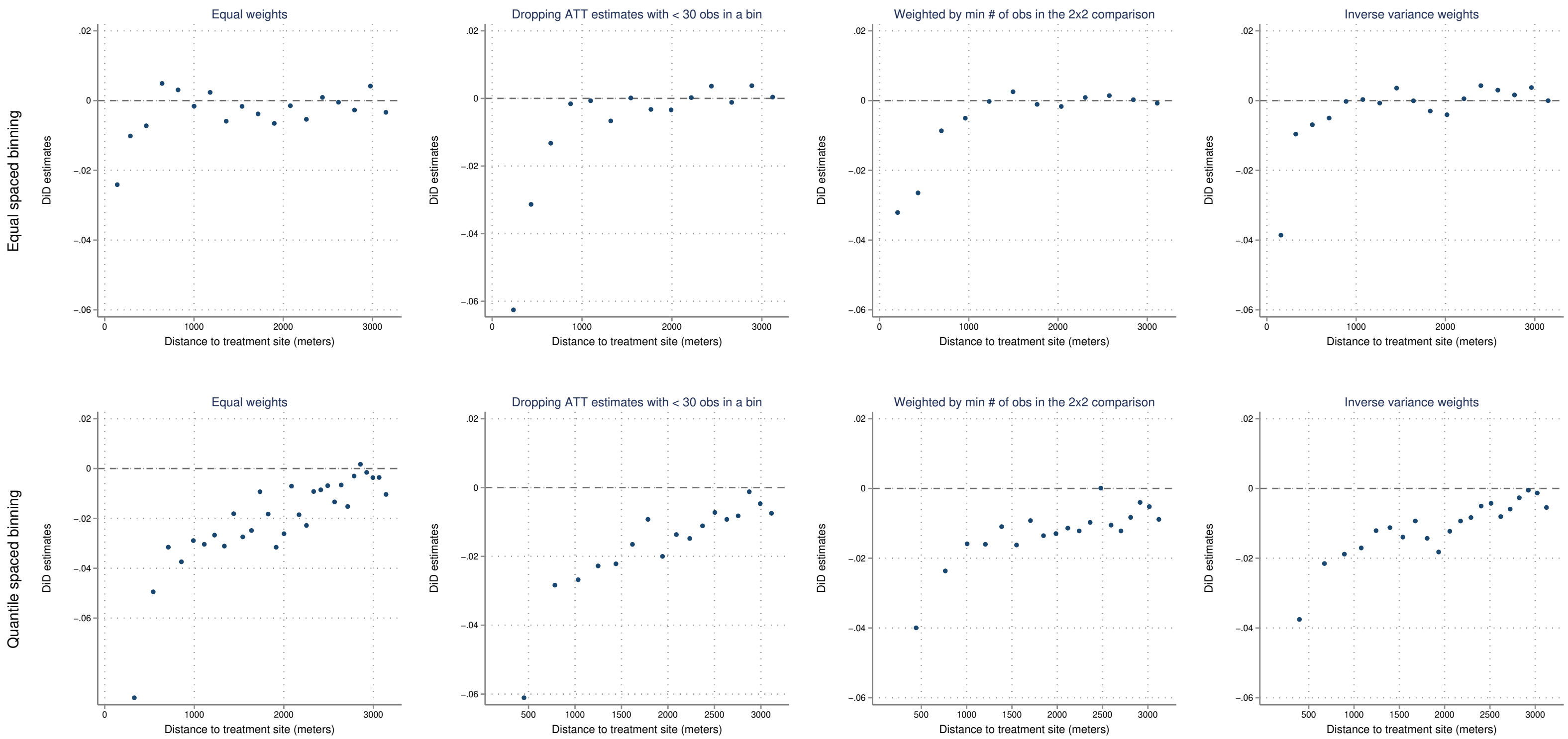
Note: This figure shows the ATT estimates as in Figure 7 under alternative binning and weighting choices. The top row show bin scatter plots using equally spaced bins and the bottom row shows plots using quantile spaced bins. The first column weights each ATT estimate equally. The second column also weights ATT estimates equally but drops ATT estimates with less than 30 observations in any of the 2×2 sample averages. The third column weights by the minimum number of housing sales across the 2×2 sample averages. The fourth column weights ATT estimates by their inverse variances.

Figure A5: LIHTC Q3 long run price effects under alternative binning and weighting choices



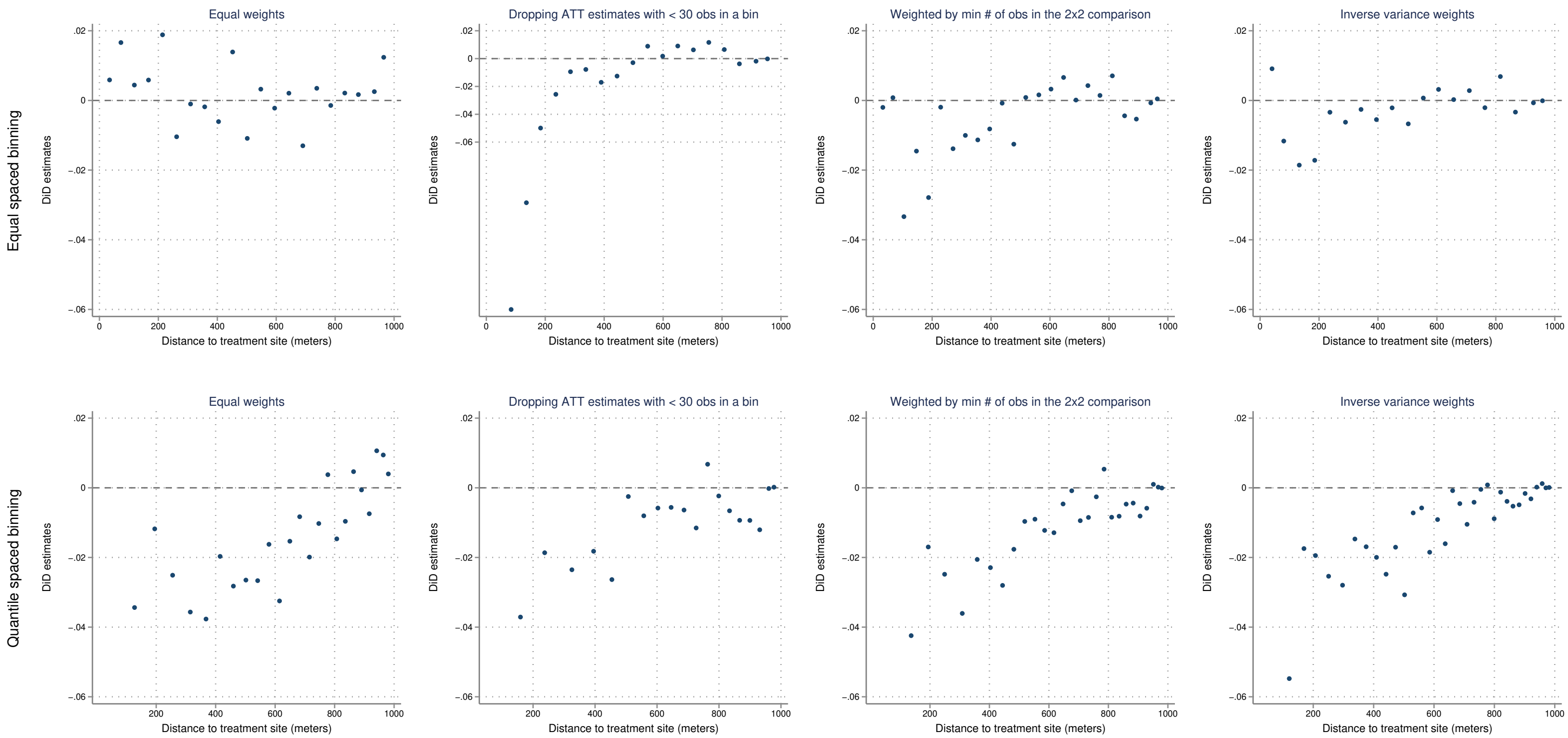
Note: This figure shows the ATT estimates as in Figure 7 under alternative binning and weighting choices. The top row show bin scatter plots using equally spaced bins and the bottom row shows plots using quantile spaced bins. The first column weights each ATT estimate equally. The second column also weights ATT estimates equally but drops ATT estimates with less than 30 observations in any of the 2×2 sample averages. The third column weights by the minimum number of housing sales across the 2×2 sample averages. The fourth column weights ATT estimates by their inverse variances.

Figure A6: LIHTC Q4 long run price effects under alternative binning and weighting choices



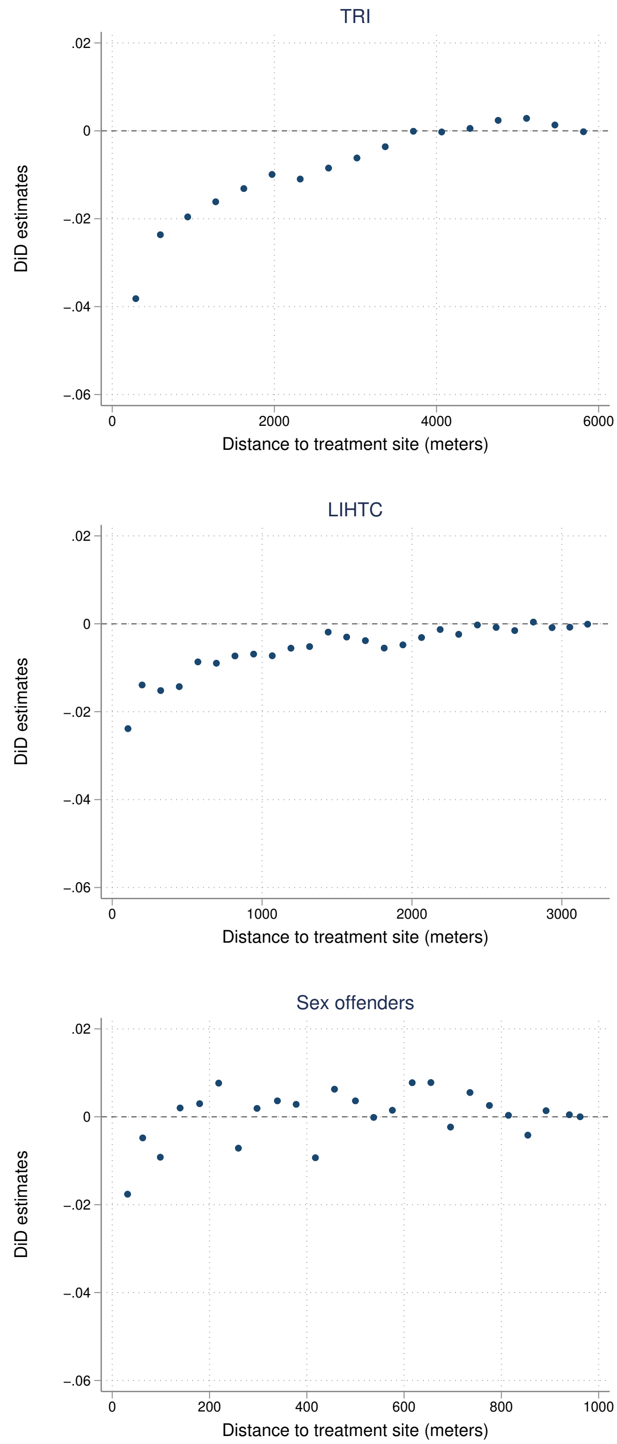
Note: This figure shows the ATT estimates as in Figure 7 under alternative binning and weighting choices. The top row show bin scatter plots using equally spaced bins and the bottom row shows plots using quantile spaced bins. The first column weights each ATT estimate equally. The second column also weights ATT estimates equally but drops ATT estimates with less than 30 observations in any of the 2×2 sample averages. The third column weights by the minimum number of housing sales across the 2×2 sample averages. The fourth column weights ATT estimates by their inverse variances.

Figure A7: Sex offender move ins long run price effects under alternative binning and weighting choices



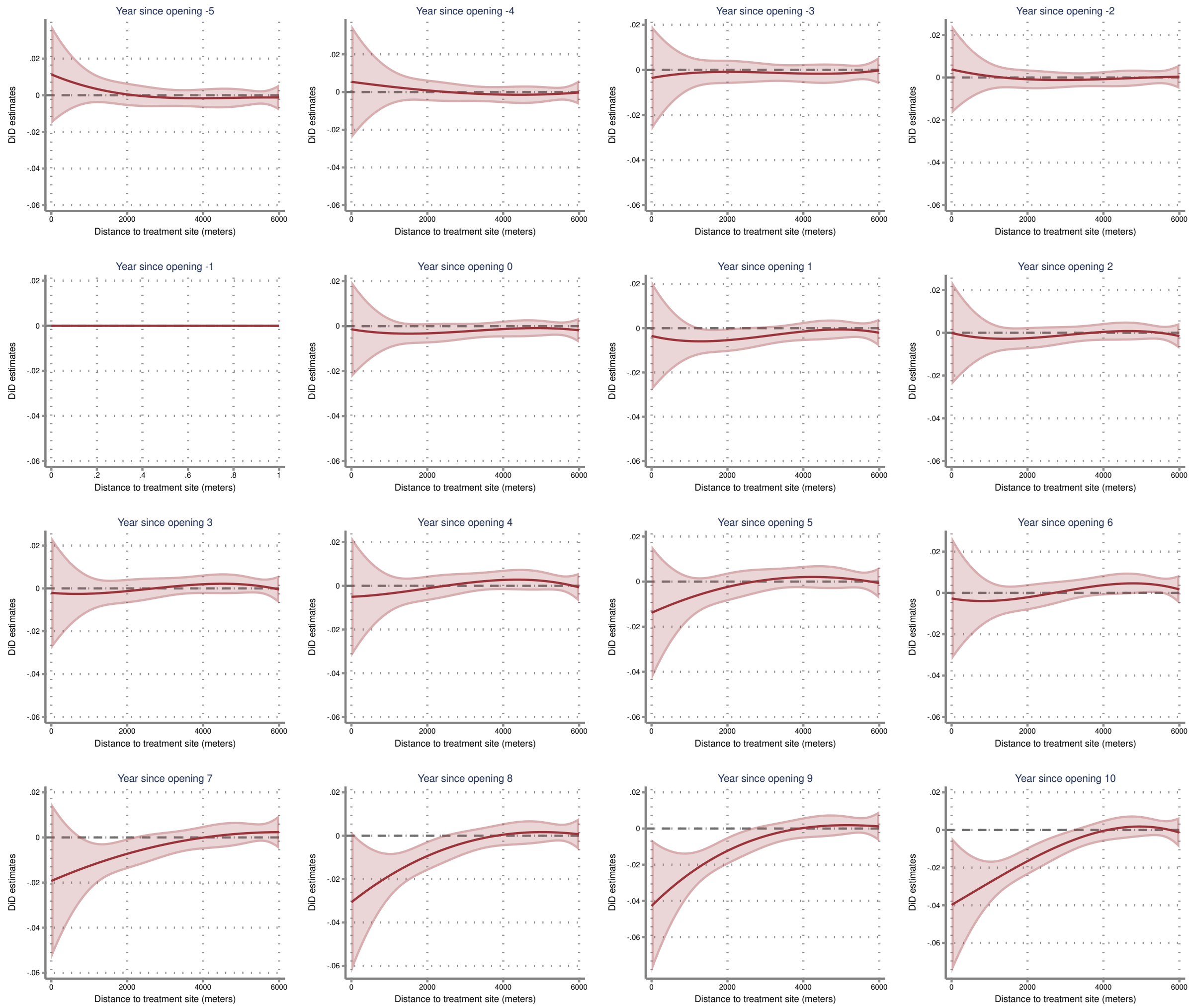
Note: This figure shows the ATT estimates as in Figure 7 under alternative binning and weighting choices. The top row show bin scatter plots using equally spaced bins and the bottom row shows plots using quantile spaced bins. The first column weights each ATT estimate equally. The second column also weights ATT estimates equally but drops ATT estimates with less than 30 observations in any of the 2×2 sample averages. The third column weights by the minimum number of housing sales across the 2×2 sample averages. The fourth column weights ATT estimates by their inverse variances.

Figure A8: Long run price effects residualizing treatment site× sale year fixed effects



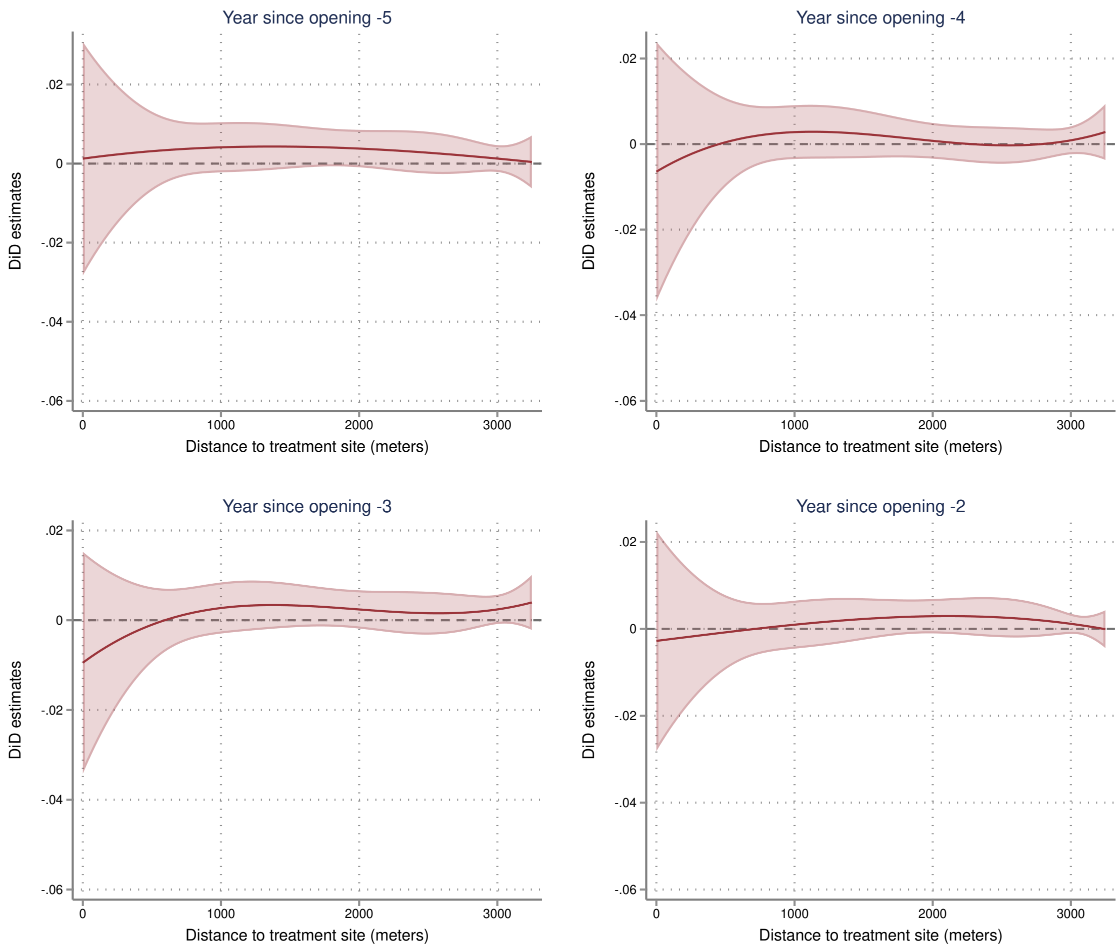
Note: This figure shows the ATT estimates as in Figure 7 estimating bin level average sale prices in the first stage with treatment site× sale year fixed effects following Cattaneo et al. (2024). As in Figure 7, the data is binned a second time for visualization and averages are weighted by the ATT estimates' inverse variances. The top row shows a binscatter plot of the estimates for TRI industrial plants. The middle row plots the estimates for LIHTC sites. The bottom plot shows estimates for sex offender move ins.

Figure A9: TRI, full continuous event study



Note: This figure shows the smoothed price effect estimates as in Figure 8 for all years. Specifically, for each year of the pre period, I form DiD comparisons at every distance bin relative to average house prices in the year before treatment. I then regress these comparisons on distance to the treatment site using the same nonparametric sieve regression procedure as in the long run price effect estimate.

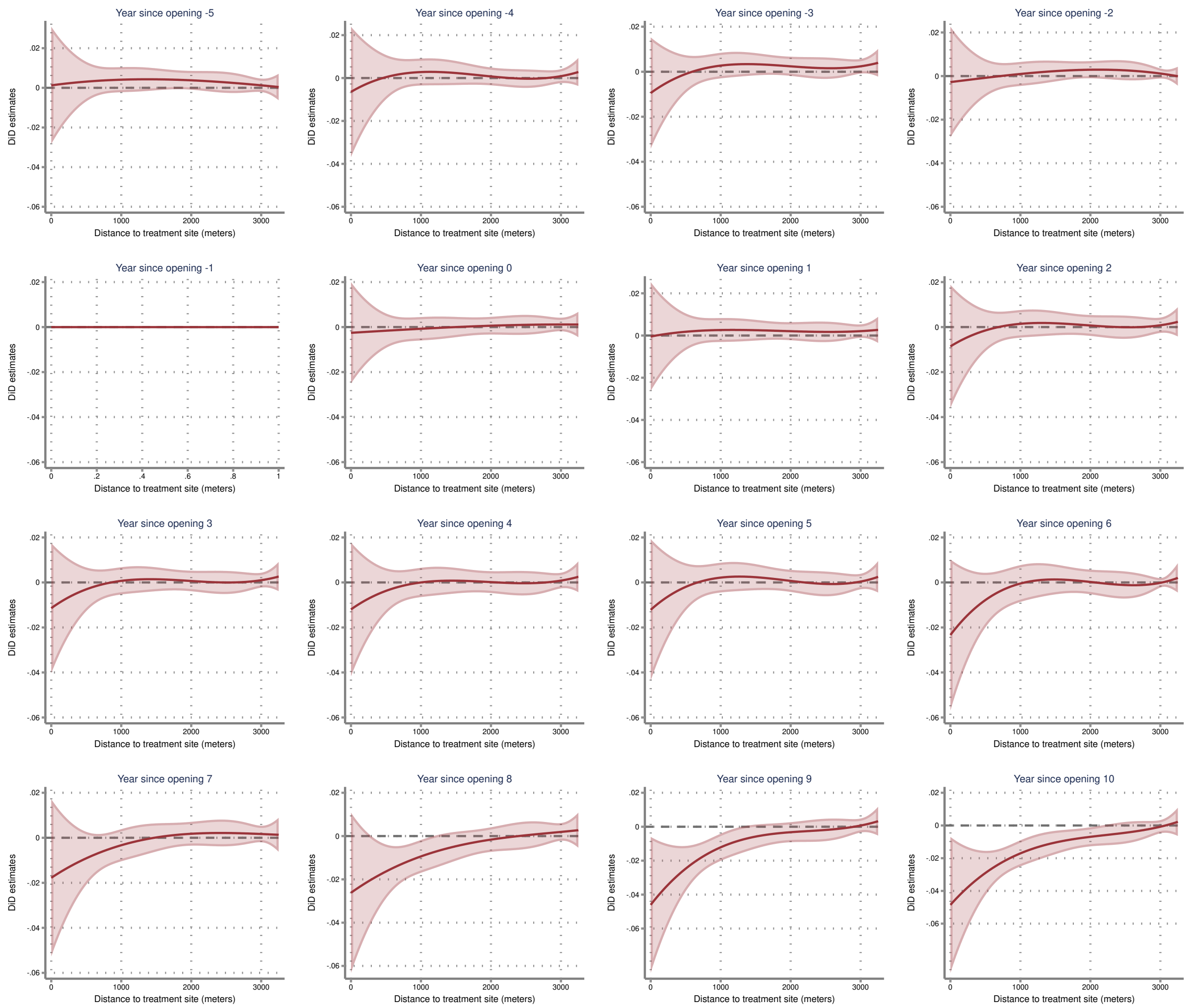
Figure A10: LIHTC developments, event study, all LIHTC sites



Note: This figure performs a similar analysis as in Figure 10 but using data prior to the treatment date. Specifically, for each year of the pre period, I form DiD comparisons at every distance bin relative to average house prices in the year before treatment. I then regress these comparisons on distance to the treatment site using the same nonparametric sieve regression procedure as in Figure 10. Appendix Figure A11 presents event study plots for all years in the post period.

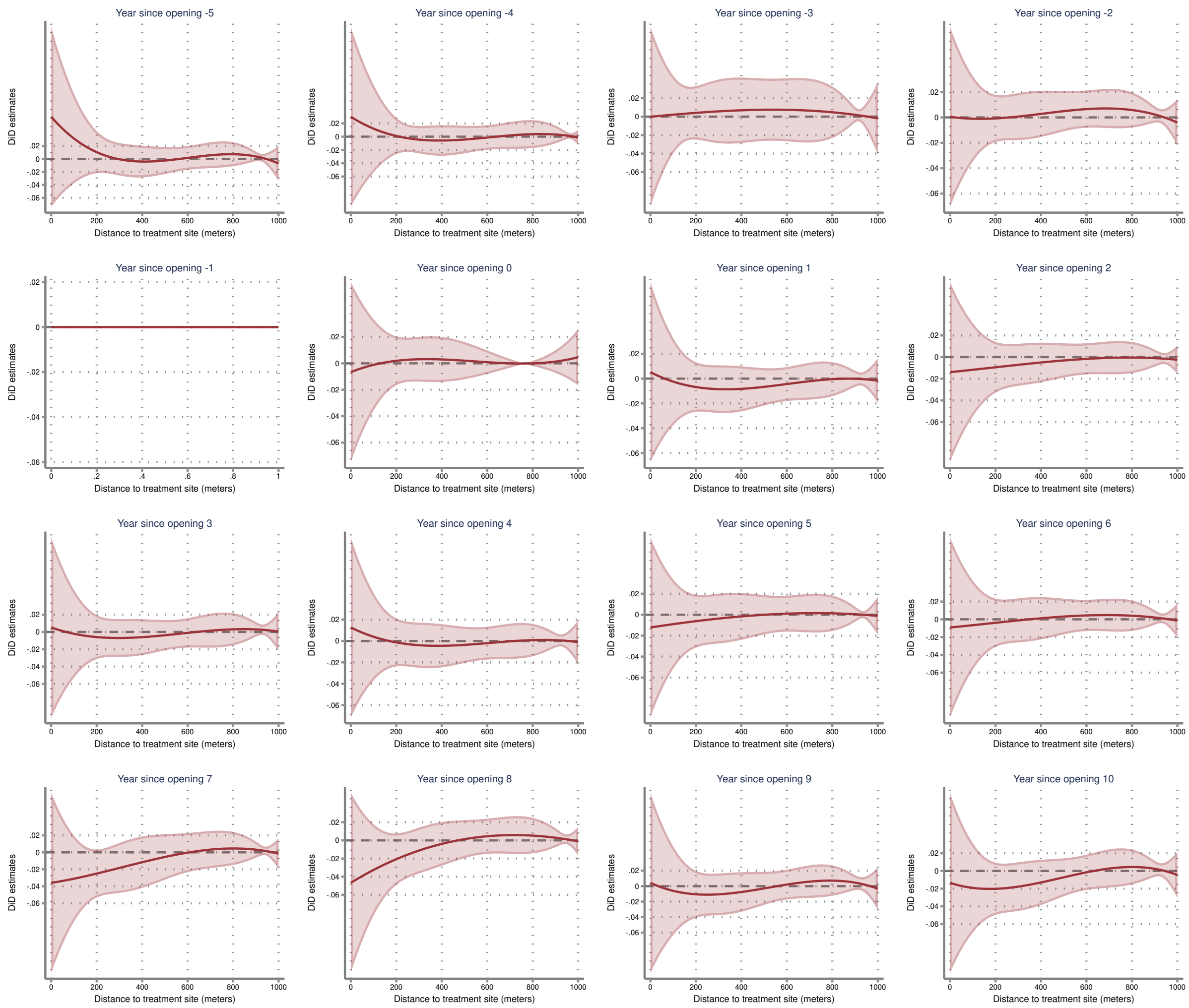
Figure A10 presents the continuous DiD event study for LIHTC developments analogous to the event study for TRI industrial plants (Figure 8). For each year of the pre period, I form DiD comparisons at every distance bin relative to average house prices in the year before treatment. I then regress these comparisons on distance to the treatment site using the same nonparametric sieve regression procedure as in Figure 10. I do not find strong visual evidence that would suggest parallel trends violations. In the top left graph, the relationship five years before the allocation of funding is essentially a flat line. Years -4 and -3 might suggest a modest increasing gradient close to the site. However, in year -2 the estimated relationship again looks like a flat line and, given the confidence bands, I can not rule out a null effect in all four pre period years. Appendix Figure A11 presents event study plots for all years in the post period.

Figure A11: LIHTC, full continuous event study



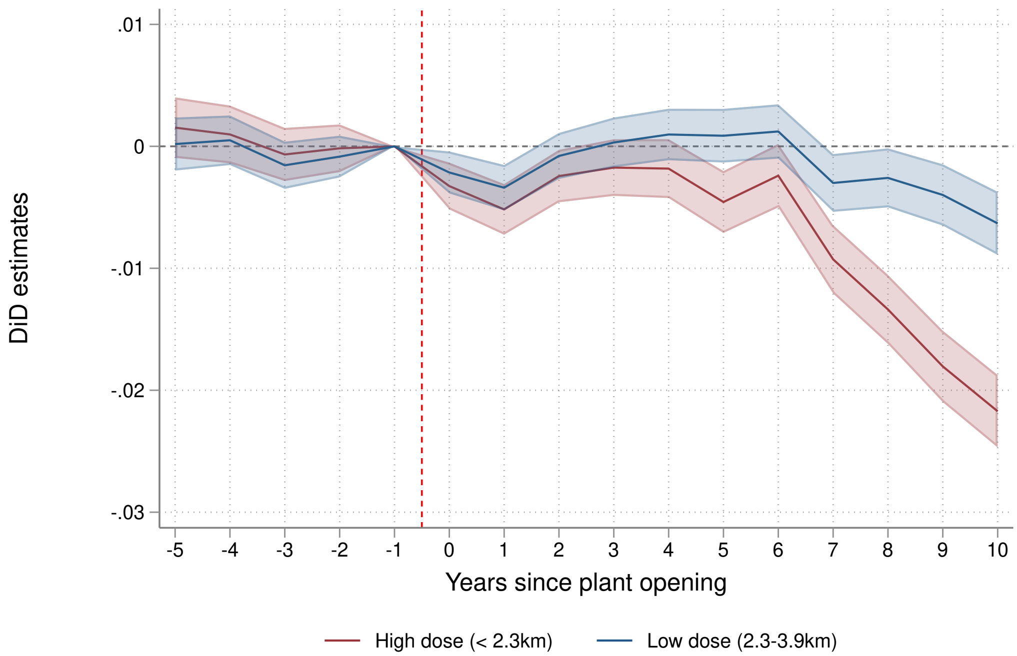
Note: This figure shows the smoothed price effect estimates as in Figure A10 for all years. Specifically, for each year of the pre period, I form DiD comparisons at every distance bin relative to average house prices in the year before treatment. I then regress these comparisons on distance to the treatment site using the same nonparametric sieve regression procedure as in the long run price effect estimate.

Figure A12: Sex offender move ins, full continuous event study



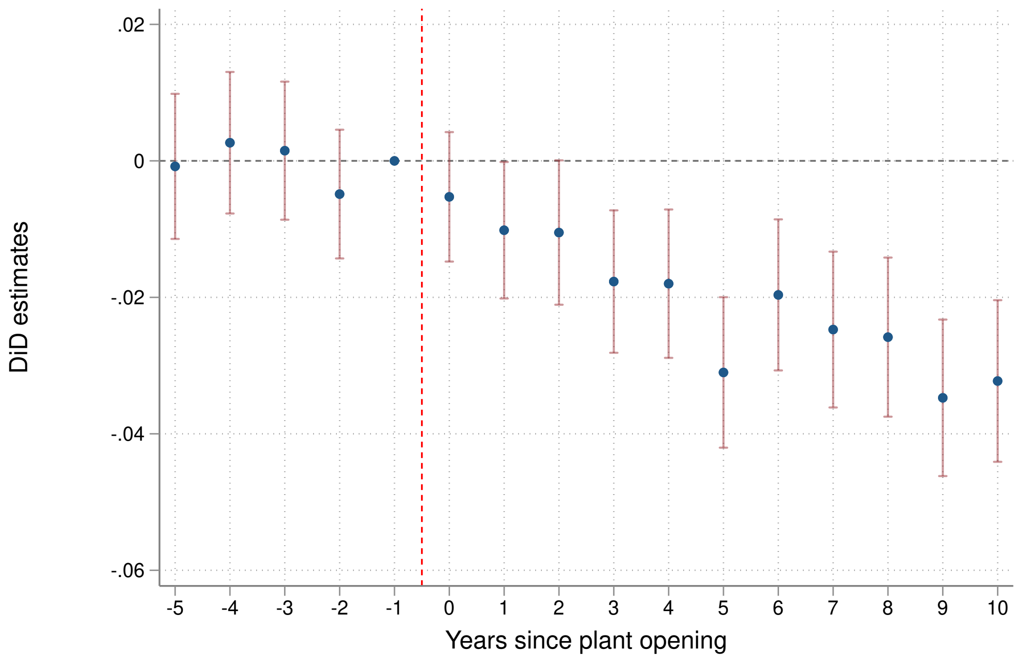
Note: This figure shows the smoothed price effect estimates as in Figure 12 for all years. Specifically, for each year of the pre period, I form DiD comparisons at every distance bin relative to average house prices in the year before treatment. I then regress these comparisons on distance to the treatment site using the same nonparametric sieve regression procedure as in the long run price effect estimate.

Figure A13: Alternative event study (Callaway et al., 2024), TRI



Note: This figure shows an event study for TRI industrial plants pooling together ATT estimates into two distance groups following Callaway et al. (2024). I split the estimates into high and low treatment dose groups at 2,307 meters, the median bin distance from a plant within the treated area. The solid red line is the estimate for "high dose" areas (<2.3km), while the blue line is the estimate among "low dose" areas (2.3-3.9km). Each point is a weighted average of the ATT estimates within the dose group. Shaded areas are 95% pointwise confidence intervals.

Figure A14: Traditional event study, TRI



Note: This figure shows the results of a traditional event study. Each point is the coefficient from a stacked DiD regression that compares price trends in each year for distance bins within 1 kilometer of a plant vs. 4–5 kilometers away from a plant. Standard errors are clustered by plant site. The traditional event study is estimated using the 4,715 plant sites with a balanced panel in the inner and outer rings in each year.

Figure A14 is based on the following regression model:

$$\log(saleprice)_{rgt} = \alpha_{gt} + \rho_{gr} + \sum_{k=-5,\, k\neq-1}^{10} \beta_k \times 1(t - t^* = k, r = 1) + \epsilon_{rgt}$$

where $\log(saleprice)_{rgt}$ is the average log sale price of houses in ring r near plant g in year t, α_{gt}, ρ_{gr} are treatment site $\times$ sale year and treatment site $\times$ ring fixed effects, and $r = 1$ if the average price is in the inner ring. β_k are the parameters of interest plotted in Figure A14.

Figures A15, A16, and A17 plot every component that goes into the aggregate local price and welfare change calculations for each type of spatial treatment. In each figure:

- Panel (a) shows the main price effect estimate comparing 6-10 years after treatment to -5 to -1 years before. The red line is the fit line from a nonparametric sieve regression of DiD bin comparisons on distance to the treatment site following Chen et al. (2024) weighting by the ATT estimates' inverse variances. The shaded red areas are the 95% uniform confidence bands. Blue dots are bin scatter plots of the comparisons following Cattaneo et al. (2024) (binning the data a second time). Confidence intervals on the binned scatter plot are calculated with standard errors clustered by treatment site.

- Panel (b) shows the derivative estimate with 95% uniform confidence bands following Chen et al. (2024), or the derivative of the red line in panel (a).

- Panel (c) estimates the relationship between the average house price and distance to the treatment site in the post period by regressing average house prices in the post period on distance to the treatment site and treatment site location specific fixed effects. To control for the fixed effects, I follow the procedure in Chen et al. (2024)'s application. Specifically, I regress housing prices on distance to the treatment site and treatment site location specific fixed effects under an initial basis choice. I then apply Chen et al. (2024) as before but using house prices minus the estimated treatment site location fixed effects from the initial regression as the outcome variable. As an additional check, the blue scatter plot also shows the estimates from `binsreg` with covariate adjustment done following Cattaneo et al. (2024).

- Panel (d) is the PDF of owner occupied housing units on distance to the treatment site within the spatial extent of treatment effects using standard kernel density estimation.

Figure A15: TRI, components of aggregate WTP calculation

(a) Price effect estimate $\widehat{\widetilde{ATT}}_{agg}(d|d)$

(b) Derivative estimate $\frac{\partial \widehat{\widetilde{ATT}}_{agg}}{\partial d}(d|d)$

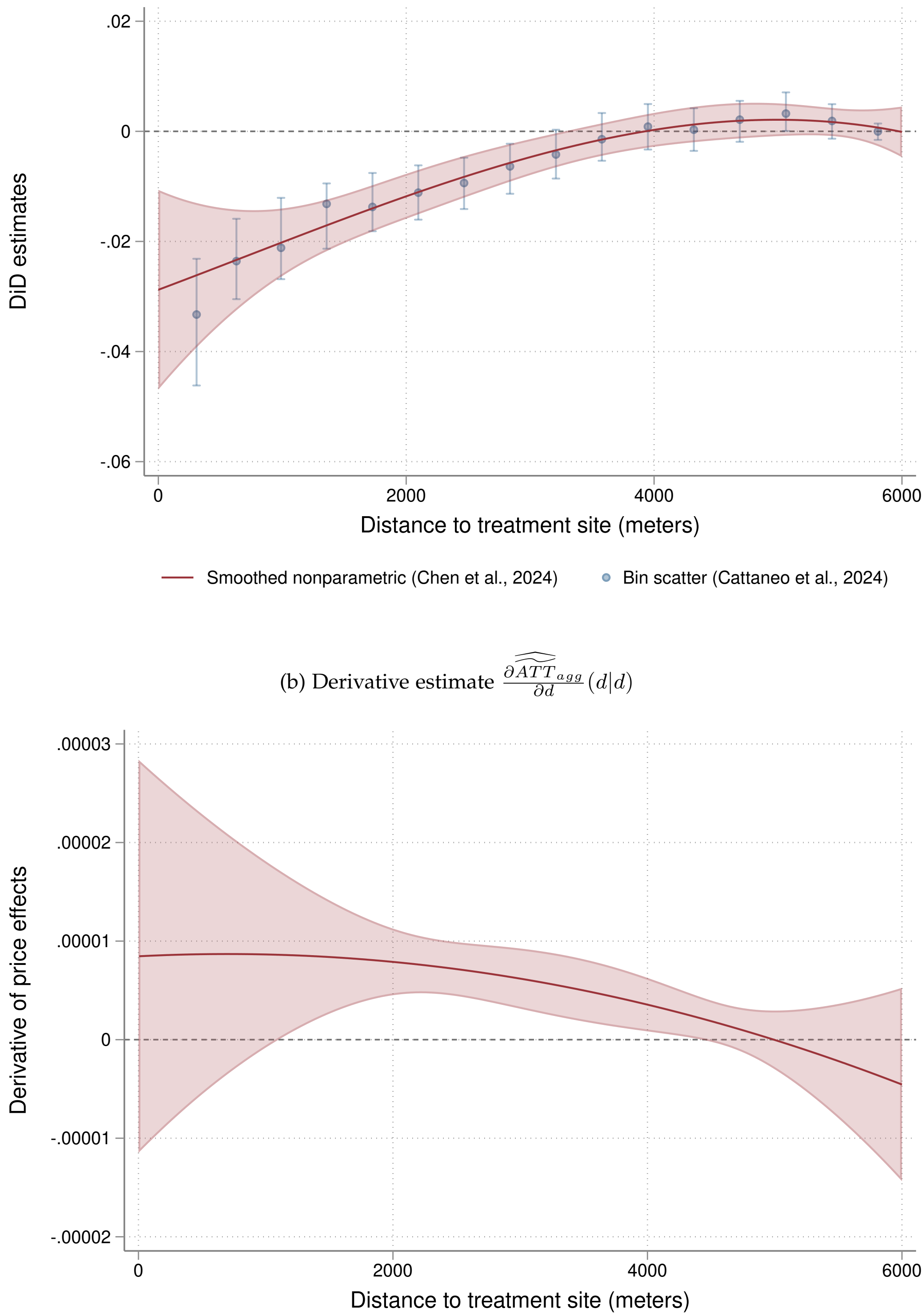
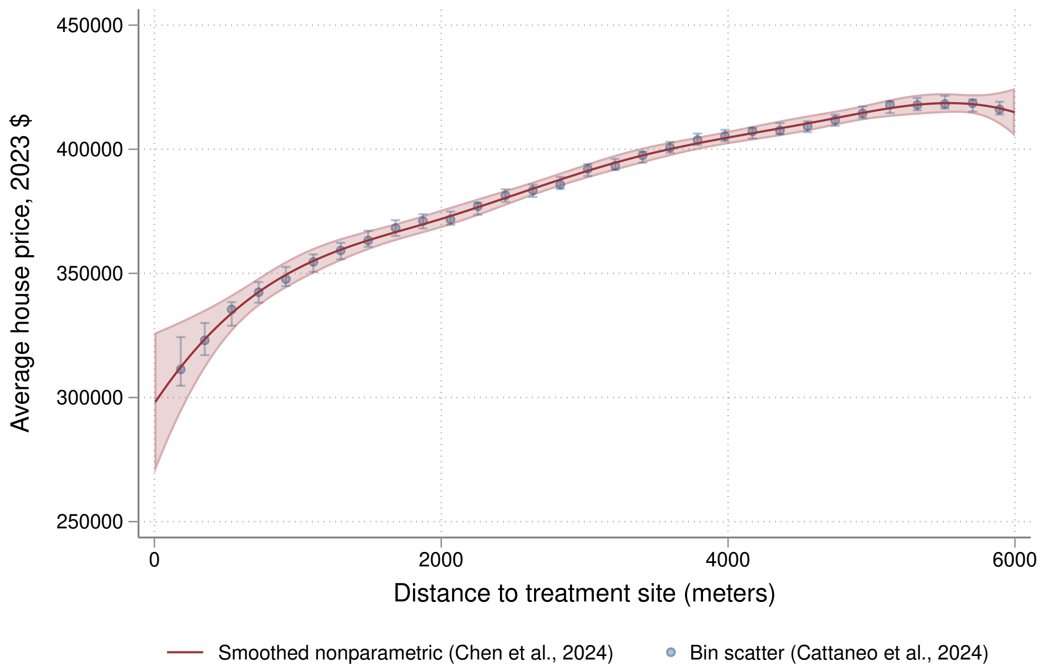


Figure A15: continued

(c) Average house price level $\widehat{\mathrm{E}[Y|D=d]}$



(d) Owner occupied housing unit density $\hat{f}(d)$

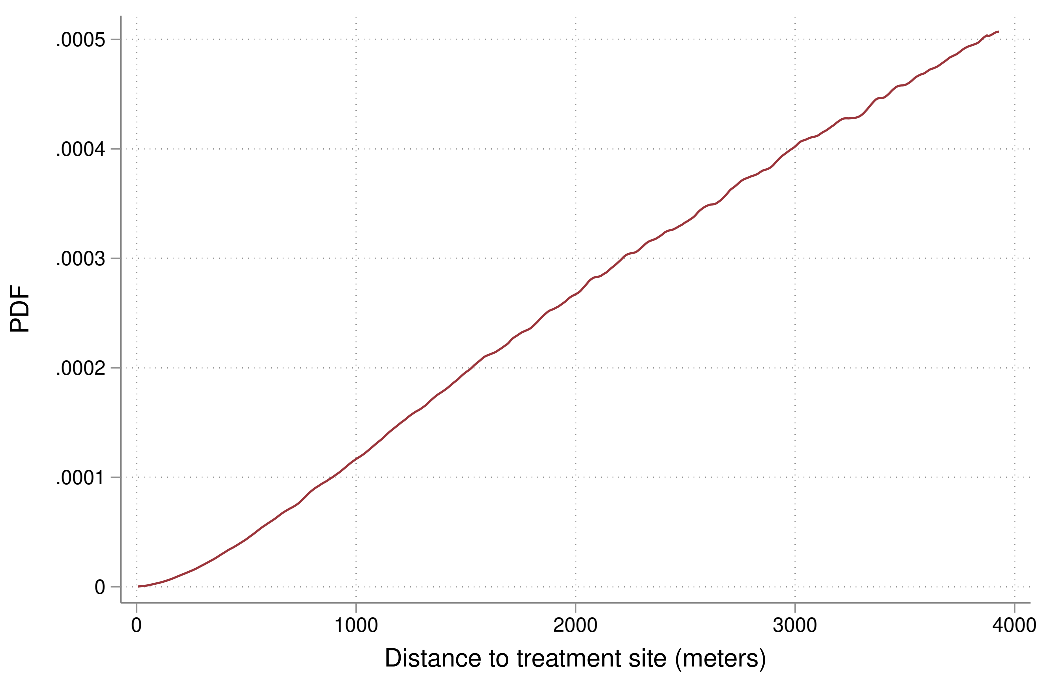
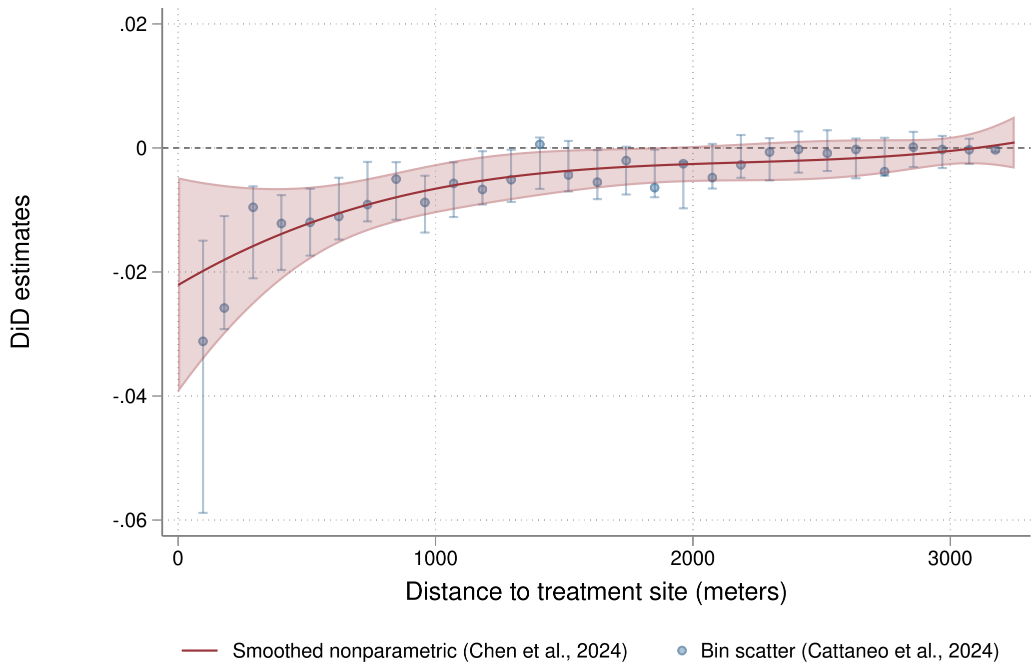


Figure A16: LIHTC, components of aggregate WTP calculation

(a) Price effect estimate $\widehat{\widetilde{ATT}}_{agg}(d|d)$



(b) Derivative estimate $\frac{\partial \widehat{\widetilde{ATT}}_{agg}}{\partial d}(d|d)$

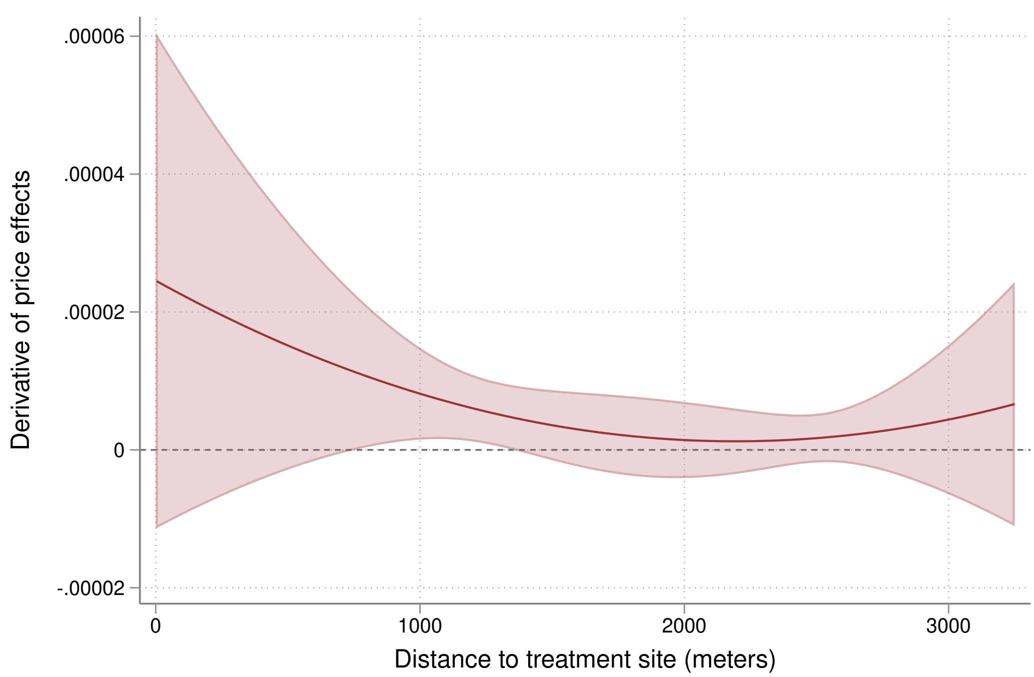
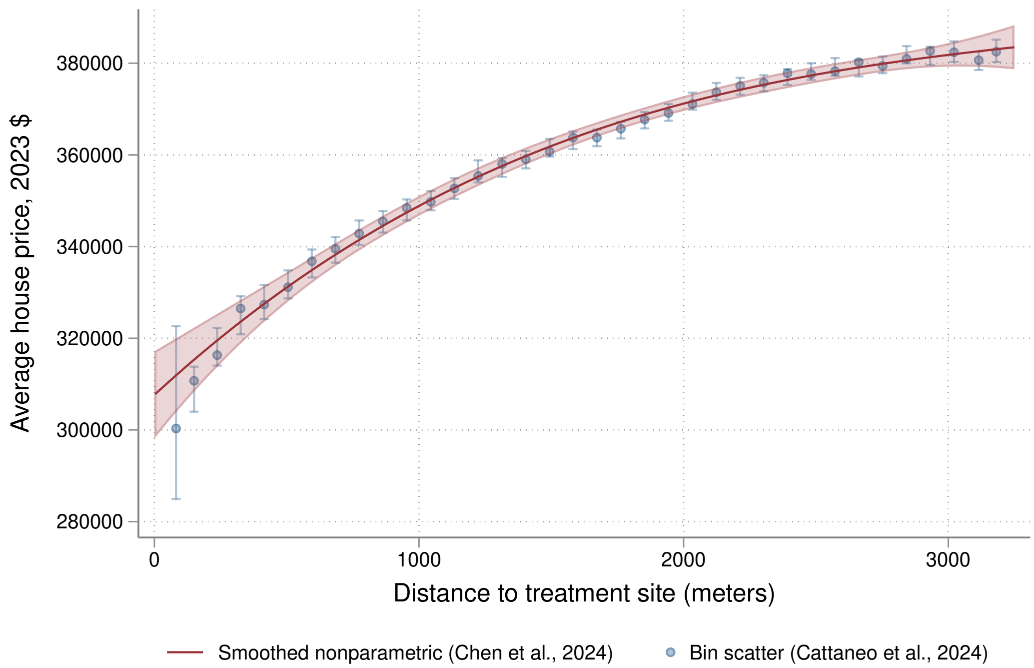


Figure A16: continued

(c) Average house price level $\widehat{E[Y|D=d]}$



(d) Owner occupied housing unit density $\hat{f}(d)$

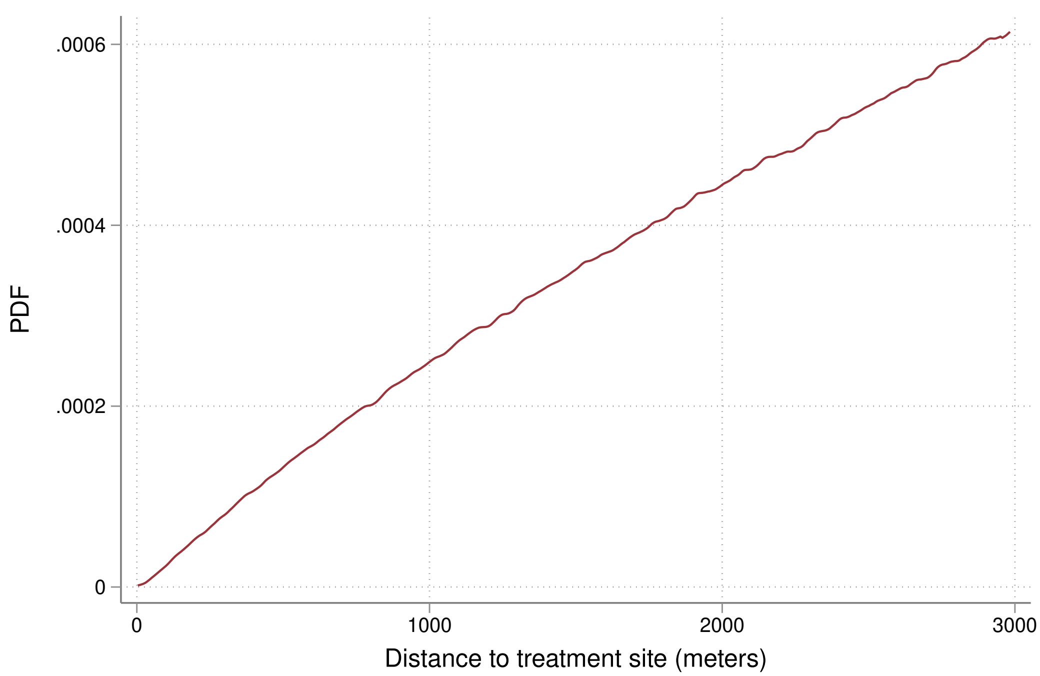
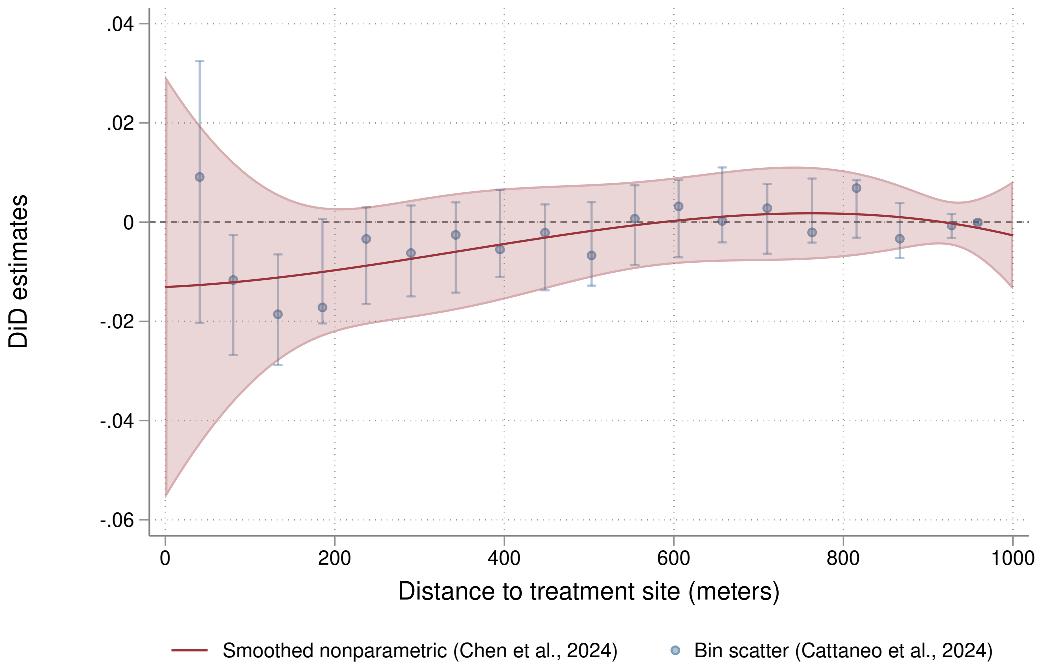


Figure A17: Sex Offenders, components of aggregate WTP calculation

(a) Price effect estimate $\widehat{\widetilde{ATT}}_{agg}(d|d)$



(b) Derivative estimate $\frac{\partial \widehat{\widetilde{ATT}}_{agg}}{\partial d}(d|d)$

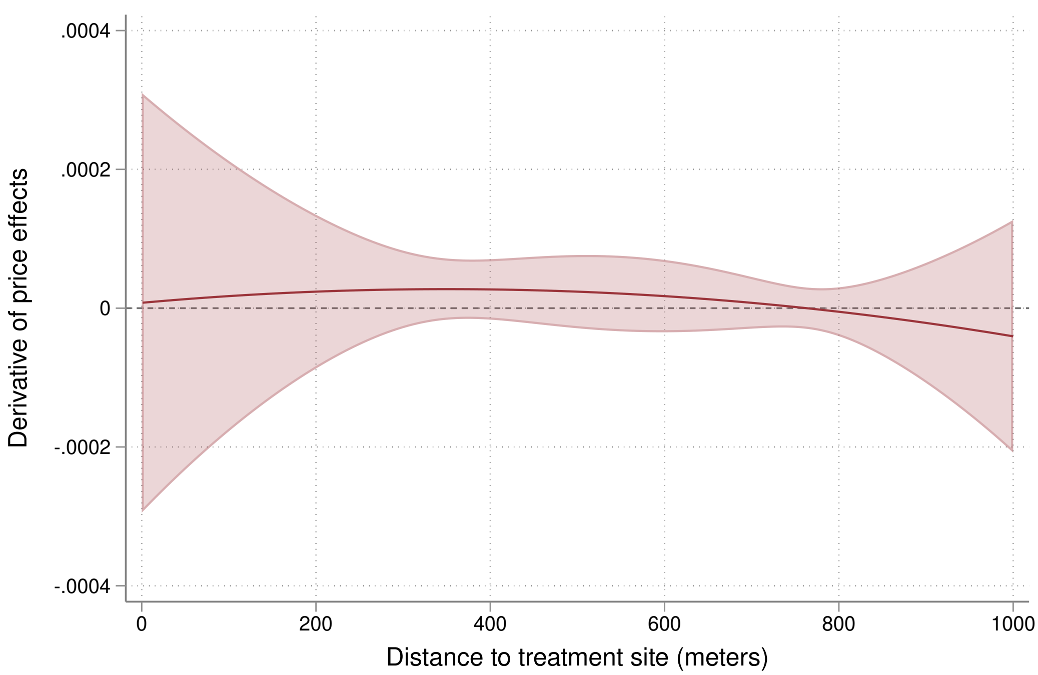
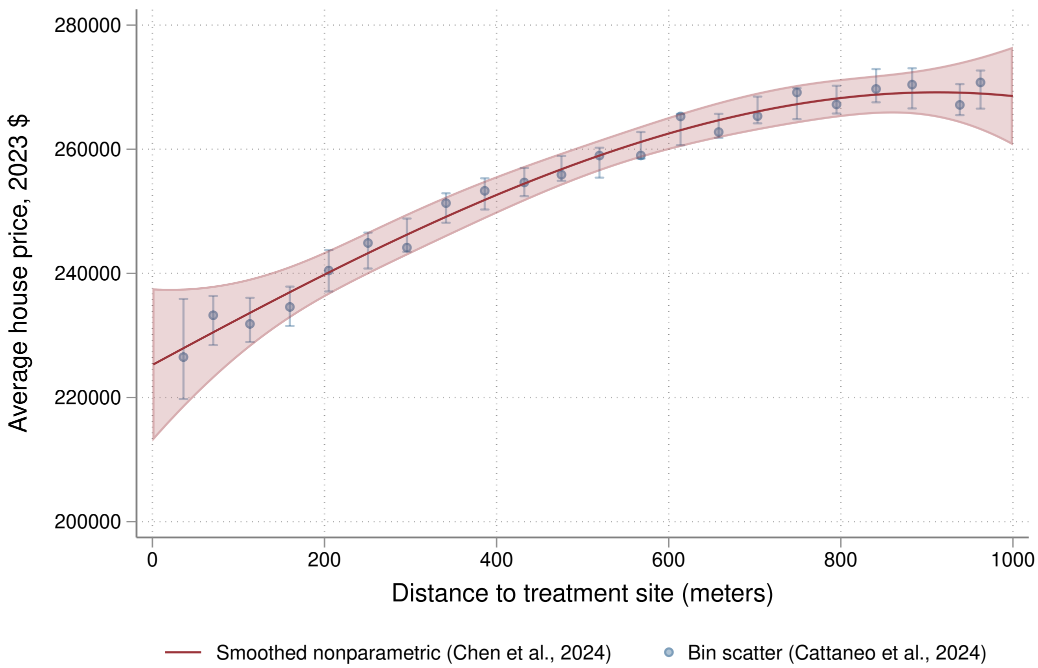
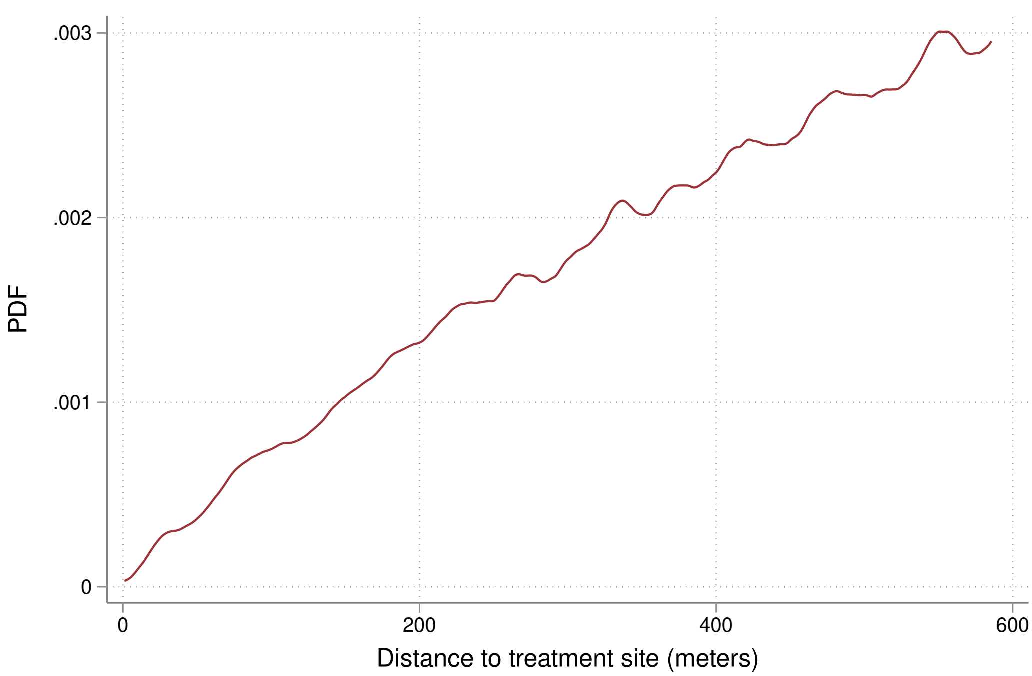


Figure A17: continued

(c) Average house price level $\widehat{\mathrm{E}[Y|D=d]}$



(d) Owner occupied housing unit density $\hat{f}(d)$



Appendix tables

A. Summary statistics

Tables A1 and A2 presents summary statistics on site locations, housing sales, and housing unit counts for LIHTC developments and sex offenders respectively.

Panel A of Table 2 reports that there are 17,895 LIHTC sites included in my estimation sample in which the typical LIHTC development was allocated funding in 2005. Panel B shows average housing prices by distance to the treated site in five equally spaced distance ranges extending out to 3,250 meters. As with TRI industrial plant locations, LIHTC developments tend to be sited in neighborhoods with lower house values compared to areas farther away from the development. The average house price is $328,051 within 650 meters of a development and increases to $381,402 for houses 2,600 to 3,250 meters away. Panel C shows the number of owner occupied housing units where, as before, there are far more housing units farther away than close in to the treated sites. There are approximately 8 million owner occupied housing units within 650 meters, while there are 57 million units 2,600 to 3,250 meters away. Compared to TRI industrial plant sites, LIHTC sites have more housing units nearby, reflecting that LIHTC developments tend to be located in more urban areas.

Table A1: Summary statistics on LIHTC developments

Panel A. Treatment sites					
Number of sites	17,895				
Average year of treatment event	2005				
	Distance to treatment site				
Panel B. Housing sales	0 to 650m	650 to 1,300m	1,300 to 1,950m	1,950 to 2,600m	2,600 to 3,250m
Average house price (2023 $)	$328,051	$342,294	$352,419	$365,661	$381,402
Number of site × house sales	4,980,723	14,428,167	22,028,002	28,499,351	34,414,231
Number of site × distance bins	49,426	67,803	68,782	68,978	69,018
Panel C. Housing unit counts					
Number of owner occupied housing units	8,077,607	22,855,868	35,401,470	46,512,737	57,218,527

Note: This table presents summary statistics on treatment sites used in the sample and the average sale prices and owner occupied housing unit counts of nearby houses by distance to the treatment site. Panel A describes 17,895 LIHTC developments from 1995-2015. Panel B shows the average house price in real 2023 dollars by distance to the treatment site. Panel C tabulates totals of owner occupied housing unit counts in nearby census blocks by distance to treatment site.

Table 3 presents the same information for sex offender move ins for North Carolina and Florida. In total, I observe 9,193 sex offender moves from 1995 to 2015 in these two states with an average move year of 2009. This average is closer to the end of the time range because I only observe the current registry of sex offenders and not a complete history of moves. Panel B shows average house prices in five equally spaced distance ranges within 1,000 meters of a sex offender's address, consistent with a smaller spatial extent as considered in Linden and Rockoff (2008) and Pope (2008). As with the other two types of treatment events, sex offenders live in areas with systematically lower house values. The average house price is $233,907 within 200 meters compared to $305,425 for houses 800 to 1,000 meters away. Panel C tabulates housing unit counts by distance to the sex offenders' addresses. In my sample areas, there are approximately 300,000 owner occupied housing units within 200 meters, while there is roughly 1.8 million units 800 to 1,000 meters away. Given the smaller spatial extent considered, for each sex offender who moves into a neighborhood, any estimated price effects will affect far fewer homes than a TRI industrial plant or a LIHTC development.

Table A2: Summary statistics on sex offender move ins

Panel A. Treatment sites					
Number of sites	9,193				
Average year of treatment event	2009				
	Distance to treatment site				
Panel B. Housing sales	0 to 200m	200 to 400m	400 to 600m	600 to 800m	800 to 1,000m
Average house price (2023 $)	$233,907	$258,563	$285,061	$296,376	$305,425
Number of site × house sales	377,047	951,218	1,377,936	1,764,400	2,103,548
Number of site × distance bins	13,059	18,666	19,830	21,354	23,334
Panel C. Housing unit counts					
Number of owner occupied housing units	300,567	762,615	1,134,528	1,478,115	1,793,093

Note: This table presents summary statistics on treatment sites used in the sample and the average sale prices and owner occupied housing unit counts of nearby houses by distance to the treatment site. Panel A describes 9,193 sex offender move ins from 1995-2015. Panel B shows the average house price in real 2023 dollars by distance to the treatment site. Panel C tabulates totals of owner occupied housing unit counts in nearby census blocks by distance to treatment site.

Table A3: Robustness check for sensitivity to choice of constant in utility function

	Value of k	$\frac{\widetilde{ATT}}{d-d_*^0}$	$\frac{\partial \widetilde{ATT}}{\partial d}$
Panel A. TRI	0	\$13.085 mil.	\$16.409 mil.
	.1	\$13.085 mil.	\$16.409 mil.
	1	\$13.087 mil.	\$16.412 mil.
	10	\$13.108 mil.	\$16.438 mil.
	100	\$13.309 mil.	\$16.692 mil.
Panel B. LIHTC	0	\$6.694 mil.	\$9.142 mil.
	.1	\$6.694 mil.	\$9.143 mil.
	1	\$6.696 mil.	\$9.147 mil.
	10	\$6.714 mil.	\$9.193 mil.
	100	\$6.884 mil.	\$9.619 mil.

Note: This table presents estimates of the aggregate WTP varying the constant used within the log transformation in the utility function, i.e. varying k in $\beta_i \log(k+d)$. The first column shows the constant used. The second column shows the welfare estimate using the rise over run estimator as the estimate for homeowner MWTP. The third column shows the welfare estimate using the ATT derivative as homeowner MWTP. For specifications with $k = 0$, the equation integrates to the next closest evaluation point to 0, which is 6 and 3 meters for TRI and LIHTC sites, respectively.

B Econometric proofs

This appendix provides proofs for the theorems on identification in Section 2 of the main text. These proofs are an adaptation of the proofs from the appendixes of Callaway et al. (2025) to fit the spatial application. I then include additional proofs for identification under a linear correlated random coefficients model and a lower bound under ATT concavity.

I begin in the same setup as Section 2. I denote by period $t = 1$ the observation prior to the plant opening and $t = 2$ the period afterwards. In the post period, an industrial plant opens at a specific location and spews toxic chemicals into the air. Housing units receive a dose of exposure to the plant indexed by their distance from the toxic plant $D = d$ (exposure decreases with more distance from the plant). Let $\mathcal{D}$ denote the continuous support of D.

Define $Y_{j,t}(d)$ as the potential outcome for house j in time period t. This is the house price (or log transformed house price) if house j was d distance away from a treatment site in time period t. The realized outcome is $Y_{j,t} = Y_{j,t}(D_j)$. To define a control group we must assume the plant has no effect on house prices beyond some distance.

Assumption 1 (Limit on the spatial extent of treatment effects). *Assume there is a known constant $\overline{d} \in (0, \infty)$ that bounds the distance that a treatment site affects house prices, so units farther than $\overline{d}$ from a plant are untreated in every time period. That is, for any $d \geq \overline{d}$, $Y_{j,t}(d) = Y_{j,t}(\overline{d})$.*

Lastly, denote $\Delta Y_j = Y_{j,2} - Y_{j,1}$ as the change over time for a housing unit.

In the DiD framework, we consider averages of individual effects. In particular,

$$ATT(d|d') = \mathrm{E}[Y_t(d) - Y_t(\overline{d})|D = d']$$

is the average effect of being d distance from the plant in time period t for units that are located d' from the plant. Similarly, define the average causal response parameter as

$$ACRT(d|d') = \frac{\partial ATT(l|d')}{\partial l}|_{l=d} = \frac{\partial \mathrm{E}[Y_t(l)|D = d']}{\partial l}|_{l=d}$$

Next, I repeat the assumptions from Section 2.

Assumption 2 (Random sampling for repeated cross sections). *Conditional on $T = t$, the data are independent and identically distributed from the distribution of (Y_t, D, T, X), for $t = 1$ and $t = 2$ with (D, X) invariant to T.*

Assumption 3 (Support and continuously differentiable treatment). *(a) The support of D is $\mathcal{D} = [0, d_U]$ with $0 < \overline{d} < d_U$, where $d_U < \infty$ is an upper bound on the spatial extent considered. Furthermore, $\mathrm{P}(D > \overline{d}) > 0$ and $dF_D(d) > 0$ for all $d \in [0, \overline{d}]$ where $dF_D(d)$ is the density of housing units by distance. (b) Assume $\mathrm{E}[\Delta Y|D = d]$ is continuously differentiable in d on $[0, \overline{d}]$.*

Assumption 4 (No anticipation and observed outcomes). *For all units, $Y_{j,t=1} = Y_{j,t=1}(\overline{d})$ and $Y_{j,t=2} = Y_{j,t=2}(D_j)$.*

Assumption 5 (Parallel trends). *For all $d \in \mathcal{D}$,*

$$\mathrm{E}[Y_{t=2}(\overline{d}) - Y_{t=1}(\overline{d})|D = d] = \mathrm{E}[Y_{t=2}(\overline{d}) - Y_{t=1}(\overline{d})|D = \overline{d}].$$

Under this setup and Assumptions 1–5, ATT is identified.

Theorem 2.1 *Under Assumptions 1 to 5, $ATT(d|d)$ is identified.*

Proof: This result and its proof are essentially the same as in the standard binary DiD setting but using the parallel trends Assumption 5 above. A similar proof is in Appendix A of Callaway et al. (2025).

$$\begin{aligned} ATT(d|d) &= \mathrm{E}[Y_2(d) - Y_2(\overline{d})|D = d] \\ &= \mathrm{E}[Y_2(d) - Y_1(\overline{d})|D = d] - \mathrm{E}[Y_2(\overline{d}) - Y_1(\overline{d})|D = d] \\ &= \mathrm{E}[Y_2(d) - Y_1(\overline{d})|D = d] - \mathrm{E}[Y_2(\overline{d}) - Y_1(\overline{d})|D = \overline{d}] \\ &= \mathrm{E}[Y_2 - Y_1|D = d] - \mathrm{E}[Y_2 - Y_1|D = \overline{d}] \end{aligned}$$

The first equality is the definition of the ATT, the second equality follows by adding and subtracting $\mathrm{E}[Y_1(\overline{d})|D = d]$, and the third equality follows by Assumption 5. Finally, note that by linearity of the expectation operator and by assuming random sampling from repeated cross sections, each term in the final equation can be estimated by the observed sample averages. □

Nonidentification of the ACRT

A key result from Callaway et al. (2025) noted in the main text is that under standard assumptions the derivative is not identified by the ATT derivative.

Theorem 2.2 *Under Assumptions 1 to 5, the ATT derivative recovers a mix of causal effect parameters and selection bias terms. In particular, for $d \in [0,\overline{d}]$,*

$$\frac{\partial \mathrm{E}[\Delta Y \mid D = d]}{\partial d} = \frac{\partial ATT(d \mid d)}{\partial d} = ACRT(d \mid d) + \underbrace{\left.\frac{\partial ATT(d \mid l)}{\partial l}\right|_{l=d}}_{\text{local selection bias}}.$$

Proof: First, notice that

$$\frac{\partial \mathrm{E}[\Delta Y_t|D = d]}{\partial d} = \frac{\partial ATT(d|d)}{\partial d} = \lim_{h\to 0} \frac{ATT(d+h|d+h) - ATT(d|d)}{h}$$

is not the $ACRT$, which is given by

$$ACRT(d|d) = \lim_{h\to 0} \frac{ATT(d+h|d) - ATT(d|d)}{h}.$$

Instead, we have

$$\begin{aligned} \lim_{h\to 0} \frac{ATT(d+h|d+h) - ATT(d|d)}{h} &= \lim_{h\to 0} \frac{ATT(d+h|d) - ATT(d|d)}{h} \\ &\quad + \lim_{h\to 0} \frac{ATT(d+h|d+h) - ATT(d+h|d)}{h} \\ &= ACRT(d|d) + \left.\frac{\partial ATT(d|l)}{\partial l}\right|_{l=d} \end{aligned}$$

□

However, we can recover identification of the $ACRT$ by the ATT derivative with additional assumptions. In particular, consider the *strong parallel trends* assumption suggested in Callaway et al. (2025).

Assumption 6 (Strong parallel trends). *For all $d \in \mathcal{D}$,*

$$\mathrm{E}[Y_t(d) - Y_t(\overline{d})] = \mathrm{E}[Y_t(d) - Y_t(\overline{d})|D = d].$$

Assumption 6 says that units at all distance levels will have the same ATT response function. This rules out treatment effect heterogeneity along the distance margin. Under this assumption, the $ACRT$ is identified by the ATT derivative.

Theorem 2.3 *Under Assumptions 1 to 6, the $ACRT(d|d)$ is identified. That is, for $d \in [0,\overline{d}]$,*

$$\begin{aligned}\frac{\partial \mathrm{E}[\Delta Y|D = d]}{\partial d} = \frac{\partial ATT(d|d)}{\partial d} &= ACRT(d|d) + \underbrace{\frac{\partial ATT(d|l)}{\partial l}|_{l=d}}_{=0} \\ &= ACRT(d|d).\end{aligned}$$

Proof:

$$\begin{aligned}\frac{\partial ATT(d|d)}{\partial d} &= \lim_{h\to 0} \frac{ATT(d+h \mid d+h) - ATT(d \mid d)}{h} \\ &= \lim_{h\to 0} \frac{\mathrm{E}[Y_t(d+h) - Y_t(\overline{d}) \mid D = d+h]}{h} - \frac{\mathrm{E}[Y_t(d) - Y_t(\overline{d}) \mid D = d]}{h} \\ &= \lim_{h\to 0} \frac{\mathrm{E}[Y_t(d+h) - Y_t(\overline{d})] - \mathrm{E}[Y_t(d) - Y_t(\overline{d})]}{h} \\ &= \lim_{h\to 0} \frac{\mathrm{E}[Y_t(d+h) - Y_t(d)]}{h} \\ &= \lim_{h\to 0} \frac{\mathrm{E}[Y_t(d+h) - Y_t(d) \mid D = d]}{h} \\ &= ACRT(d|d)\end{aligned}$$

The first equality follows from the definition of the derivative. The second equality follows from the definition of the ATT. The third equality follows from the strong parallel trends assumption. The fourth equality uses the linearity of the expectation operator. The fifth equality uses the strong parallel trends assumption again. In the last line, the result follows by taking the limit as $h \to 0$ and the definition of the $ACRT$. □

In the next few results, I consider alternative assumptions on treatment effect heterogeneity to identify or bound the $ACRT$. First, I show the $ACRT$ is identified by the rise over the run in the special case that units' potential outcomes are linear. Second, I show that an alternative assumption of concave counterfactual ATT response functions imply that the estimator under linearity is less than or equal to the true $ACRT$.

Point identifying the $ACRT$ by assuming linearity

Assumption 7 (Linear potential outcomes). *Suppose there exists a known constant $d^0 \in (0, \overline{d}]$ such that d^0 is the minimum distance where spillovers from the plant no longer impact house prices and assume a housing unit j's potential outcomes is piecewise linear in distance to the treatment site. That is, for $D_j \in \mathcal{D}$ we have*

$$Y_{j,t} = Y_{j,t}(d^0) + \gamma_j \min\{D_j - d^0, 0\}$$

where γ_j is a unit-specific slope coefficient.

This assumption says that housing units' potential outcomes follow a linear correlated random coefficients model. Under this model, the average of the unit-specific slope coefficients γ_j is the average causal response $ACRT$.

The following theorem shows that under Assumption 7 the alternative "rise over run" estimator identifies the $ACRT$.

Theorem 2.4 *Under Assumptions 1 to 5 and Assumption 7, the $ACRT$ is identified. In particular, for $d \in [0, d^0)$,*

$$\frac{ATT(d|d)}{d - d^0} = ACRT(d|d).$$

Proof: First, recall the definition of the ATT function:

$$ATT(d|d) = \mathrm{E}[Y_t(d) - Y_t(\bar{d})|D = d].$$

Now plugging in the linear potential outcomes from Assumption 7,

$$\begin{aligned} ATT(d|d) &= \mathrm{E}[Y_t(d^0) + \gamma\ (d - d^0) - Y_t(d^0)|D = d] \quad \text{(By A7)} \\ \frac{ATT(d|d)}{d - d^0} &= \mathrm{E}[\gamma|D = d] = ACRT(d|d) \end{aligned}$$

$\square$

In the main text, I emphasized that this result requires a researcher to know d^0. However, under mild regularity assumptions, d^0 is identified given Assumption 1 and the ATT function, so a researcher does not need to know d^0 ex ante. For example, a researcher could assume the unit-specific slopes are positive $\gamma_j \geq 0$ because the polluting plant is a disamenity (the x-axis on the slope is inverted, so γ is positive with distance from the plant). Then, intuitively, d^0 will be identified by where, starting from $\bar{d}$ and moving in the direction of the treatment site, the ATT first becomes nonzero. The following lemma formalizes this intuition.

Lemma B.1 *Suppose Assumptions 1 to 5 and Assumption 7. In addition, suppose price effects are negative and occur with positive probability in the affected geographic area. That is, assume $\gamma_j \geq 0$ (a.s.) and $\mathrm{P}(\gamma > 0|D = d) > 0$ for all $d \in [0, d^0)$. Then,*

$$d^0 = \sup_d \{d \in \mathcal{D} | ATT(d|d) \neq 0\}.$$

Proof: In Assumption 7, d^0 is a constant such that for all j

$$Y_{j,t} = Y_{j,t}(d^0) + \gamma_j \min\{D_j - d^0, 0\}.$$

We want to show

$$d^0 = \sup_d \{d \in \mathcal{D} | ATT(d|d) \neq 0\}.$$

First, we will show that d^0 an upper bound on the set $\{d : ATT(d|d) \neq 0\}$. For all $d' \geq d^0$, we have

$$\begin{aligned} ATT(d'|d') &= \mathrm{E}[Y_t(d') - Y_t(\bar{d})|D = d'] \\ &= \mathrm{E}[Y_t(d^0) + \gamma \min\{d' - d^0, 0\} - (Y_t(d^0) + \gamma \min\{\bar{d} - d^0, 0\})|D = d'] \\ &= \mathrm{E}[Y_t(d^0) - Y_t(d^0)|D = d'] \\ &= 0 \end{aligned}$$

where the first line follows from the definition of the ATT, the second line plugs in Assumption 7, and the third line follows because $d' \geq d^0$ and $\overline{d} \geq d^0$. Hence, there is no $d' \geq d^0$ in the set $\{d : ATT(d|d) \neq 0\}$, so d^0 is an upper bound.

Second, we will show $ATT(d|d) \neq 0$ for all $d \in [0,d^0)$. Consider a distance $d' \in [0,d^0)$. We have

$$\begin{aligned}
ATT(d'|d') &= \mathrm{E}[Y_t(d') - Y_t(\overline{d})|D = d'] \\
&= \mathrm{E}[Y_t(d^0) + \gamma \min\{d' - d^0, 0\} - (Y_t(d^0) + \gamma \min\{\overline{d} - d^0, 0\})|D = d'] \\
&= \mathrm{E}[\gamma(d' - d^0)|D = d'] = (d' - d^0)\mathrm{E}[\gamma|D = d'] \\
&< 0.
\end{aligned}$$

As above, the first line plugs in the definition of the ATT and the second line plugs in Assumption 7. The third line is algebra. The inequality in the final line follows because $(d' - d^0)$ is negative and we assumed $\gamma_j \geq 0$ (a.s.) and $\mathrm{P}(\gamma > 0|D = d) > 0$ for all $d \in [0,d^0)$, so $\mathrm{E}[\gamma|D = d']$ is positive. Hence, for all $d' \in [0, d^0)$ we have $ATT(d'|d') \neq 0$.

Putting together the two results, over the support of d we have $\{d : ATT(d|d) \neq 0\} = [0, d^0)$. The supremum of an interval $[0, d^0)$ is d^0, so $d^0 = \sup_d\{d \in \mathcal{D}|ATT(d|d) \neq 0\}$. □

Finally, I note that the same result may apply under other mild restrictions on individual effects so long as the ATT becomes nonzero at d^0, e.g. for an amenity, $\gamma_j \leq 0$ (a.s.) and $\mathrm{P}(\gamma < 0|D = d) > 0$. In other words, as long as individual price effects do not follow extreme pathological cases such as the nonzero individual price effects only occurring with 0 probability or the nonzero individual price effects exactly averaging out to 0.

A. *Bounding the $ACRT$ by assuming concavity*

Building off the intuition from the linear potential outcomes case, define the distance threshold

$$d^0_* := \sup_d\{d \in \mathcal{D}|ATT(d|d) \neq 0\}.$$

Assumption 8 (Concave ATT response functions). *Assume the ATT functions for each treatment group subpopulation $d' \in [0,\overline{d}]$ are concave on the interval $[d', \overline{d}]$. That is, for any $\lambda \in [0,1]$ we have*

$$ATT((1-\lambda)d' + \lambda\overline{d}|d') \geq (1-\lambda)ATT(d'|d') + \lambda ATT(\overline{d}|d').$$

The next theorem shows the rise over the run is less than the $ACRT$ if the counterfactual ATT response functions are concave. This follows from the "rooftop" theorem for 1-dimensional concave functions.

Theorem 2.5 *Suppose Assumptions 1 to 5 and Assumption 8. Then, for $d \in [0, d^0_*)$, we have*

$$\frac{ATT(d|d)}{d - \overline{d}} \leq \frac{ATT(d|d)}{d - d^0_*} \leq ACRT(d|d).$$

Proof: The result follows from the "rooftop" theorem for 1-dimensional concave differentiable functions. Assumption 8 implies that for $d' \in [0, d^0_*)$ we also have concavity on the subdomain $[d', d^0_*]$,

$$ATT((1-\lambda)d' + \lambda d^0_*|d') \geq (1-\lambda)ATT(d'|d') + \lambda ATT(d^0_*|d').$$

Rearranging terms,

$$ATT(d' + \lambda(d_*^0 - d')|d') - ATT(d'|d') \geq \lambda(ATT(d_*^0|d') - ATT(d'|d')).$$

Dividing both sides by λ,

$$\frac{ATT(d' + \lambda(d_*^0 - d')|d') - ATT(d'|d')}{\lambda} \geq ATT(d_*^0|d') - ATT(d'|d').$$

Then,

$$(d_*^0 - d')\frac{ATT(d' + \lambda(d_*^0 - d')|d') - ATT(d'|d')}{\lambda(d_*^0 - d')} \geq ATT(d_*^0|d') - ATT(d'|d').$$

Taking the limit as $\lambda \to 0$,

$$(d_*^0 - d')ACRT(d'|d') \geq ATT(d_*^0|d') - ATT(d'|d').$$

Rearranging terms and noting that $ATT(d_*^0|d') = 0$, establishes the middle inequality:

$$\frac{ATT(d'|d')}{d' - d_*^0} \leq ACRT(d'|d').$$

For the other inequality, notice that Assumption 8 implies that $ATT(d'|d') \leq 0$ for any $d' \in [0, d_*^0)$, otherwise $ATT(d|d')$ would not be concave on $[d', \overline{d}]$. Because $ATT(d'|d') \leq 0$ and $\overline{d} \geq d_*^0$, we have

$$\frac{ATT(d'|d')}{d' - \overline{d}} \leq \frac{ATT(d'|d')}{d' - d_*^0} \leq ACRT(d'|d'),$$

which holds for any $d' \in [0, d_*^0)$. □

B. *Bounding the $ACRT$ by assuming convexity*

The main application that motivates the analysis is a disamenity, but researchers also often evaluate amenities with positive price effects. In this case, a natural shape restriction may instead be that price effects are positive, decreasing with distance from the treatment site, and convex.

Assumption (Convex ATT). *Assume the ATT functions for each treatment group subpopulation $d' \in [0, \bar{d}]$ are convex on the interval $[d', \bar{d}]$. That is, for any $\lambda \in [0,1]$ we have*

$$ATT((1-\lambda)d' + \lambda\bar{d}|d') \leq (1-\lambda)ATT(d'|d') + \lambda ATT(\bar{d}|d').$$

This restriction leads to the opposite inequality on the $ACRT$.

Theorem B.2 *Suppose Assumptions 1 to 5 and Assumption Convex ATT. Then, for $d \in [0, d_*^0)$, we have*

$$ACRT(d|d) \leq \frac{ATT(d|d)}{d - d_*^0}.$$

Proof: The same argument as in first part of the proof of Theorem 2.5 instead under Assumption Convex ATT, i.e. flipping the inequality sign, proves this result. □

In the case of an amenity with convex shape, the $ACRT$ will be negative as the positive price effect dissipates with distance from the treatment site. In the above inequality, the $ACRT$ is more negative than the rise over the run. Thus, the rise over run is again a conservative bound in this case.

C Heteroskedasticity by distance to the treatment site

This appendix section considers the rationale for weighting ATT estimates to improve precision by providing a few arguments for why we should expect heteroskedasticity to be a significant issue in this context.

In the main results of this paper, I report estimates weighting by an estimate of the subgroup ATTs' inverse variances. For an ATT estimated around a site g by bin level average house prices,

$$\widehat{ATT}(g,2,d) = (\hat{\bar{Y}}_{g,2,d} - \hat{\bar{Y}}_{g,1,d}) - (\hat{\bar{Y}}_{g,2,\bar{d}} - \hat{\bar{Y}}_{g,1,\bar{d}})$$

where $\hat{\bar{Y}}_{g,t,d}$ is a sample bin mean with estimated variance $\hat{\sigma}^2_{g,t,d}$. I suggest an estimator for the ATT estimate's variance:

$$var(\widehat{ATT(g,t,d)}) = \hat{\sigma}^2_{g,2,d} + \hat{\sigma}^2_{g,1,d} + \hat{\sigma}^2_{g,2,\bar{d}} + \hat{\sigma}^2_{g,1,\bar{d}}$$

where all terms can be estimated using the standard errors of the means in the data. Then, the regression weights are

$$\hat{w}(g,2,d) = \frac{1/var(AT\hat{T}(g,2,d))}{\sum_{g \in \mathcal{G}} 1/var(AT\hat{T}(g,2,d))}.$$

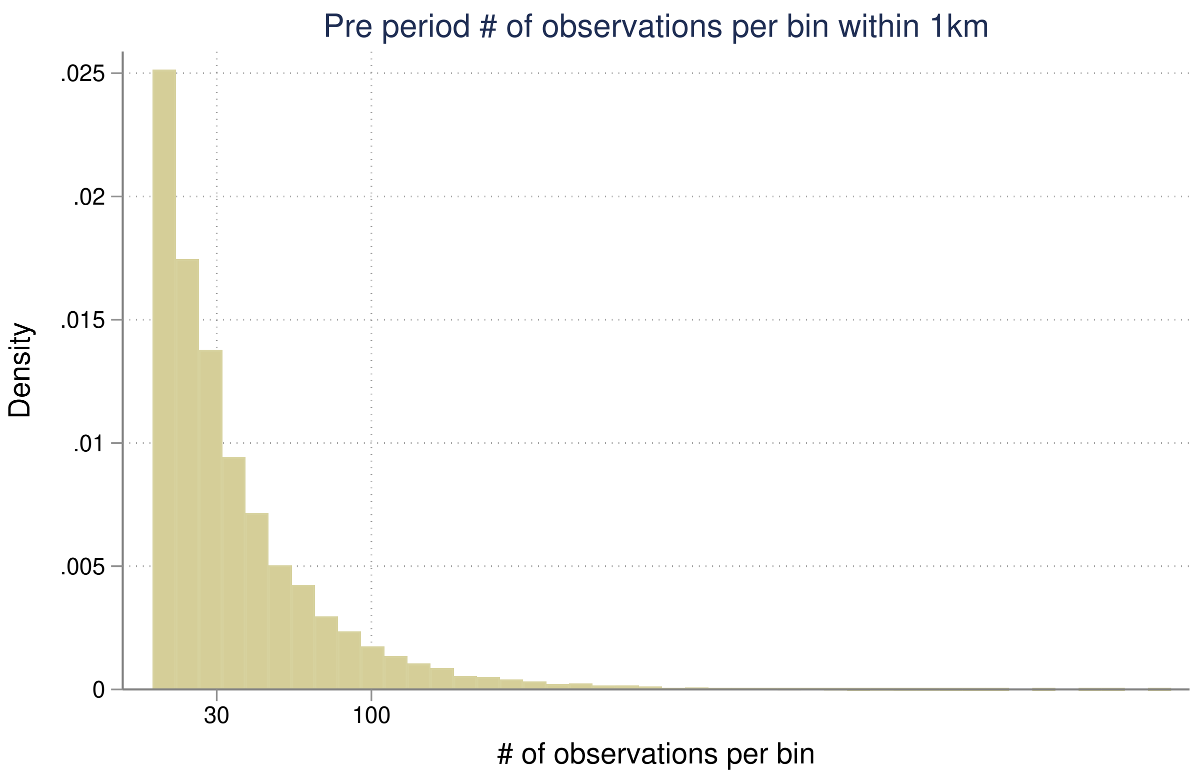
While weighting to target precision can sometimes do more harm than good, Solon et al. (2015) provides some practical guidance for when weighting is more likely to be beneficial. In particular, if the within group sample size is highly variable and small for some groups, then weighting in proportion to the within group sample size can significantly improve precision.

I provide three pieces of evidence for weighting to target precision in this context. First, I argue that we should expect the data generating process to exhibit heteroskedasticity on proximity to the site d. I then show evidence that housing sales are in fact sparsely populated close to the TRI industrial plant sites. Finally, I follow the suggestion in Solon et al. (2015) to test for the presence of heteroskedasticity using a modified Breusch-Pagan test. This test strongly indicates the presence of heteroskedasticity in this setting.

The area of a circle scales geometrically with its radius, suggesting we should expect the number of observations to be larger for distance bins farther away from the treatment site. Consider a sequence of radii of a circle $\{d_1, d_2, \ldots, d_L\}$ that characterize the distance bins used in the main paper. Consider these bins as equally spaced so $d_2 - d_1 = m, d_3 - d_2 = m$, and so on where m is constant. The area of a ring of a circle is $\pi(d^2_{i+1} - d^2_i) = \pi((d_i + m)^2 - d^2_i) = \pi(d^2_i + 2d_i m + m^2 - d^2_i) = \pi(2d_i m + m^2) = \pi 2 d_i m + \pi m^2$. Hence, the area of the distance bins will scale at a linear rate. If on average housing sales are distributed approximately uniform across geographic space, we would expect the number of observations per bin to also scale linearly. Panel B of Table 1 is suggestive of this linear relationship as the number of observations increases for the distance ranges farther away from the treated sites. Similarly, the PDF of housing units used in the welfare calculation shows a linear relationship (Figure 9).

Figure C1 is a PDF of the number of observations per bin within 1 kilometer of a TRI industrial plant in the pre period. A significant share of the distance bins have a small number of housing sales underlying the bin level average house price. For this application of spatial DiD based around a point, it is the case that the within group sample size is highly variable and small for many groups.

Figure C1: PDF number of observations in the pre period by distance to the treated site, TRI



Note: This figure shows the PDF of the number of observations in the pre period for distance bins within 1 kilometer of a TRI industrial plant.

We can also test for the presence of heteroskedasticity empirically. Solon et al. (2015) suggest a modified Breusch-Pagan test. In particular, for the ATT estimates surrounding the sample of TRI industrial plant sites I regress,

$$\widehat{ATT}(g,t,d) = \alpha_0 + \alpha_1 d_{g,d} + u_{g,d}$$

and then get the predicted residuals $\hat{u}_{g,d}$. In a second stage, take the pre period number of obs in the bin as the within group sample size $J_{g,d}$, and regress $\hat{u}^2_{g,d}$ on the inverse of the within group sample size

$$\hat{u}^2_{g,d} = \beta_0 + \beta_1 \frac{1}{J_{g,d}} + \epsilon_{g,d}.$$

The significance of the t-stat for the coefficient β_1 indicates whether there is significant evidence of heteroskedasticity. For TRI industrial plants, the estimated coefficient from the second regression is $\hat{\beta}_1 = 1.14$ with $t = 173.9$, which suggests a high degree of heteroskedasticity.

D Proofs for results in hedonic model section

I restate the results from Section 3 and provide proofs.

Proposition 1 (Rosen (1974)'s first order condition and the $ACRT$). *Suppose Assumptions 1 to 5, Assumption 9, and the standard hedonic model hold. Then*

$$\mathrm{E}\Big[\frac{\partial u_{t=2}}{\partial d}|D=d\Big] = \mathrm{E}\Big[\frac{\partial Y_{t=2}}{\partial d}|D=d\Big] = ACRT(d|d).$$

Furthermore, Section 2 suggested three alternative assumptions for identifying or bounding the $ACRT$:
(a) If strong parallel trends holds (Assumption 6), then

$$\mathrm{E}\Big[\frac{\partial u_2}{\partial d}|D=d\Big] = \frac{\partial ATT}{\partial d}(d|d).$$

(b) If instead linear potential outcomes holds (Assumption 7), then

$$\mathrm{E}\Big[\frac{\partial u_2}{\partial d}|D=d\Big] = \frac{ATT(d|d)}{d-d^0}.$$

(c) If instead concave ATT functions holds (Assumption 8), then

$$\mathrm{E}\Big[\frac{\partial u_2}{\partial d}|D=d\Big] \geq \frac{ATT(d|d)}{d-d^0_*}.$$

Proof: To start, take expectations on both sides of Rosen (1974)'s FOC given by Equation (2) in the main text.

$$\mathrm{E}\Big[\frac{\partial u_2(d,\mathbf{z})}{\partial d}|D=d\Big] = \mathrm{E}\Big[\frac{\partial Y_2(d,\mathbf{z})}{\partial d}|D=d\Big]$$

where the condition holds in the post period $t=2$. I will show that $ACRT(d|d) = \mathrm{E}[\frac{\partial Y_2(d,\mathbf{z})}{\partial d}|D=d]$. Starting with the definition of the $ACRT$,

$$\begin{aligned} ACRT(d|d) &= \frac{\partial ATT(l|d)}{\partial l}\bigg|_{l=d} = \frac{\partial}{\partial l}[ATT(l|d)]\bigg|_{l=d} \\ &= \frac{\partial}{\partial l}\left[\mathrm{E}[Y_2(l) - Y_2(\bar{d}) \mid D=d]\right]\bigg|_{l=d} \\ &= \frac{\partial}{\partial l}\left[\mathrm{E}[Y_2(l) \mid D=d]\right]\bigg|_{l=d} \end{aligned}$$

where the first and second lines follow from the definitions of the $ACRT$ and ATT, and the final line follows because $Y_2(\bar{d})$ does not depend on l. Now, using Assumption 9,

$$\frac{\partial}{\partial l}\left[\mathrm{E}[Y_2(l) \mid D=d]\right]\bigg|_{l=d} = \frac{\partial}{\partial l}[\mathrm{E}[m(l,\mathbf{z}) + h_2(\mathbf{z})|D=d]]|_{l=d}$$

It follows that

$$\frac{\partial}{\partial l}[\mathrm{E}[m(l,\mathbf{z}) + h_2(\mathbf{z})|D=d]]|_{l=d} = \frac{\partial}{\partial l}[\mathrm{E}[m(l,\mathbf{z})|D=d]]|_{l=d}$$

because $h_2(.)$ does not depend on l. Finally, we can interchange the partial derivative and the expectation because of the regularity assumptions in Assumption 3:

$$\begin{aligned} \frac{\partial}{\partial l}\left[\mathrm{E}[m(l,\mathbf{z}) \mid D=d]\right]\bigg|_{l=d} &= \mathrm{E}\left[\frac{\partial m(d,\mathbf{z})}{\partial d}\bigg| D=d\right] \\ &= \mathrm{E}\left[\frac{\partial Y_2(d)}{\partial d}\bigg| D=d\right] \end{aligned}$$

where the last line follows from Assumption 9. Hence,

$$\mathrm{E}\Big[\frac{\partial u_2}{\partial d}|D=d\Big] = \mathrm{E}\Big[\frac{\partial Y_t}{\partial d}|D=d\Big] = ACRT(d|d).$$

The subparts (a), (b), and (c) follow immediately from the theorems in Section 2. Adding Assumption 6 to the assumptions noted at the beginning of the lemma, Part (a) follows from Theorem 2.3. Instead adding Assumption 7 to the assumptions noted at the beginning of the lemma, Part (b) follows from Theorem 2.4. Instead adding Assumption 8 to the assumptions noted at the beginning of the lemma, Part (c) follows from Theorem 2.5. □

Lemma 3.1 *Suppose Assumptions 1 to 5, Assumption 9, the standard hedonic model hold, and homeowners' utility functions are log linear according to Equation (3). Then*

$$\mathrm{E}[WTP|D=d] = (1+d)ACRT(d|d)\log\left(\frac{1+d_*^0}{1+d}\right).$$

As in Proposition 1(a),(b), and (c), depending on the assumption imposed, one of the two $ACRT$ estimators from Section 2 used in place of the $ACRT$ in the right hand side expression will identify or lower bound $\mathrm{E}[WTP|D=d]$.

Proof: This result follows from Proposition 1 and the assumed functional form on homeowner's utility functions in Equation (3). In the main text, the assumed functional form yields

$$WTP_i = \beta_{i,d}\log\left(\frac{1+d_*^0}{1+d_i^*}\right) = (1+d_i^*)\frac{\partial Y_{j,t}}{\partial d}\log\left(\frac{1+d_*^0}{1+d_i^*}\right).$$

Then, taking expectations conditional on d,

$$E[WTP|D=d] = (1+d)\mathrm{E}[\frac{\partial Y_t}{\partial d}|D=d]\log\left(\frac{1+d_*^0}{1+d}\right) = (1+d)ACRT(d|d)\log\left(\frac{1+d_*^0}{1+d}\right)$$

where the last equality follows by Proposition 1. □

Proposition 2 (Aggregate willingness to pay). *Suppose Assumptions 1 to 5, Assumption 9, the standard hedonic model hold, and homeowners' utility functions are log linear according to Equation (3). Then the aggregate willingness to pay for the plant not to exist is*

$$\text{Agg WTP} = N_H\int_0^{d_*^0}(1+d)ACRT(d|d)\log\left(\frac{1+d_*^0}{1+d}\right)f(d)\,\mathrm{d}d. \tag{5}$$

As in Proposition 1 and Lemma 3.1, depending on the assumption imposed, one of the two $ACRT$ estimators from Section 2 used in place of the $ACRT$ in the right hand side expression will identify or lower bound Agg WTP.

Proof: This result follows by integrating homeowners' WTPs over $[0,d_*^0]$ and applying Lemma 3.1. Note that this assumes the functional form on homeowner's utility functions in Equation (3). In the main text, we had

$$\text{Agg WTP} = \sum_i WTP_i = N_H\int_0^{d_*^0}\mathrm{E}[WTP|D=d]f(d)\,\mathrm{d}d.$$

Applying Lemma 3.1,

$$\text{Agg WTP} = N_H \int_0^{d_*^0} (1+d) ACRT(d|d) \log\left(\frac{1+d_*^0}{1+d}\right) f(d)\, \mathrm{d}d.$$

$\square$

Theorem 3.2 (Bounds on Agg WTP). *Suppose Assumptions 1 to 5, Assumption 9, the standard hedonic model hold, and homeowners' utility functions are log linear according to Equation (3). Under concave ATT functions (Assumption 8), homeowners' aggregate willingness to pay is bounded:*

$$\begin{aligned} &N_H \int_0^{d_*^0} (1+d)\frac{ATT(d|d)}{d-d_*^0} \log\left(\frac{1+d_*^0}{1+d}\right) f(d)\, \mathrm{d}d \quad \leq \text{Agg WTP} \\ &\leq -N_H \int_0^{d_*^0} ATT(d|d) f(d)\, \mathrm{d}d. \end{aligned}$$

Proof: I start with the lower bound. The lower bound follows from Theorem 2.5, the parametric assumption on homeowners' utility functions in Equation (3), and the properties of integrals. Under Assumption 8, by Theorem 2.5 we have $ACRT(d|d) \geq \frac{ATT(d|d)}{d-d_*^0}$ for all $d \in [0, d_*^0)$. Because $(1+d)\log\left(\frac{1+d_*^0}{1+d}\right)$ is positive and using the properties of integrals,

$$\begin{aligned} &N_H \int_0^{d_*^0} (1+d)\frac{ATT(d|d)}{d-d_*^0} \log\left(\frac{1+d_*^0}{1+d}\right) f(d)\, \mathrm{d}d. \\ &\leq N_H \int_0^{d_*^0} (1+d) ACRT(d|d) \log\left(\frac{1+d_*^0}{1+d}\right) f(d)\, \mathrm{d}d \\ &= \text{Agg WTP} \end{aligned}$$

where the last line follows from Proposition 2.

I now make the argument for the upper bound. First, note that the canonical hedonic model assumes homeowners are in equilibrium after making a decision from a complete, continuous product choice set. That is, homeowners can freely choose any house across the support of $\mathbf{z}$ to maximize utility. In hedonic equilibrium, I will show that the price difference is greater than each homeowner i's willingness to pay for the plant to not be there. That is, $Y_t(\overline{d}, \mathbf{z_i^*}) - Y_t(d_i^*, \mathbf{z_i^*}) \geq WTP_i$ for any i and $d_i^* \in [0, \overline{d}]$ where other characteristics $\mathbf{z_i^*}$ are fixed at each homeowner's equilibrium choice.

Assume to the contrary that $Y_t(\overline{d}, \mathbf{z_i^*}) - Y_t(d_i^*, \mathbf{z_i^*}) < WTP_i$ for some i. A complete, continuous product choice set implies the homeowner could choose $(\overline{d}, \mathbf{z_i^*})$ at the price $Y_t(\overline{d}, \mathbf{z_i^*})$. From this alternative product bundle they would receive utility $WTP_i - [Y_t(\overline{d}, \mathbf{z_i^*}) - Y_t(d_i^*, \mathbf{z_i^*})]$ relative to their bundle chosen in equilibrium. But $WTP_i - [Y_t(\overline{d}, \mathbf{z_i^*}) - Y_t(d_i^*, \mathbf{z_i^*})] > 0$ because $Y_t(\overline{d}, \mathbf{z_i^*}) - Y_t(d_i^*, \mathbf{z_i^*}) < WTP_i$. Hence, they would have a profitable deviation from their chosen bundle by choosing a house with characteristics $(\overline{d}, \mathbf{z}_i^*)$, and this can not represent a hedonic equilibrium. Therefore, in equilibrium, we must have $Y_t(\overline{d}, \mathbf{z_i^*}) - Y_t(d_i^*, \mathbf{z_i^*}) \geq WTP_i$ for all i.

The ATT identifies average price differences in the post period. $ATT(d|d) = \mathrm{E}[Y_{t=2}(d) - Y_{t=2}(\overline{d})|D = d]$. Notice that

$$\begin{aligned} Y_{t=2}(\overline{d}, \mathbf{z_i^*}) - Y_{t=2}(d_i^*, \mathbf{z_i^*}) &\geq WTP_i \ \ \forall\, i \\ \implies \mathrm{E}[Y_{t=2}(\overline{d}, \mathbf{z}) - Y_{t=2}(d, \mathbf{z})|D = d] &\geq \mathrm{E}[WTP|D = d] \\ \mathrm{E}[Y_{t=2}(\overline{d}) - Y_{t=2}(d)|D = d] &\geq \mathrm{E}[WTP|D = d] \\ -\mathrm{E}[Y_{t=2}(d) - Y_{t=2}(\overline{d})|D = d] &\geq \mathrm{E}[WTP|D = d] \\ -ATT(d|d) &\geq \mathrm{E}[WTP|D = d] \end{aligned}$$

where the first line follows taking expectations on both sides, the second line follows because a change in d holding all other characteristics constant is the potential outcomes, the third line factors out -1, and the final line is by the definition of the ATT. By the properties of integrals,

$$\begin{aligned} & - N_H \int_0^{d_*^0} ATT(d|d) f(d)\, \mathrm{d}d \\ & \geq N_H \int_0^{d_*^0} \mathrm{E}[WTP|D = d] f(d)\, \mathrm{d}d \\ & = \text{Agg WTP}. \end{aligned}$$

Finally, putting together the two bounds,

$$\begin{aligned} & N_H \int_0^{d_*^0} (1 + d) \frac{ATT(d|d)}{d - d_*^0} \log\left(\frac{1 + d_*^0}{1 + d}\right) f(d)\, \mathrm{d}d \quad \leq \text{Agg WTP} \\ & \leq -N_H \int_0^{d_*^0} ATT(d|d) f(d)\, \mathrm{d}d. \end{aligned}$$

□

E Sharpness of lower bound

This appendix discusses sharpness of the lower bound on the $ACRT$. I show the bound in Theorem 2.5 is sharp by showing that for any observed data it is possible to construct potential outcomes such that the bound holds with equality.

To start, I define a notion of sharpness. Let P denote the distribution of observable data. The researcher observes house prices in the pre and post period and houses' distances from the treatment site, (Y_1, Y_2, D). Let P^* denote the distribution of primitives. For Theorem 2.5, I take the potential outcomes and treatments $(Y_t(.), D)$ as primitives. There is some mapping $T()$ that transforms primitives into observable data, so that $T(P^*) = P$. The parameter of interest is the ACRT. Let $\theta(P^*)$ be a function that maps primitives to the ACRT. A lower bound function $lb(P)$ maps the observable data to a bound on the parameter of interest. In this case, the lower bound maps the observable data to the "rise over run" estimand. The lower bound is sharp if for every P, there exists some set of primitives P^{**} consistent with the data and assumptions that achieves the bound such that $T(P^{**}) = P$ and $lb(P) = \theta(P^{**})$. For any observed house prices and distances, I want to show there exists potential outcomes that generate the observed data, satisfy Assumptions 1-5 and Assumption 8, and yield $\frac{ATT(d|d)}{d-d^0_*} = ACRT(d|d)$ for $d \in [0,d^0_*)$.

For each house j on $d \in [0,d^0_*)$, define potential outcomes by

$$\begin{aligned} Y_{j,t=1}(d) &= \gamma_j(d - d^0_*) + \kappa_j \\ Y_{j,t=2}(d) &= \gamma_j(d - d^0_*) + \kappa_j + \Delta \end{aligned} \tag{E1}$$

where γ_j and κ_j are unit-specific slope and intercept coefficients and Δ is a constant time trend common across units. In the pre period, all units have realized outcome $Y_{j,t=1} = Y_{j,t=1}(d^0_*) = \kappa_j$. With two equations and two unknowns, we can set $\kappa_j := Y_{j,t=1}$ and set γ_j such that $\gamma_j := \frac{Y_{j,t=2} - Y_{j,t=1} - \Delta}{(D_j - d^0_*)}$ so that the potential outcomes generate any observed data in the pre or post period. Thus, for any P we can find a P^{**} where the potential outcomes are consistent with the data and follow a linear correlated random coefficients model.

The lower bound function takes the observed data and outputs the "rise over run" estimand, $\frac{ATT(d|d)}{d-d^0}$, and I will show under P^{**} the lower bound equals the parameter of interest. Under (E1), the ATT is

$$\begin{aligned} ATT(d|d) &= \mathrm{E}[Y_2(d) - Y_2(\overline{d})|D = d] \\ &= \mathrm{E}[Y_2(d) - Y_2(d^0_*)|D = d] \\ &= \mathrm{E}[\gamma d - \gamma d^0_* |D = d] \\ &= \mathrm{E}[\gamma|D = d](d - d^0_*) \end{aligned} \tag{E2}$$

Thus, the "rise over run" estimand is

$$\frac{ATT(d|d)}{d - d^0_*} = \mathrm{E}[\gamma|D = d]. \tag{E3}$$

The parameter of interest is the $ACRT$. Under (E1),

$$\begin{aligned} ACRT(d|d) &= \frac{\partial ATT(l|D = d)}{\partial l}|_{l=d} \\ &= \frac{\partial}{\partial l}\left[\mathrm{E}[\gamma|D = d](l - d^0_*)\right]|_{l=d} = \mathrm{E}[\gamma|D = d]. \end{aligned} \tag{E4}$$

Thus, the lower bound in the middle inequality in Theorem 2.5 is sharp.